wattox

MPCS Series

1500 V Power Conversion System Commissioning Guide

Legal Information

Trademarks

All other trademarks or registered trademarks mentioned in this documentation are the property of their respective owners. Unauthorized use of these trademarks by third parties for any purpose without written authorization could violate the rights of their owners.

Disclaimer of Liability

Due to continuous updates and improvements of products and technologies, the content of this documentation may not fully match the actual products. In the event of any discrepancies, the actual products shall prevail.

The contents are subject to change without notice due to product upgrade.

Waste Disposal

The storage, use, and disposal of this product (including optional accessories) must comply with local laws and regulations.

Qualified Personnel

The product/system described in this documentation may be operated only by personnel qualified for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified personnel can identify the risks of the product/system and prevent possible dangers.

Proper Use of the Product

Proper transportation, storage, assembly, installation, commissioning, operation, and maintenance are required to ensure the safe operation of the product without any problems. The required ambient conditions must be met. All operations must follow the guidelines provided in this documentation.

Preface

Introduction

The power conversion system (PCS) exchanges energy between the storage battery and the grid to charge and discharge the storage battery in a controlled way.
In a grid-connected system, the PCS implements load shifting, active power reserve, and reactive power support. In addition, the PCS supports various charge and discharge modes, such as constant voltage, constant current, and constant power.

This guide describes the software tools, processes, and specific steps of commissioning of the product, as well as troubleshooting and fault codes related to the PCS.

Revision History

Date	Version	Description
September 2025	A02	Made minor corrections.
September, 2025	A01	Added 1.725 MW model information.
March 2025	A00	Initial release

Warranty disclaimer

We provide warranty service within the warranty period (as specified in your order) for any fault or damage that is not caused by improper operation of the user. Maintenance fees will be charged after the warranty period expires.

Even within the warranty period, maintenance will be charged for the following product damage:

- Damage caused by operations not following the instructions in the guide
- Damage caused by fire, flood, or abnormal voltage
- Damage caused by using the product for unintended functions
- Damage caused by using the product outside the specified scope
- Damage or secondary damage caused by force majeure (natural disaster, earthquake, and lightning strike)

The maintenance fee will be charged according to the latest Price List. If otherwise agreed upon, the terms and conditions in the agreement shall prevail.

For details, see the Warranty Card.

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1 General Safety Rules

1.1 Safety Instructions

Safety Disclaimer

- This section describes the safety precautions that help you use this product correctly. Before operating the equipment, read through the guide and comprehend all the safety instructions. Failure to comply with the safety precautions may result in death, serious injury, or equipment damage.
- "Danger", "Warning", and "Caution" items in this guide do not indicate all safety precautions that need to be followed; instead, they just supplement the safety precautions.
- Use this product according to the designated environment requirements. Damage caused by improper use is not covered by warranty.
- Wattox shall take no responsibility for any personal injuries or property damage caused by violation to instructions in this manual and improper usage.

Safety Levels and Definitions

indicates that failure to comply with the notice will result in severe physical injuries or even death.

indicates that failure to comply with the notice may result in severe physical injuries or even death.

indicates that failure to comply with the notice may result in minor or moderate physical injuries or equipment damage.

Safety Warning Signs

To prevent misuse or accidents caused by approach of unrelated personnel during installation, daily maintenance, or inspection of the PCS, observe the following:

- Set up a visible sign near the upstream/downstream switch of the power conversion system to prevent accidents caused by inadvertent switch-on.
- Set up a warning sign or safety belt near the operating area.
- Always pull out the key to the cabinet door after maintanence is done.

Requirements on the evacuation passage

DANGER

- During maintenance or other operations of the energy storage converter, keep the the evacuation passage clear of obstacles.
- Do not occupy the evacuation passage.

Safety Instructions

- Drawings in the guide are sometimes shown without covers or protective guards. Remember to install the covers or protective guards as specified first, and then perform operations in accordance with the instructions.
- The drawings in the guide are shown for illustration only and may be different from the product you purchased.
- Operators must take mechanical precautions to protect personal safety and wear protective equipment, such as anti-smashing shoes, safety clothing, safety glasses, protective gloves, and protective sleeves.

Unpacking

WARNING

- Do not install the equipment if you find damage, rust, or signs of use on the equipment or accessories upon unpacking.
- Do not install the equipment if you find water seepage or missing or damaged components upon unpacking.
- Do not install the equipment if you find the packing list does not conform to the equipment you received.

CAUTION

- Check whether the packing is intact and whether there is damage, water seepage, dampness, and deformation before unpacking.
- Unpack the package in accordance with the package sequence. Do not hit the package with force.
- Check whether there is damage, rust, or signs of bumping on the surface of the equipment and equipment accessories before unpacking.
- Check whether the package contents are consistent with the packing list before unpacking.

Storage and Transportation

- Large-scale or heavy equipment must be transported by qualified professionals using specialized hoisting equipment. Failure to comply may result in personal injuries or equipment damage.
- Before hoisting the equipment, ensure that components such as the front cover and terminal blocks are secured firmly with screws. Loosely-connected components may fall off and result in personal injury or equipment damage.
- Never stand or stay below the equipment when the equipment is lifted by hoisting equipment.
- When hoisting the equipment with a steel rope, ensure the equipment is hoisted at a constant speed without suffering from vibration or shock. Do not turn the equipment over or let the equipment stay hanging in the air. Failure to comply may result in personal injuries or equipment damage.

- Handle the equipment with care during transportation and mind your steps to prevent personal injuries or equipment damage.
- Store and transport the equipment based on the storage and transportation requirements. Failure to comply will result in equipment damage.
- Avoid storing and transporting the product in environments with water splash, rain, direct sunlight, strong electric field, strong magnetic field, and strong vibration.
- Store the product for no more than 3 months. For longer storage periods, implement stricter protective measures and conduct necessary inspections.
- Package the product strictly before transporting it, and use an enclosed container for long-distance transportation.
- Never transport the product with devices or materials that may damage or negatively impact the product.

Installation

- The equipment must be operated only by professionals with electrical knowledge. Non-professionals are not allowed to operate on the equipment.

- Before installation, read this guide, especially the safety precautions, carefully.
- Do not install this equipment in places with strong electric or magnetic fields.
- Before installation, check that the mechanical strength of the installation site can bear the weight of the equipment. Failure to comply will result in mechanical hazards.
- Do not wear loose clothes or accessories during installation. Failure to comply may result in an electric shock.
- When installing the equipment in a closed environment (such as a cabinet or casing), use a cooling device (such as a fan or air conditioner) to cool the environment down to the required temperature. Failure to comply may result in equipment over-temperature or a fire.
- Do not retrofit the equipment.
- Do not fiddle with the bolts used to fix equipment components or the bolts marked in red.
- When installing the product in a cabinet or final assembly, make sure that the enclosure of the cabinet or final assembly provides adequate fire prevention, electrical protection, and mechanical protection conforming to relevant IEC standards and local laws and regulations.
- Before installing devices with strong electromagnetic interference, such as a transformer, install a shielding device for the equipment to prevent malfunction.
- Install the equipment on metal or other incombustible objects. Keep the equipment away from combustible objects. Failure to comply will result in fire.

- Cover the air inlet and outlet of the equipment with a piece of cloth or paper during installation. This is to prevent unwanted objects such as metal chippings, oil, and water from falling into the equipment and causing faults. After installation, remove the cloth or paper on top of the equipment to prevent over-temperature caused by poor ventilation due to blocked ventilation holes.

Wiring

- Equipment installation, wiring, maintenance, inspection, or parts replacement must be performed only by professionals.
- Before wiring, disconnect all the power supplies of the equipment. Wait for at least the time designated on the equipment warning label before further operations because residual voltage still exists after power-off. Measure the DC voltage of the main circuit and make sure that it is below the safety voltage. Failure to comply will result in an electric shock.
- Do not perform wiring, remove the equipment cover, or touch the circuit board with power ON. Failure to comply will result in an electric shock.
- Make sure that the equipment is reliably grounded. Failure to comply will result in an electric shock.

- Do not connect the input power supply to the output end of the equipment. Failure to comply will result in equipment damage or even a fire.
- Wiring cables must meet diameter and shielding requirements. The shielding layer of the shielded cable must be reliably grounded at both ends.
- Fix the terminal screws with the tightening torque specified in the user guide. Improper tightening torque may overheat or damage the connecting part, resulting in a fire.
- After wiring is done, check that all cables are connected properly and no screws, washers or exposed cables are left inside the equipment. Failure to comply may result in electric shock or equipment damage.

- Use shielded twisted pairs for the control circuit. Connect the shield to the grounding terminal of the equipment for grounding purpose. Failure to comply can result in equipment malfunction.

Power-on

- Before power-on, check that the equipment is installed and wired properly.
- Check that the power supply meets equipment requirements before power-on to prevent equipment damage or fire accident.
- After power-on, do not open the cabinet door or protective cover of the equipment, touch any terminal, or disassemble any unit or component of the equipment. Failure to comply will result in electric shocks.

- Perform a trial run after wiring and parameter setting to ensure the equipment operates safely. Failure to comply may result in personal injuries or equipment damage.
- Before power-on, check that the rated voltage of the equipment is consistent with that of the power supply. Failure to comply may result in fire accident.

Operation

- The equipment must be operated only by professionals. Failure to comply can result in death or personal injury.
- Do not touch any connecting terminals or disassemble any unit or component of the equipment during operation. Failure to comply will result in electric shocks.

- Do not touch the equipment casing, fan, or resistor with bare hands to feel the temperature. Failure to comply may result in personal injuries.
- Prevent metal or other objects from falling into the equipment during operation. Failure to comply may result in a fire or equipment damage.

Maintenance

- Equipment installation, wiring, maintenance, inspection, or parts replacement must be performed only by professionals.
- Do not maintain the equipment with power on. Failure to comply can result in electric shock.
- Before maintenance, disconnect all the power supplies of the equipment and wait for at least the time designated on the equipment warning label.
- When maintaining the equipment, ensure that the power conversion system has been disconnected from the battery.

- Perform routine and periodic inspection and maintenance on the equipment in accordance with maintenance requirements and keep a maintenance record.

Repair

- Equipment installation, wiring, maintenance, inspection, or parts replacement must be performed only by professionals.
- Do not maintain the equipment with power on. Failure to comply can result in electric shock.
- Before inspection or repair, cut off the equipment power supply and wait for at least the time designated on the equipment warning label, and use a multimeter to check that the energy storage capacitor has no voltage.
- When maintaining the equipment, ensure that the power conversion system has been disconnected from the battery.

- Require for repair services according to the product warranty agreement.
- When the fuse is blown or the circuit breaker or earth leakage current breaker (ELCB) trips, cut off all equipment power supplies, wait for at least the time designated on the equipment warning label before power-on or further operations. Before power-on, check for short circuits before power-on or further operations. Failure to comply may result in equipment damage, physical injuries, or even death.
- When the equipment is faulty or damaged, the troubleshooting and repair work must be performed by professionals that follow the repair instructions, with repair records kept properly.
- Replace quick-wear parts of the equipment in according with the replacement instructions.
- Do not operate on damaged equipment. Failure to comply may result in death, personal injury, or severe equipment damage.
- After the equipment is replaced, check the wiring and set parameters again.

Disposal
WARNING • Dispose of retired equipment by following local regulations or standards. Failure to comply may result in property damage, personal injuries, or even death. • Recycle retired equipment by following industry waste disposal standards to avoid environmental pollution.

Safety Labels

For safe equipment operation and maintenance, comply with the safety labels on the equipment. Do not damage or remove the safety labels. See the following table for descriptions of the safety labels.

Safety Labels	Description
	• The equipment must be serviced by professionals only. • During power-on, connect the control power supply first, and then the main power supply. • During power-off, disconnect the main power supply first, and then the control power supply. • Before maintenance, cut off all power supplies for 60 minutes for complete discharge, and then open the cabinet door. • Do not touch the equipment when power is on. Failure to comply may result in burns. • Read the maintenance guide before maintenance.
	During hoisting, wrap the straps with some flexible materials to avoid cabinet damage.
	Wear ear protector to prevent noise during equipment running.

Safety Labels	Description
	After installation and commissioning, ensure that the door screws are installed properly during product operation. Ensure personal safety.
	This is the protective earth (PE) terminal, which must be grounded firmly to ensure personnel safety.

1.2 Industrial Information Security

This product provides an interface to connect to the network and transmit data through the network interface. To protect factories, systems, machines, and networks from cyber attacks, appropriate industrial information security protection mechanisms need to be implemented to ensure the safe operation of factories, systems, machines, and networks.

The customer is responsible for providing and continuously ensuring a secure connection between the product and the customer's network or any other network, to prevent unauthorized access to their factory, systems, machines, and networks. The system or machine may only be connected to the corporate network or the Internet when a secure connection is established and proper security measures are in place (for example, using antivirus software and installing firewalls).

Wattox continuously develops and improves its products and solutions to enhance safety. It is strongly recommended that you update your products promptly and always use the latest product versions.

Malware (such as viruses, Trojans, and worms) can bring the device into an unsafe operating state, resulting in death, serious injury, and property damage. Observe the following precautions strictly:

- Always use the latest software version. If the product version is no longer supported or the latest program version is not applied, customers are at increased risk of cyberattacks.
- Implement and maintain appropriate security measures (including but not limited to deploying anti-virus software, firewall, WAF, IPS/IDS, situational awareness system, ID verification, and data encryption) to prevent files in the removable storage device from being damaged by malware and to protect products, networks, systems, and interfaces from unauthorized access, disruption, intrusion, data leakage, or information theft.
- Check all safety-related interfaces and settings after commissioning.

2 Product Information

2.1 Working Principle

Main circuit principle

There is one input on the battery side and three-phase three-wire output on the output side. The schematic for the main power is shown below (only for reference).

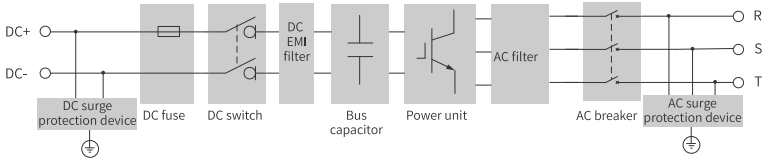


Figure 2-1 Power topology (1725 kW single-branch model)

Operating mode

The PCS can run in grid-connected and off-grid modes.

- When the PCS runs in grid-connected mode, its AC side is connected to the grid and its DC side is connected to the storage battery. The PCS is used with the grid scheduling system for load shifting. The PCS can charge and discharge the storage battery in constant voltage, constant current, or constant power mode, depending on the selected operation mode.
- When the PCS runs in off-grid mode, its AC side is connected to the storage battery. In this mode, the PCS can output three-phase AC voltage with a fixed frequency and an RMS value for continuous power supply to loads connected to its AC side.

2.2 Model description

$$\underset{①}{\text{MPCS}} - \underset{②}{1725} - \underset{③}{\text{S}}$$

① Wattox PCS	② 1725 kW	③ Single Out

2.3 Technical Specifications

Table 2–1 Technical specifications of 1725 kW model

Items		Parameter
DC side	Maximum bus voltage (VDC)	1500
	Maximum current (A)	1807
	Operating voltage range (VDC)	1000 to 1500
	Number of input channels	1
AC grid-connected	Rated power (kW)	1725(@45°C)
	Max. output power (kW)	1898(@35°C)
	Rated output current (A)	1444 (@45°C)
	Maximum output current (A)	1588 (@35°C)
	AC connection mode	Three-phase three-wire, IT, angle form
	Isolation method	No isolation
	Reactive power range	-105% to +105%
	Rated grid voltage (VAC)	690
	Allowable grid voltage (VAC)	621 to 759
	Rated grid frequency (Hz)	50
	Allowable grid frequency deviation (Hz)	±5
	Total harmonic distortion of the current (%)	<3% (rated power)
	AC-side DC component	< 0.5%
	Charge and discharge switchover time (ms)	< 30

Items		Parameter
AC off-grid	Rated output voltage (VAC)	690
	Rated output power (kW)	1725 (@45℃)
AC off-grid	Voltage variation	<1% U_N (linear load balance)
	Degree of voltage unbalance	< 2% or ≤ 4% (short-term)
	Voltage total harmonic distortion rate	< 3% (linear balanced load)
	Rated output frequency (Hz)	50
System	Cable inlet mode	Bottom-in for the DC cable; rear-out for the AC cable
	Overload capacity	110% overload for long term (35℃); 120% overload for 1 minute (35℃)
	Power control accuracy	Error < 1% (after correction)
	Standby energy consumption	≤ 0.5% x Rated power
	No-load energy consumption	≤ 0.8% x Rated power
	Allowable ambient temperature (℃)	-35℃ to +60℃ (derating required for temperature above +45℃)
	Allowable relative humidity (RH)	0% to 100% (non-condensing)
	Transportation/Storage environment temperature (℃)	-40 to +70
	Transportation/Storage relative humidity (RH)	20% to 95%
	Maximum operating altitude (m)	The PCS can work at an altitude no more than 5000 m. Derating is required when the PCS works at an altitude above 2000 m. The actual altitude must be set to obtain the running power curve. For details, contact our technical engineers.
	Noise (dBA)	≤ 85
	Dimensions (W x D x H, in mm)	1080 × 870× 2175
	Weight (kg)	< 1500
	IP rating	Electrical airlock: IP 65; air filter: IP34
	Cooling method	Air-water cooling
	Coolant	Ethylene glycol
Communication	Communication interface	RS485, CAN, EtherCAT, and Ethernet
	Communication protocols	Standard: Modbus-RTU, MODBUS-TCP, and CAN2.0B

CAUTION

The insulation resistance test and dielectric strength test cannot be performed on field for the PCS. Otherwise, the PCS may be damaged.

2.4 Components

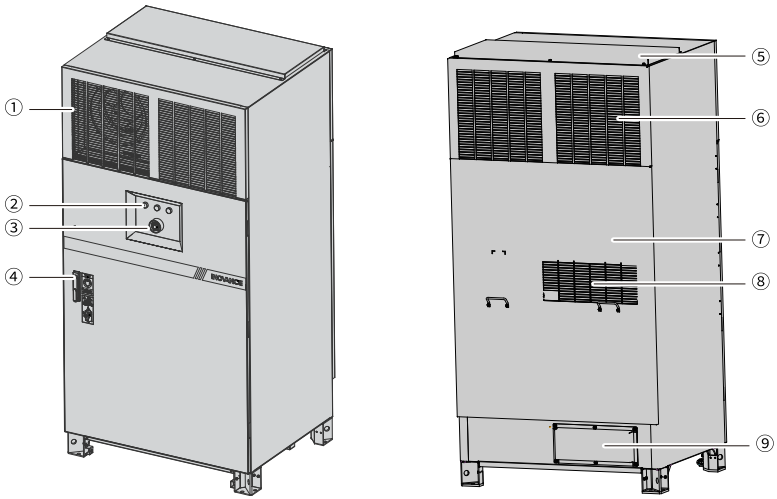


Figure 2-4 Component diagram (1725 kW model)

Table 2–3 Component description (1725 kW model)

No.	Name	Description
①	Fan front cover	Ventilate the fan and prevent incoming of foreign objects.
②	Indicator	• White: Power indicator. When any of the PCS battery voltage, bus voltage, and AC voltage is greater than 36 V, the indicator turns on, indicating that the PCS has been energized and there is an electric shock risk. • Green: Running indicator. When the PCS runs, the indicator turns on. • Red: Fault/alarm indicator. When the PCS is abnormal, the indicator turns on, indicating that the PCS is faulty and you need to locate and remove the fault according to *"8.1 Insulation detection function" on page 112*.
③	Emergency stop button (EPO)	Press this switch in case of emergency to power off the PCS. When a restart is needed after fault is cleared, rotate the switch clockwise to release the switch.

No.	Name	Description
④	Door lock	Open and close the door.
⑤	Top cover	Top outlet
⑥	Fan rear cover	Ventilate the fan and prevent incoming of foreign objects.
⑦	Reactor air outlet hole	Ventilate and prevent incoming of foreign objects.
⑧	Handle	Used for handling and moving.
⑨	AC-side cover	AC outlet

2.5 Product Dimensions

1725 kW model

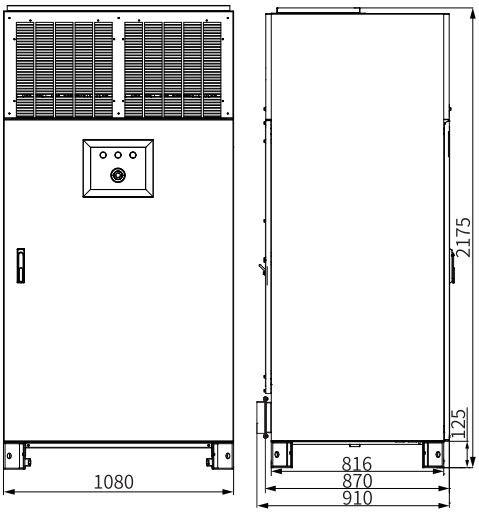
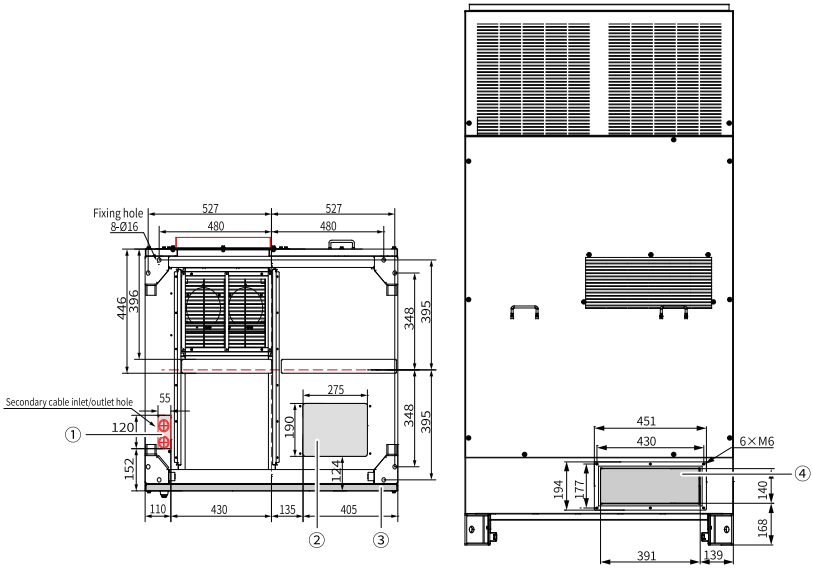


Figure 2-6 Product dimensions for 1725 kW model (unit: mm)



Top view Rear side view

Figure 2-7 Bridging busbar mounting hole of DC cable inlet hole and AC cable inlet hole (unit: mm)

Table 2–5 Bridging busbar mounting hole of DC cable inlet hole and AC cable inlet hole

No.	Name	No.	Name
①	Customer cable inlet holes	③	Front door
②	DC cable inlet hole	④	AC bridging busbar mounting hole[1]

Note

[1]: The AC outlet power cable cover must be removed.

2.6 System Overview

The power conversion system is mainly used for power conversion between the power grid and the energy storage battery, with the exchange process being monitored and managed. The following figure shows the system composition.

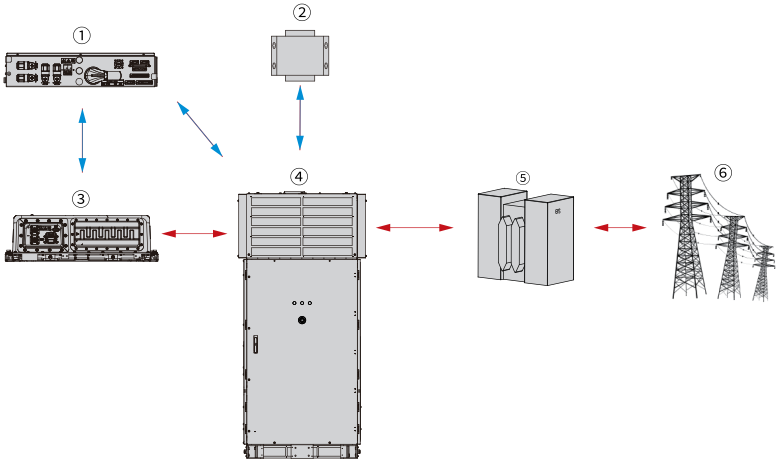


Figure 2-10 Diagram of the power conversion system

Table 2–7 Description of the power conversion system

No.	Name	Description
①	BMS	Battery management system, which is used to manage, protect and monitor the storage battery
②	EMS	Energy management system, which is used for energy dispatching and monitoring of the entire system
③	Battery	All types of storage batteries
④	PCS	Power conversion system, used for energy exchange between the battery and power grid.
⑤	Isolation transformer	You can configure based on actual system needs.
⑥	Power grid	To run the PCS in grid-connected mode, the AC side must be connected to the grid, while the DC side can be connected to the storage battery.

Note

The red cable is the power cable and the blue cable is the communication monitoring cable.

No.	Name	Description
③	Battery	All types of storage batteries
④	PCS	Power conversion system, used for energy exchange between the battery and power grid.
⑤	Isolation transformer	You can configure based on actual system needs.
⑥	Power grid	To run the PCS in grid-connected mode, the AC side must be connected to the grid, while the DC side can be connected to the storage battery.

Note

The red cable is the power cable and the blue cable is the communication monitoring cable.

3 Basic Functions

3.1 Introduction to Operation Status

The PCS can be in the initializing, self-test, stop, soft start, running, fault, and test states. The following figure shows the state switching principle.

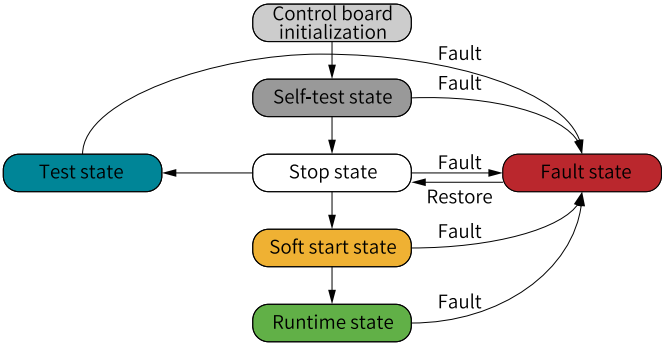


Figure 3-1 State switchover principles

1. When the PCS is powered on and enters the initial stop state, the control system completes a self-test to verify the integrity of the control and sensor system.
2. The MCU starts normally, and the PCS enters the press to stop state. In this state, the PCS blocks the drive and turns off the AC/DC switch.
3. When the host controller issues the power-on command, the PCS enters the soft-start state and performs the AC/DC relay close and soft-start process.
4. After successful soft start, the PCS enters the running state.

State description

- **Initializing**
  When no instruction operation or scheduling is performed on the PCS after it is connected to the AC/DC power, the PCS enters the initializing state after first power-on. In this state, the PCS performs zero current drift detection and grid status detection. After the detection is completed, the PCS automatically enters the self-test state.

- **Self-test**
  When no instruction operation or scheduling is performed on the PCS after it is connected to the AC/DC power, the PCS enters the self-test state after first power-on. In this state, the PCS performs self-test on the following items based on the self-test command word: GFCI, IGBT block, insulation impedance, and heat dissipation of components such as the fan and water pump. Self-test is performed only once after power-on. If no self-test command word is enabled, the PCS waits for a period of time and then enters the stop state.

- **Stop**
  When no instruction operation or scheduling is performed on the PCS after it is connected to the AC/DC power, the PCS enters the press to stop state after initial stop.

  When no fault or alarm is reported, if the PCS receives a local or remote instruction operation or scheduling in stop state, it changes to the running state if the working conditions are met.

  If the PCS receives a stop instruction after it starts running, it changes to the press to stop state.

- **Soft start**
  When the PCS receives an instruction operation or scheduling, it changes to the soft start state. In this state, the PCS checks the AC/DC relay voltage difference and relay adhesion in different working modes and states to complete soft start.

- **Running**
  After AC/DC connection is normal, the PCS enters the running state after soft start is successful. The PCS operates according to the commands and scheduling of the host controller.

  If a fault such as stop occurs, the PCS disconnects the AC/DC relay, reports the alarm information, and changes to the fault state.

- **Fault**
  When the energy storage system is faulty, the PCS stops working, and the AC/DC side switch is disconnected immediately to disconnect the main circuit of the device from the battery, the power grid, and the load.

  The system continuously monitors whether the fault is cleared. If the fault is not cleared, the faulty state remains. If the fault is cleared, wait for a period of time and exit the faulty state. If the manual fault recovery function is activated, the fault state will not disappear until you manually reset the system or press the start/stop button.

- **Testing**
  When no instruction operation or scheduling is performed on the PCS after it is connected to the AC/DC power, the PCS enters the press to stop state after initial stop.

  When no fault or alarm is reported, if the PCS receives a instruction operation or scheduling from the HMI or host controller in stop state, it changes to the test state if test is triggered on the host controller.

3.2 On-grid Operation

The PCS is equipped with the complete on-grid operation function, which enables flexible adaption of charge and discharge control of different batteries. The charge/discharge mode and command are set by the host controller or HMI.

In grid-connected mode, the PCS can convert the electrical energy that enters the battery into AC power and transmits the AC power to the grid. Alternatively, the PCS converts the electrical energy of the grid into DC power to charge the battery. The PCS can operate in charge and discharge states in grid-connected mode according to scheduling commands. The PCS supports the following working modes: constant power PQ mode, constant DC voltage AFE mode, and constant DC current mode.

Mode	Description
Constant power charge/ discharge	In this working mode, the active power for charge and discharge can be adjusted. The PCS charges and discharges the battery based on the predefined power, which can be set locally or remotely.
Constant DC voltage charge mode	In this working mode, the charging mode changes from constant power charging to constant voltage charging when the battery voltage rises to the predefined value during the charging process. After that, the value predefined for constant voltage charging (1150 VDC to 1500 VDC) takes effect, instead of the value predefined for constant power charging.
Constant DC charge and discharge mode	In this operating mode, the PCS has the ability to adjust the charge and discharge current. The PCS charges and discharges according to the set current (the value range is -IN to +IN, and the negative value indicates charging). The current can be set locally or remotely.

All these grid-connected operation modes support both charge and discharge. These modes are applicable to usage scenarios, such as load shifting and smooth grid-connected for new energy.

3.3 Off-grid Operation

The power conversion system supports off-grid operation mode. It continuously provides power supply to AC-side loads during off-grid operation.

In off-grid operation, the DC side of the power conversion system is connected to the battery. The system can serve as the AC voltage source and output three-phase AC voltage with fixed frequency and effective value. The voltage is adjustable within the range of –15% to +10%, and the frequency is 50 Hz.

Droop control can be applied in off-grid mode to support multiple parallel PCSs, flow can be load balanced to these parallel PCSs in any case.

In grid-connected mode, a PCS can directly switch between the charging and discharging states, without a need for state transition.

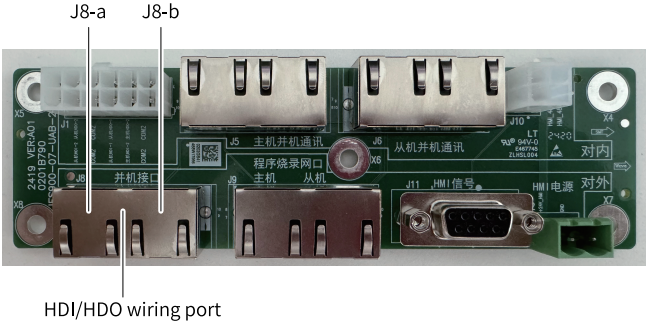
3.4 Carrier Synchronization

When the PCSs are deployed in parallel mode, you must configure the carrier synchronization function and ensure that the main power supply is disconnected and the

bus and power unit are not energized. Open the lower cabinet door of the slave and find the user adapter board to commission the carrier synchronization function.

Carrier synchronization wiring terminals

The HDI/HDO wiring ports are the J8 ports on the user adapter board. The two PCSs are connected through a network cable to implement carrier synchronization.



The following table describes the mapping between pins and carrier synchronization signals.

Table 3–1 Mapping between pins and carrier synchronization signals in parallel mode

Terminal No.	Pin No.	Name	Port	Function
J8	1	PANEL_CANH_1	J8-a	The two PCSs in parallel mode provide output through dual network ports. For details about wiring, see "Wiring procedure" below.
	2	PANEL_CANL_1		
	4	HDO_1+		
	7	HDO_2+		
	5, 8	COM2		
	3, 6	GND		
	17	PE		
	9	PANEL_CANH_3	J8-b	
	10	PANEL_CANL_3		
	12	HDO_1+		
	15	HDO_2+		
	13, 16	COM2		
	11, 14	GND		
	18	PE		

Wiring procedure

1. Confirm the carrier synchronization terminal wiring according to the following figure.

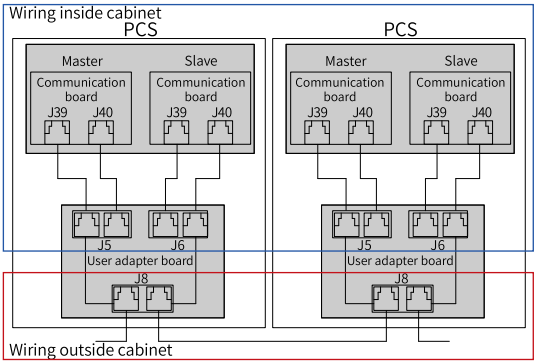


Figure 3-2 Carrier synchronization connection

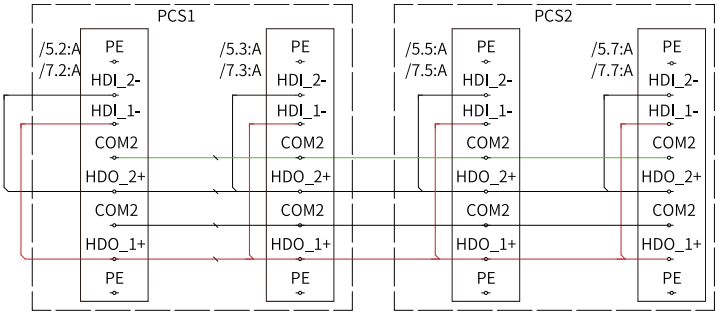


Figure 3-3 Carrier signal connection

2. Access the "Manufacturer" (group F8) page and set the master/slave relationship in parallel mode through the "Local role" parameter. Set "Local role" to "Parallel master" if the PCS functions as the master. Set "Local role" to "Parallel slave" if the PCS functions as the slave.
   You can configure only one master and multiple slaves in a bus. When "Local role" is set "Parallel slave", "Slave delay compen val" must be set to "-800".

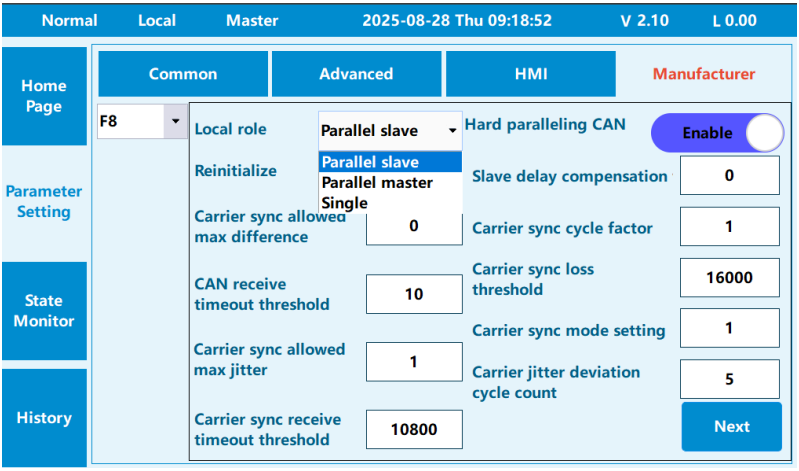


Figure 3-4 Setting the master/slave relationship in parallel mode

3. Click "Next" to check the local carrier synchronization status. If the carrier synchronization status of the master is ongoing or successful, the carrier synchronization function is configured successfully.

3.5 Droop Control

The droop control function is provided by the grid-forming PCS and works in grid-connected and off-grid modes.

When the power grid exists normally, you can use the droop control function to schedule the active power and reactive power of the parallel system as planned. When the power grid does not exist, you can use the droop control function to establish an AC power supply for loads. In this case, the amplitude and frequency of the AC voltage source are automatically adjusted by the load attributes within a certain range.

The droop control function has obvious advantages in the distributed microgrid capable of adopting a balanced energy mix. It can stabilize system running especially in microgrid systems with a large number of rotary devices such as water power and diesel.

3.6 LVRT

Note

According to the Requirements for generating plants to be connected in parallel with distribution networks (EN 50549-2), medium and large power stations shall have low voltage ride through (LVRT) capability.

When different types of faults or interference occur in the power system, which causes voltage drops at grid connection port of a power station, the power station can remain in service within a certain voltage drop range and time interval.

When the AC port voltage falls at or above the voltage contour lines as shown in the following figure, the equipment shall operate continuously without disconnection. The equipment must meet the following requirements for LVRT:

- When the voltage of the AC port falls to 0, the PCS can operate continuously in grid-connected mode for 150 ms.
- When the voltage of the AC port falls to 20% of the rated voltage, the PCS can operate continuously in grid-connected mode for 625 ms.
- When the AC port voltage drops to 90% of the rated voltage, the device can run continuously for 2s without disconnecting from the grid.
- When the AC port voltage drops below the voltage contour lines, the device can be disconnected from the grid.

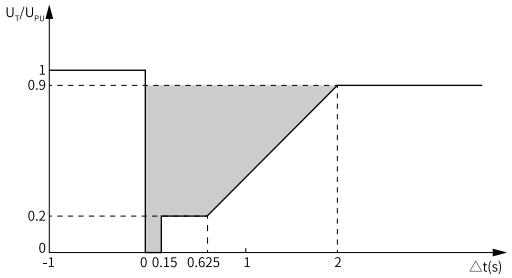


Figure 3-5 LVRT capability of medium- and large-sized power stations

Note

- If a three-phase or two-phase short circuit fault occurs on the AC port of the device, the LVRT test voltage must be the line voltage of the AC port.
- If a single-phase grounding short circuit fault occurs on the AC port of the device, the LVRT test voltage must be the phase voltage of the AC port.

3.7 HVRT

Note

According to the Requirements for generating plants to be connected in parallel with distribution networks (EN 50549-2), the power stations should be able to operate according to the regulations within a certain voltage range of grid-connected port.

When accidents or interference occur in the power system, which causes AC port voltage of the equipment increases, the equipment remain in service within a certain voltage increasing range and time interval.

When the AC port voltage falls at or below the voltage contour lines as shown in the following figure, the equipment shall operate continuously without disconnection. The equipment must meet the following requirements for HVRT:

- When the voltage of the AC port rises to 1.3 times of the rated voltage, the PCS can operate continuously in grid-connected mode for 0.5s.
- When the AC port voltage increases to 125% of the rated voltage, the device can run continuously for 1s without disconnecting from the grid.
- When the AC port voltage increases to 120% of the rated voltage, the device can run continuously for 10s without disconnecting from the grid.
- When the AC port voltage rises above the voltage contour lines, the device can be disconnected from the grid.

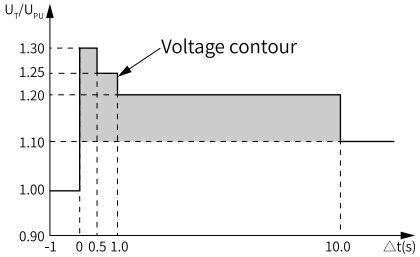


Figure 3-6 HVRT capability of medium- and large-sized power stations

3.8 Temperature-based derating curve

The temperature derating curve changes according to the altitude. The following figure shows the temperature derating curve when the altitude is below 2000 m.

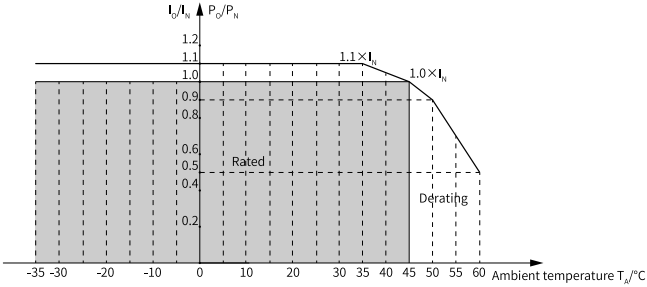


Figure 3-7 Temperature-based derating curve

3.9 Altitude-based Derating Curve

Ambient temperature derating by altitude

Altitude	25°C	30°C	35°C	40°C	45°C	50°C	55°C	60°C
2000 m	110%	110%	110%	105%	100%	90%	70%	50% to 0%
2500 m	110%	110%	105%	100%	95%	85%	65%	45% to 0%
3000 m	110%	105%	100%	95%	90%	80%	60%	40% to 0%
3500 m	105%	100%	95%	90%	85%	75%	55%	35% to 0%
4000 m	100%	95%	90%	85%	80%	70%	50%	30% to 0%
4500 m	95%	90%	85%	80%	75%	65%	45%	25% to 0%
5000 m	90%	85%	80%	75%	70%	60%	40%	20% to 0%

3.10 Protective Functions

The PCS provides comprehensive protection functions to ensure secure operations of the PCS in case of input voltage or grid abnormalities. The PCS continues to generate power in grid-connected mode when the abnormalities disappear.

Protective Functions	Description
DC input voltage exceeding the limit	When the input voltage is lower than 1000 VDC or greater than 1500 VDC, the PCS reports a fault and disconnect the DC contactor.
Module / Ambient overtemperature protection	The PCS with a high-accuracy temperature sensor can monitor the internal temperature of the module or device in real time. When the temperature is higher than the specified threshold, the MCU sends a command to stop the PCS, protecting stable operations of the device.
Output overcurrent protection	When the output current reaches the overcurrent threshold, the PCS stops. After troubleshooting, the PCS can be powered on normally.
IGBT fault	When an IGBT fault occurs, the hardware protection circuit identifies the IGBT fault signal and stops wave output.
Grid voltage error detection	When the grid voltage amplitude is abnormal, the system can quickly detect the fault (undervoltage protection threshold: -20% to 0 of rated voltage, overvoltage protection threshold 0% to 30% of rated voltage, response time less than 100 ms).

Protective Functions	Description
Large current control deviation	When the deviation between the current loop reference and the current loop feedback exceeds the threshold and lasts longer than the specified period, the system sets the current control deviation flag to 1. Otherwise, the system clears the flag.
Grid frequency error detection	When the mains voltage frequency is abnormal, the system can quickly detect the fault (undervoltage protection threshold: -2 Hz to 0 Hz based on rated frequency, overfrequency protection threshold: 0 Hz to 5 Hz based on rated frequency, response time: Less than 200 ms).
Unbalanced AC voltage	When the grid imbalance calculated by the PCS exceeds the allowable range, the PCS stops working.
LVRT tolerance	When the PCS is used as the PQ source and when a voltage drop is detected, the PCS can operate continuously for a period of time without stopping. After the voltage is restored, the PCS can resume operation immediately.
HVRT tolerance	When the PCS is used as the PQ source and when a voltage rise is detected, the PCS can operate continuously for a period of time without stopping. After the voltage is restored, the PCS can resume operation immediately.
Anti-islanding protection	When the grid voltage is 0 or the grid voltage exceeds the allowable range, the PCS stops working. In anti-islanding protection mode, the high voltage inside the PCS is still present. If inspection and maintenance are performed, power off the system and ensure safety before operation.
Resonance suppression	The PCS is equipped with wide-band resonance suppression function.
Grid reverse sequence protection	This function can detect the grid reverse sequence and prevent the PCS from powering on and report the corresponding fault.
Grid phase loss protection	The PCS can be switched off immediately and cannot be powered on after the grid phase loss is detected. The power-off judgment criteria is based on the grid voltage abnormality detection criteria.
Grid overvoltage	• 1.1Un: 11s • 1.22Un: 1.1s
Grid undervoltage	• 0.9Un: 2.1s • 0.5Un: 1.3s
Off-grid overvoltage	1.2Un: 0.1s
Off-grid undervoltage	0.8Un: 0.1s
Output overload	The PCS has overload capacity. It can be used for a long time under 110% load and for a minute under 120% load. Exceeding this overload curve, the PCS stops.
Battery current fault protection	When the battery current exceeds the set value, the PCS stops and the DC contactor trips.
Fan fault	The PCS provides the automatic fan detection function. If a fan is faulty, the PCS can quickly send the fault information to the MCU.

Protective Functions	Description
AD sampling fault	When the PCS detects that the zero deviation of the sampling channel exceeds the allowable range during self-test, the PCS sends an alarm signal.
Battery reverse connection protection	After the DC side battery is connected inversely, the DC contactor is not allowed to close and the fault is triggered.
Battery voltage fault protection	When the battery voltage is higher than the overvoltage or lower than the undervoltage thresholds, the PCS stops and the DC contactor trips.
DC half-bus deviation too large	When the average upper and lower half bus deviation exceeds the protection threshold, the PCS triggers stop protection.
Bus soft start failed	The PCS starts detecting the voltage difference between the battery and the bus capacitor if the DC pre-charge contactor is switched on during start. If the voltage difference is higher than the specified value, the PCS stops wave output and disconnects the AC/DC relay.
Insulation resistance fault	When the insulation impedance is lower than the threshold, the low insulation impedance fault flag is set. The wave is stopped, and the AC/DC relay is disconnected.
DC component beyond the limit on the AC side	When the DC current component of any phase exceeds the threshold, the fault flag is set and the wave is stopped, and the AC/DC relay is disconnected.
Soft start failed on the AC side	In the DC power-on mode, if any phase (three-phase grid voltage RMS - inverter voltage RMS) exceeds the threshold after the DC circuit breaker is closed, the AC soft start fails. The wave is stopped, and the AC/DC relay is disconnected.

When the above protection is triggered, the PCS sends an alarm signal and displays the fault type on the monitoring interface. The PCS can detect the exceptions and respond quickly.

3.11 Insulation detection function

Overview

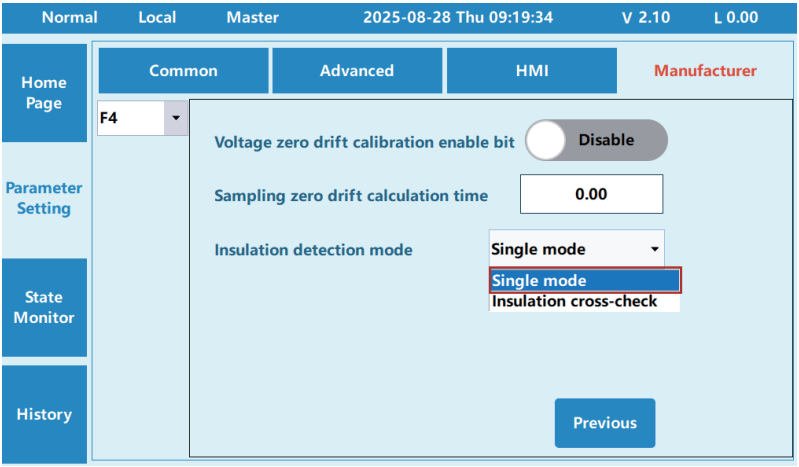
The insulation resistance value is criteria for assessing the insulation performance of electrical systems. It is also critical for the safety of electrical equipment and personnel.

Generally, active or passive insulation detection technology is adopted in the energy field. The unbalanced bridge (passive) method has been widely used because it has high detection accuracy, is easy to operate, and can measure the insulation resistance of the positive bus and negative bus in real time.

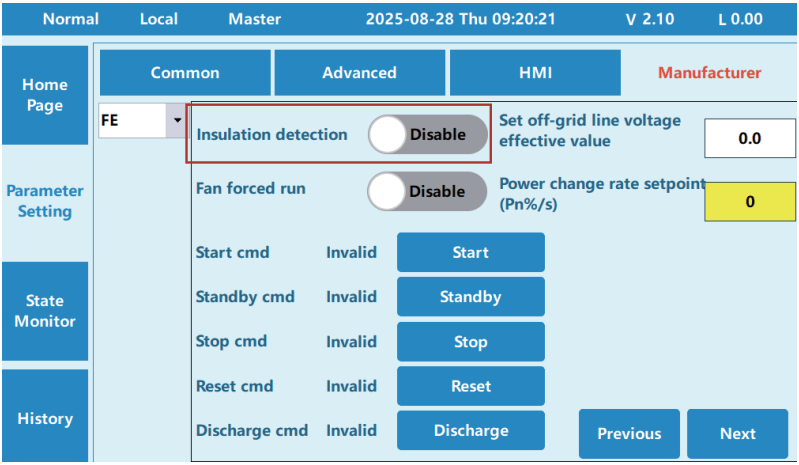
Single and dual DC models adopt the passive insulation detection method. When the battery voltage is greater than 540 V, insulation detection in single mode and mutual insulation detection of master and slave in parallel mode are supported.

Single mode

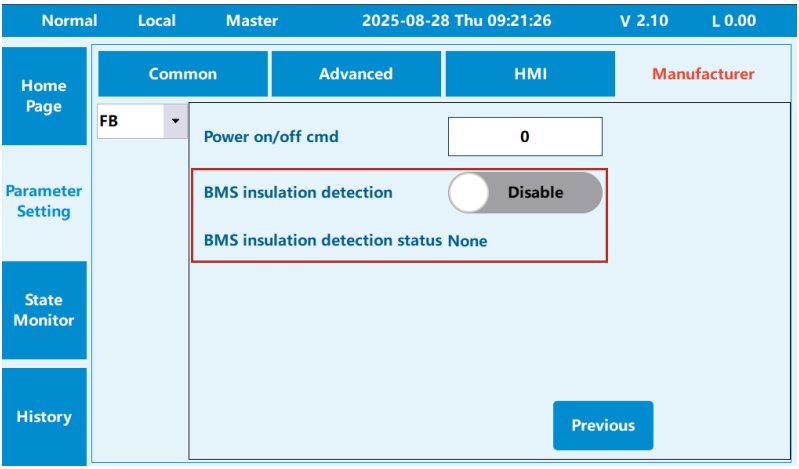
Mode selection: Set the insulation detection mode (F4.19) to 0 (single mode).



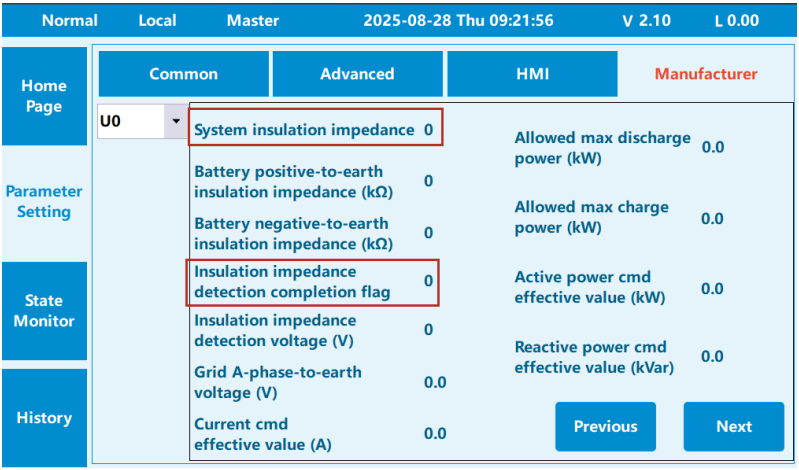
Detection enabling: Set insulation detection (FE.07) to 1 (enable).



BMS detection configuration: If the PCS is connected to the BMS that supports insulation detection, set FB.51 to 1 to enable BMS insulation detection.



Detection result: After insulation detection of the local device is complete, the insulation impedance detection completion flag (U00.82) is set to 1 and the detection result is displayed in U00.73.

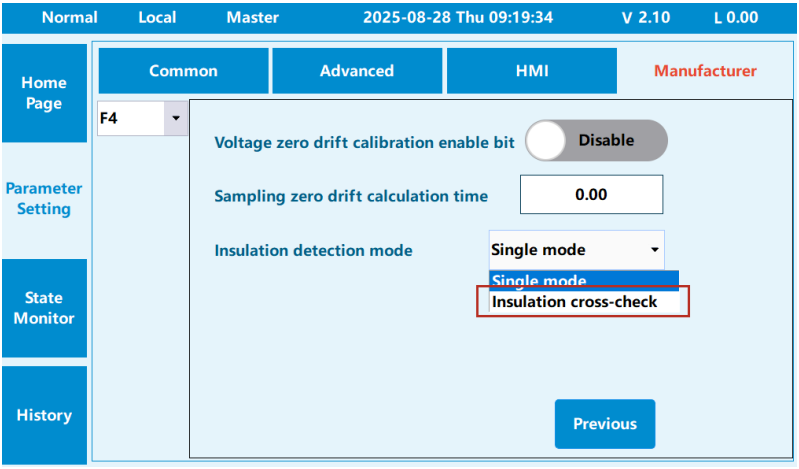


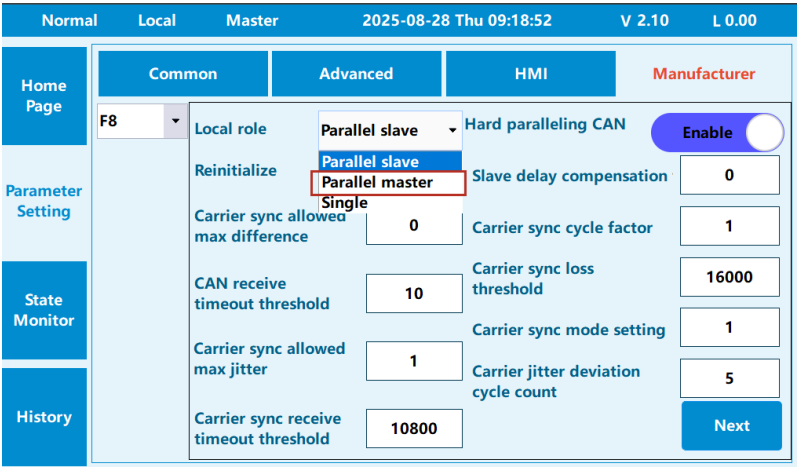
CAUTION

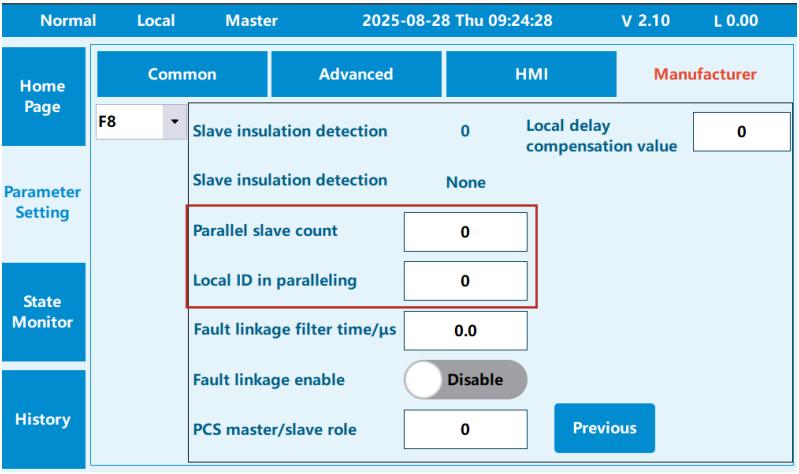
Currently, only the system insulation impedance is displayed. The grounding insulation impedance of the battery positive and negative electrodes is not separately displayed.

Mutual insulation detection

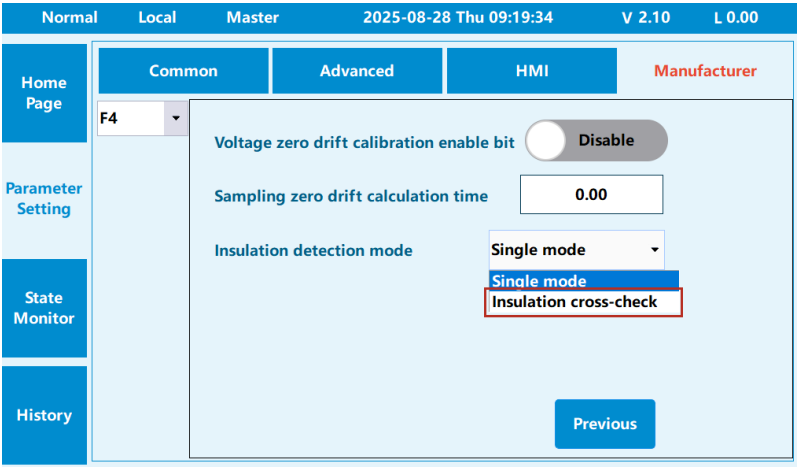
Parallel master configuration: Set the insulation detection mode (F4.19) to 1 (mutual insulation detection), set the Local role (F8.00) to 1 (parallel master), set CAN communication for hardware parallel (F8.26) to 1 (enable), and specify the total number of slaves in parallel mode (F8.41) and the number of the local device in parallel mode (F8.42), which defaults to 0 for the master. If the PCS is connected to the BMS that supports insulation detection, set FB.51 to 1 to enable BMS insulation detection.

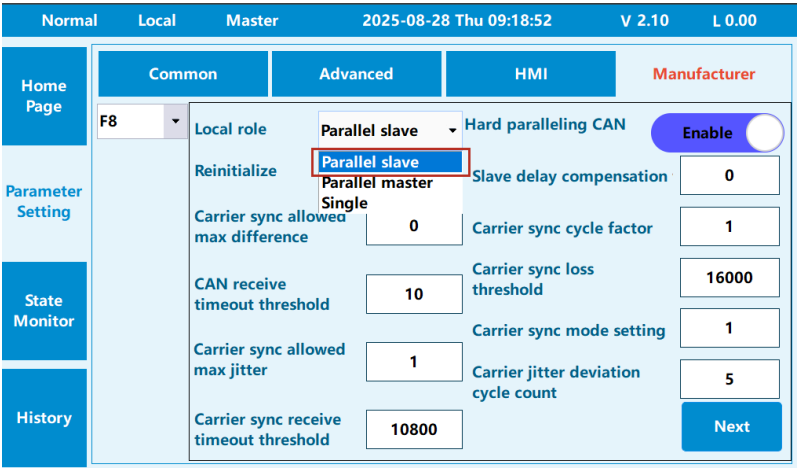


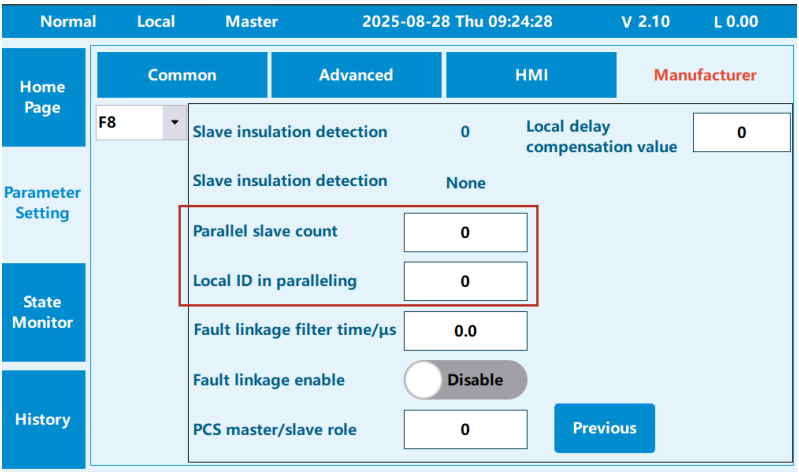




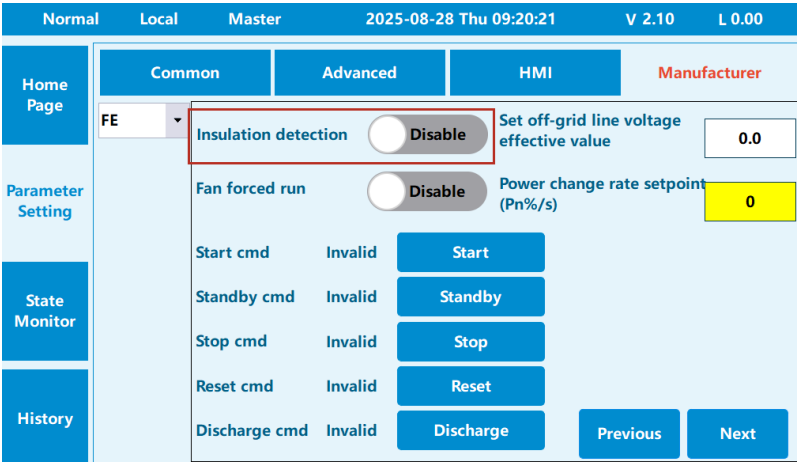
Parallel slave configuration: Set the insulation detection mode (F4.19) to 1 (mutual insulation detection), set the Local role (F8.00) to 0 (parallel slave), set CAN communication for hardware parallel (F8.26) to 1 (enable), and specify the number of the local device in parallel mode (F8.42), which starts from 1 for the slave. If the PCS is connected to the BMS that supports insulation detection, set FB.51 to 1 to enable BMS insulation detection.







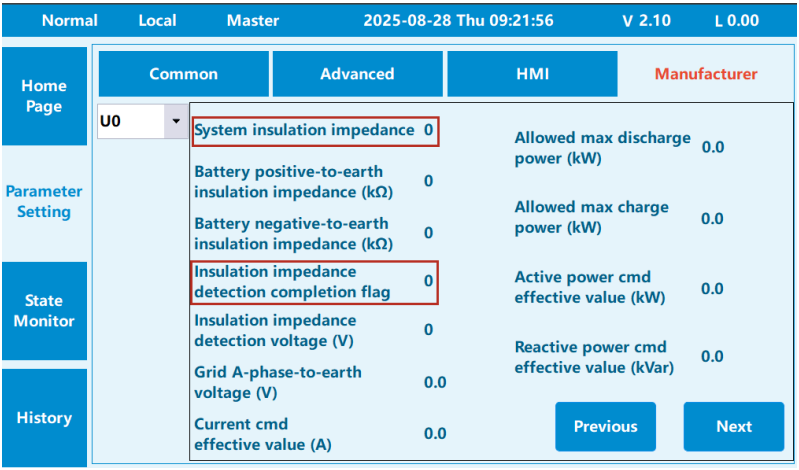
Detection enabling: Set insulation detection (FE.07) to 1 (enable).



CAUTION

If mutual insulation detection is selected for all parallel devices, all the devices start mutual insulation detection only when FE.07 is set to 1 for the parallel master to send a mutual insulation detection command.

Detection result: After insulation detection of the local device is complete, the insulation impedance detection completion flag (U00.82) is set to 1 and the detection result is displayed in U00.73.



Currently, only the system insulation impedance is displayed. The grounding insulation impedance of the battery positive and negative electrodes is not separately displayed.

4 Commissioning Tools

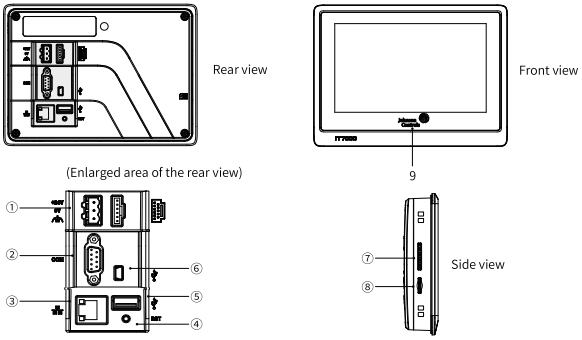
4.1 Commissioning Tool List

Table 4–1 Commissioning tool list

No.	Tool	Description
1	HMI	IT7070E, used to control a single PCS and display the running data of the PCS in real time
2	RS485 DB9 communication cable	Used to connect the RS485 port built in the DB9 socket of IT7070E to the RS485 port on the user adapter board of the PCS
3	iFA Drive	Used to monitor common parameters of the PCS, display and record faults, and record the waves with a black box. It is connected to SOP-20 through the USB flash drive. For details, see the Help center guide of iFA Drive.
4	SOP-20	Used to connect a PC to the PCS
5	IESTool communication commissioning software	Configuration commissioning tool specially designed for energy storage products. It is suitable for configuration commissioning of IES1000 V1.3 and later versions. The main functions include remote commissioning of the SCL configuration and the PCS. For details, see the Help center guide of IESTool.
6	Wireshark communication analysis software	Network packet analysis software, formerly known as Ethereal. Wireshark uses WinPCAP as the port to directly exchange data packets with the NIC. For details, see the user guide at **https://www.wireshark.org/docs/wsug_html_chunked/**.
7	Commissioning PC	Used for software upgrade and maintenance
8	Screwdriver suite	Regular tool used in the site
9	38-piece sleeve suite	Regular tool used in the site
10	Multimeter	Used for electric charge detection in the site
11	Insulation meter	Used to test the insulation of positive and negative cables
12	Clip-on ammeter	Used to measure the wire current
13	Phase sequence tester	Used to test the sequence of inlet wires
14	Insulation gloves	Used for protection during electric charge detection

4.2 HMI

4.2.1 HMI Components



No.	Name	Function Description	No.	Name	Description
①	Power supply port	Used to input 24 VDC power to the HMI (One power terminal for this port is delivered with the HMI.)	⑥	Mini USB port	Secondary port for USB communication, used for PC file download and user program commissioning
②	DB9 male	Used for communication between the HMI and peripheral devices; includes three built-in serial ports COM1, COM2, and COM3 COM1: RS422 or RS485 COM2: RS232 COM3: RS485	⑦	Firmware upgrade port	For firmware upgrade
③	Ethernet Terminal	Ethernet communication port (RJ45) that used to access the PC or PLC with LAN port	⑧	Micro SD card slot	Mounts a Micro SD card for data read/write and project file download

No.	Name	Function Description	No.	Name	Description
④	RESET key	Used to restore to default settings	⑨	Power supply indicator	ON: normal Blinking: unstable OFF: power is off
⑤	USB Type A port	Main port for USB communication, used to read and write data in USB flash drives and connect to a mouse, printer, or other USB devices	/	/	/

4.2.2 Connection

Use a RS485 DB9 communication cable to connect the PCS user adapter board and the HMI.

4.2.3 Page Display

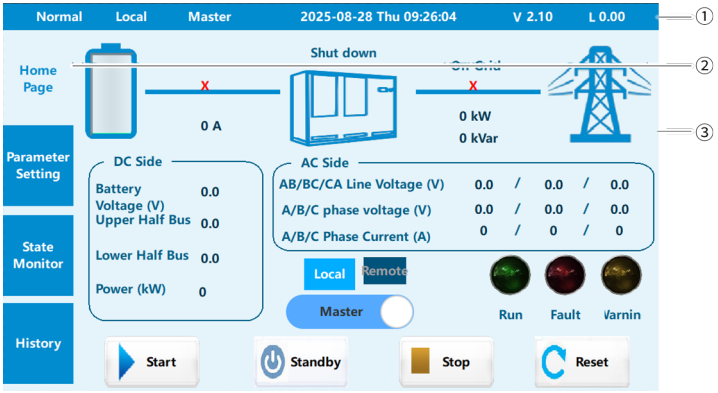


Figure 4-1 HMI page layout

No.	Item	Detailed Description
①	Top column	Displays the communication state, command source, time, and PCS model.
②	Menu bar	Switches among Level-1 menus: Home Page, Parameter Setting, State Monitor, and History Record.
③	Content display area	Displays the content of the current menu.

5 Commissioning and Operation

5.1 Commissioning Flowchart

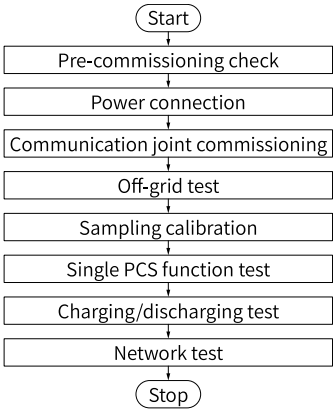


Figure 5-1 Commissioning flowchart

5.2 Safety Precautions

Note

- Read and follow the safety cautions prior in this manual.
- Only qualified professionals are allowed to operate the equipment.
- The equipment is typically used in industrial applications with high voltage. Its parts become live during operation. Therefore, unauthorized removal of required covers, improper use or operation, or poor maintenance can lead to property loss, severe physical injury, or even death.
- Ensure that the all-in-one equipment and other components are installed and connected in accordance with the national technical rules and other applicable regional regulations.
- Ensure that the equipment is properly grounded.
- Cover or isolate the adjacent components that are still live.
- If the fuse in a branch circuit trips, the fault current may have been cut off. To reduce the risk of fire and electric shock, inspect the conductive parts and other components in the cabinet and replace damaged parts as required. If any fuse trips, figure out the cause and rectify the fault.

5.3 Pre-commissioning Check

No.	Item	Checked
1	For the installation and operation environments, see the requirements on the installation environment and site.	☐
2	The power conversion system is fixed to the rack properly.	☐
3	The power conversion system is grounded properly.	☐
4	The grounding busbar, DC positive/negative busbar, and busbar on the AC side are connected properly.	☐
5	The voltage on the both ends of the battery is lower than 1500 VDC and in the range of the corresponding model specifications. The positive and negative polarities are connected correctly.	☐
6	The voltage on the AC side is in the range of model technical specifications.	☐
7	There is no sign of damage or scratching on the cable. The air filter of the air vent is intact.	☐
8	Perform insulation test on the circuit between the battery components to the DC input side of the power conversion system to ensure no grounding fault exists.	☐

5.4 Power-on

Start the device by strictly following the procedure. Failure to comply can result in damage or malfunction of the device.

1. Check that the AC and DC sides of the PCS are connected properly (AC side: 690 VAC power supply or grid; DC side: Battery power supply).
2. After the power supply on the AC and DC sides is on, the PCS is powered on. If the power indicator of the PCS is steady on, the PCS is successfully powered on. By default, the PCS is in the stop state and the main switches on the AC side and DC side are off.

5.5 Communication Joint Commissioning

If one of the following occurs, restore factory settings before you perform communication joint commissioning:

- The software is upgraded before communication joint commissioning.

- The software is upgraded during communication joint commissioning due to a communication issue. (Record communication parameters before you restore factory settings.)
- The detection board and control board are replaced during the commissioning process. (Record communication parameters and the calibration parameters in the FG group before you replace the detection board. Check whether the model code is correctly set before you restore factory settings.)

To restore factory settings, do as follows:

1. Go to the "Manufacturer" page and select the FF group.
2. Select "Restore factory param" to "Restore default"

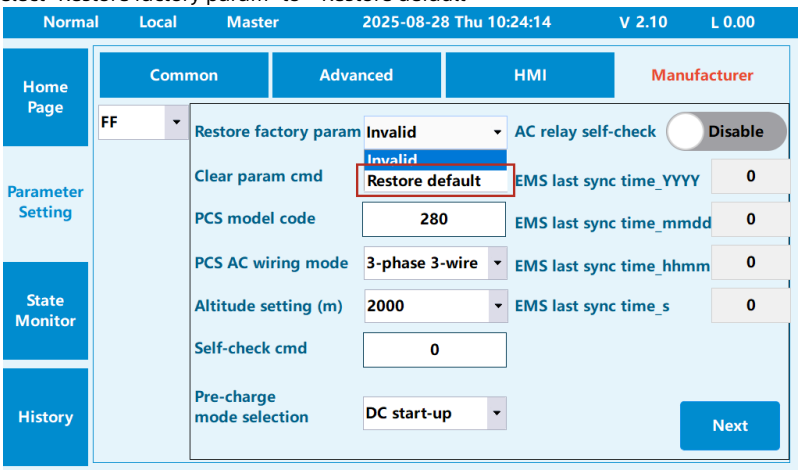


Figure 5-2 Manufacturer

3. Wait for a few seconds and "Restore default" resumes to "Invalid".

5.5.1 EMS Communication Setting

Note

Modifications to parameters in groups FA and FB take effect only after the smart PCS is re-powered. The control power must be disconnected during the re-powering process. That is, the indicator on the control board turns off.

Modbus RTU (RS485)

1. Go to the "Manufacturer" page and select the FA group.
2. Set "EMS communication protocol" to "RTU" and enable communication control.
3. Check the local address, baud rate, data bit, checksum, and stop bit of the PCS with EMS and ensure they match the EMS communication settings, as shown in the figure below.

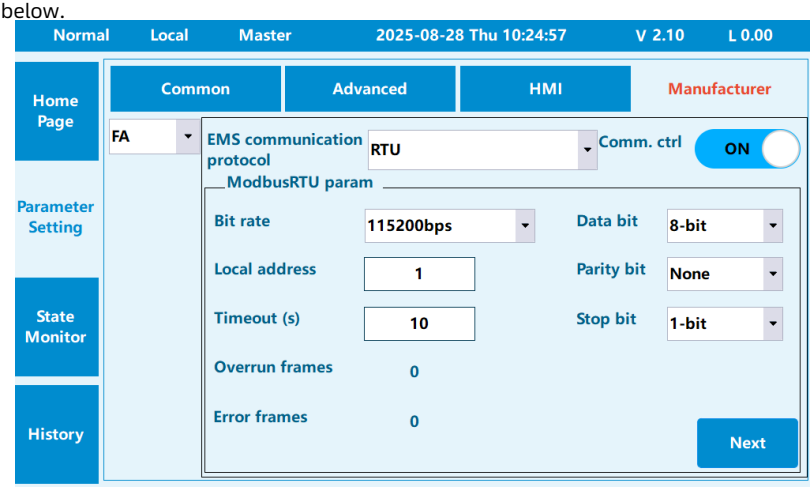


Figure 5-3 Manufacturer

4. Based on the model, check and confirm that the EMS RS485+ and RS485- communication cables are reliably connected.
5. Based on the model, set the corresponding DIP switch on the PCS communication board to ON.

Modbus TCP

1. Go to the "Manufacturer" page and select the FA group.
2. Set "EMS communication protocol" to "TCP" and enable communication control.

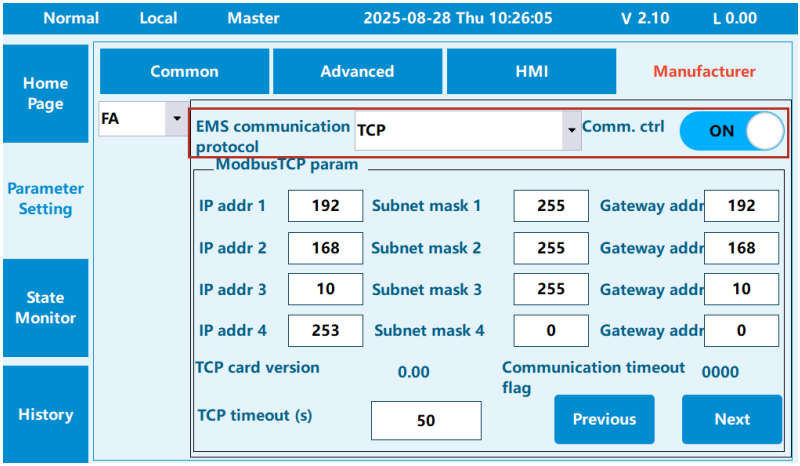


Figure 5-4 Manufacturer

3. Check whether the expansion card version is displayed (① in the figure below) and whether the status is online. If the status is offline, verify the card installation position and restart the PCS.

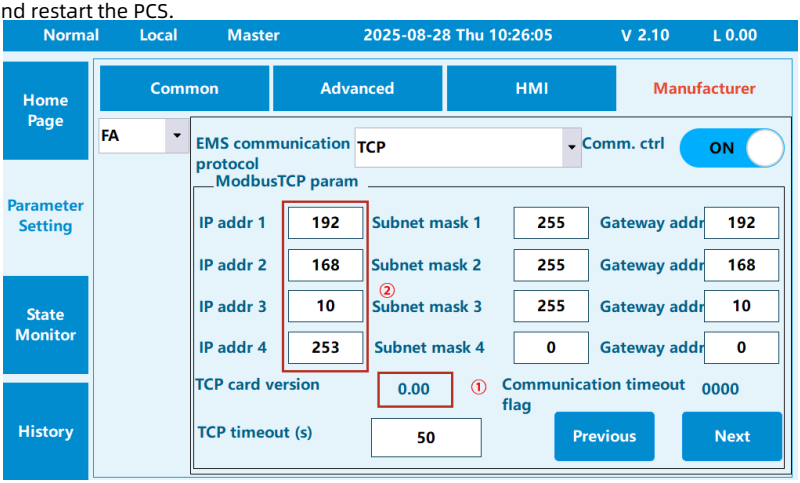


Figure 5-5 Manufacturer

4. Check the IP address (② in the figure above) and other settings with EMS, ensuring they match the host controller communication settings.
5. Check and confirm that the network cable is securely connected to the TCP expansion card and relay ports.

5.5.2 BMS Communication Setting

Modbus RTU (RS485)

1. Select the "Manufacturer" tab and select the FB group.
2. Set "BMS communication protocol" to "RTU" and enable communication control.
3. Check the local address, baud rate, data bit, checksum, and stop bit of the PCS with BMS and ensure they match the BMS communication settings, as shown in the figure below.

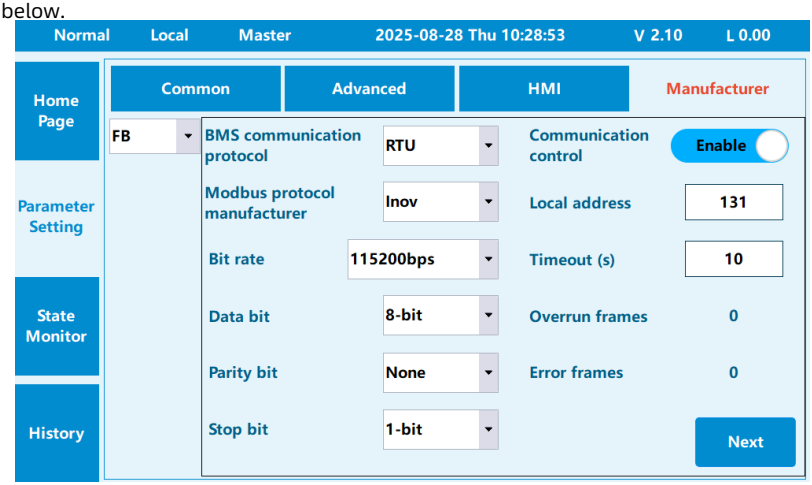


Figure 5-6 Manufacturer

4. Based on the model, check and confirm that the BMS RS485+ and RS485- communication cables are reliably connected.
5. Based on the model, set the corresponding DIP switch on the PCS communication board to ON.

CAN

1. Go to the "Manufacturer" page and select the FC group.
2. Set "BMS communication protocol" to "CAN" and enable communication control.
3. Ensure that the settings of CAN protocol manufacturer, CAN baud rate, and BMS address match the BMS equipment manufacturer, as shown in the figure below.

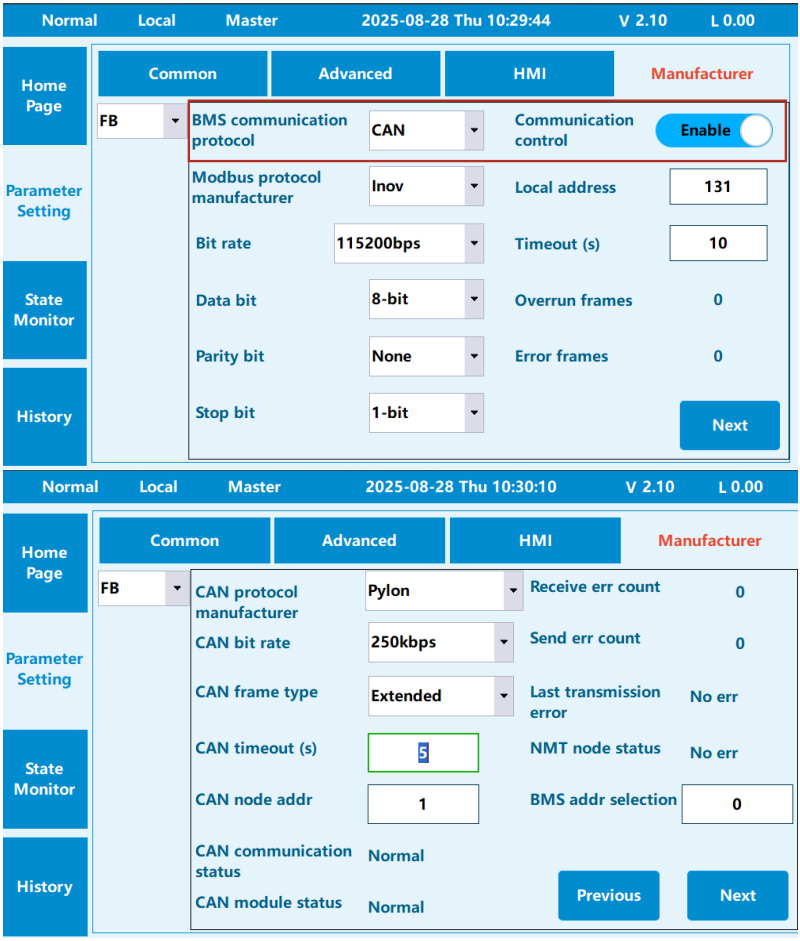


Figure 5-7 Manufacturer

4. Based on the model, check and confirm that the BMS B_H and B_L communication cables are reliably connected to the interfaces.
5. Based on the model, set the corresponding DIP switch on the PCS communication board to ON.

5.6 Off-grid Test

1. Connect the DC side (DC side: Battery) of the PCS properly. Power on the DC side of the PCS after the wiring is completed.
2. Check the communication status in the top-left corner of the HMI homepage to confirm normal communication.

3. On the parameter setting page, set "Run Mode" to "Off-Grid". Set the off-grid voltage to 690 VAC and frequency to 50 Hz.
4. Return to the HMI homepage, click the "Start" command button. Confirm the Run indicator is steady green, indicating successful operation.

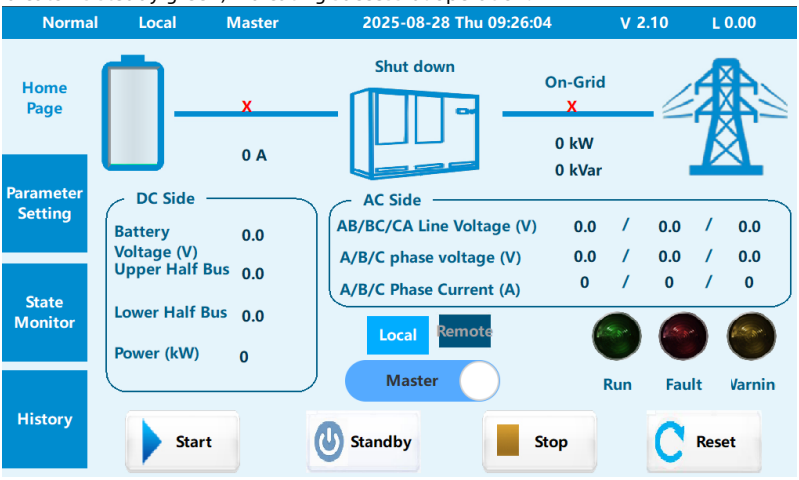


Figure 5-8 HMI homepage

5. Run for 10 minutes in off-grid mode. If no fault occurs, the off-grid test is completed.

The following table describes parameters related to commissioning in the IDS background.

Table 5–1 Parameters related to commissioning in the IDS background

Param.	Name	Value Range	Default	Unit	Change Method	Effective Mode
FE.00	Command source	0: HMI 1: EMS 2: Software (Reserved)	1	-	Change at stop	Real time
FE.01	PCS operation mode	0: On-grid mode 1: Off-grid mode	0	-	Change at stop	Real time
FE.02	PCS on-grid control mode	0: PQ mode 1: Constant DC current mode 2: Constant DC voltage mode	0	-	Change at stop	Real time
FE.20	Start command	0: Disabled 1: Send the start command	0	-	Real time	/
FE.21	Stop command	0: Disabled 1: Send the stop command	0	-	Real time	/

5.7 Sampling Calibration

CAUTION

When you perform calibration operations, do not measure or touch the masked areas on the product diagram. You can only perform measurement through the AC and DC detection terminals. Failure to comply may cause electric shocks.

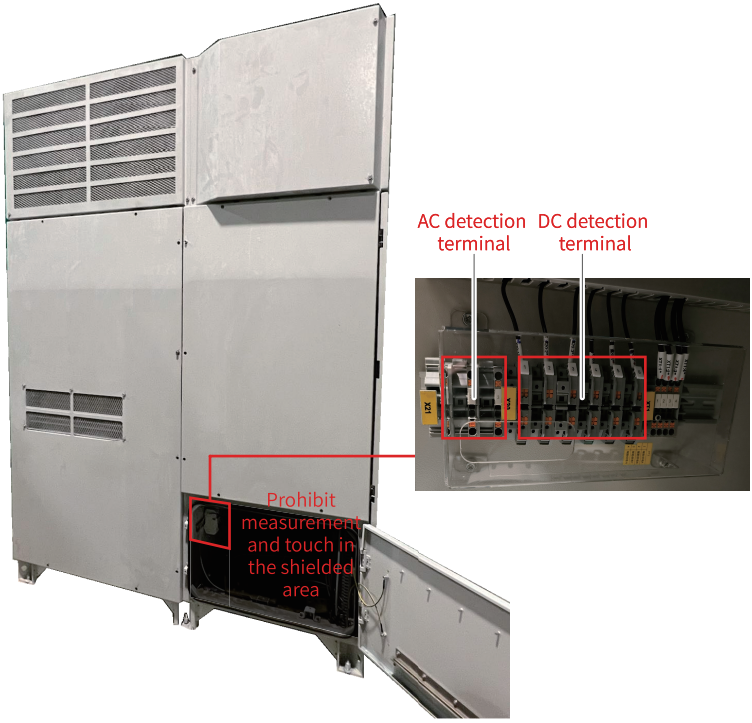
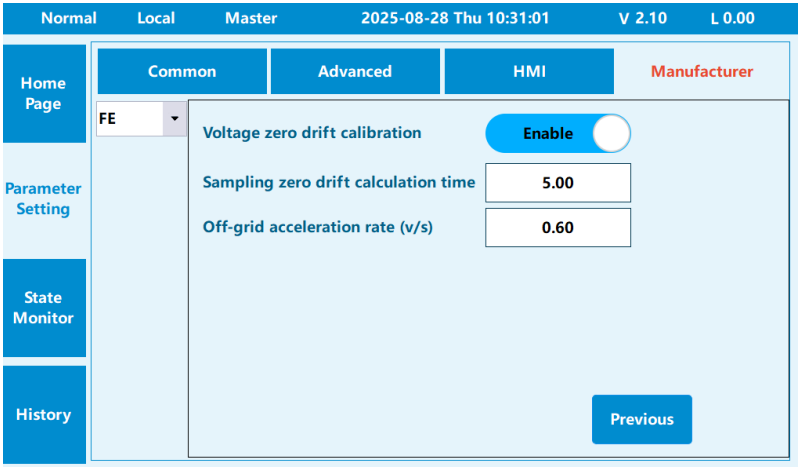


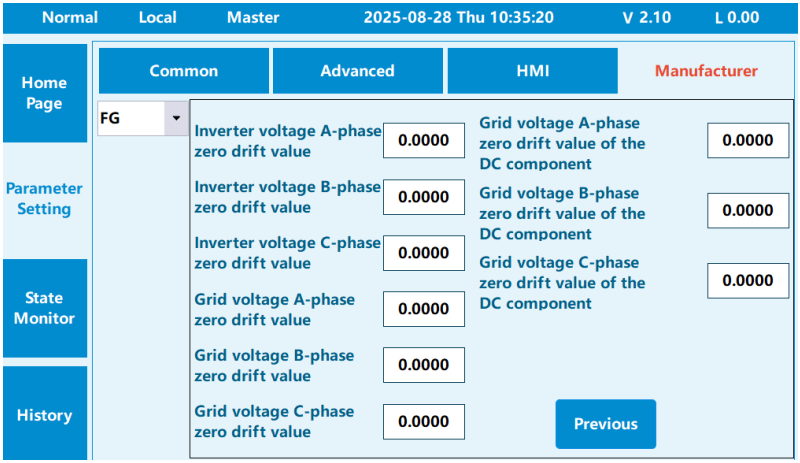
Figure 5-9 Area where internal operations are prohibited

5.7.1 Zero Drift Calibration

1. Go to "Parameter Setting" > "Manufacturer".
2. Perform the calibration when there is no AC-side voltage. Set "Voltage zero drift calibration" in the FE group to "Enable".



3. Check whether parameters in the FG group are in the reasonable range.
   After zero drift calibration, the AC-side voltage displayed on the homepage shall be about 0 V, with a deviation not greater than 2 V.



You can also contact Wattox service engineers to obtain the Calibration calculator.

5.7.2 Voltage Calibration

Calibration is required in case of control deviations, output accuracy lower than expectation, control exceptions, and fault reporting by mistake. You do not need to perform calibration operations if the PCS on the site has been calibrated (The FG coefficient is not 1.) or calibration is not definitely required on the site.

Preparation before calibration

Contact the Wattox engineer to obtain the Calibration calculator.

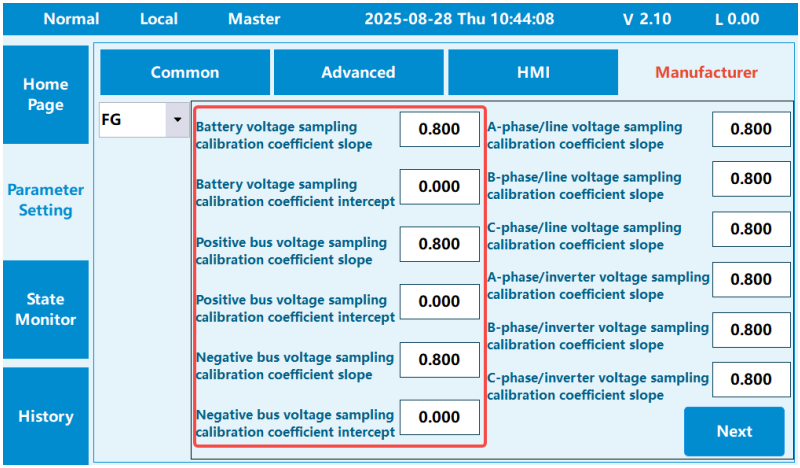
Calibrating the upper/lower half-bus voltage and battery voltage

During calibration, ensure that the span of the multimeter is within the measuring range. To calibrate the upper/lower half-bus voltage and battery voltage, measure the upper/lower half-bus voltage and battery voltage of two groups and calibrate the values through linear least squares fitting by using the following formula: y = kx + b.

Check whether the upper/lower half-bus voltage displayed on the HMI deviates from the actual voltage. For a single-circuit device, measure the voltage between pins 1 and 2 of X20 as the upper half-bus voltage and between pins 2 and 3 of X20 as the lower half-bus voltage. For a dual-circuit device, measure the voltage between pins 1 and 2 of X20 as the upper half-bus voltage of the master, between pins 2 and 3 of X20 as the lower half-bus voltage of the master, between pins 4 and 5 of X20 as the upper half-bus voltage of the slave, and between pins 5 and 6 of X20 as the lower half-bus voltage of the slave.

If deviations are found, calibrate the PCS as follows:

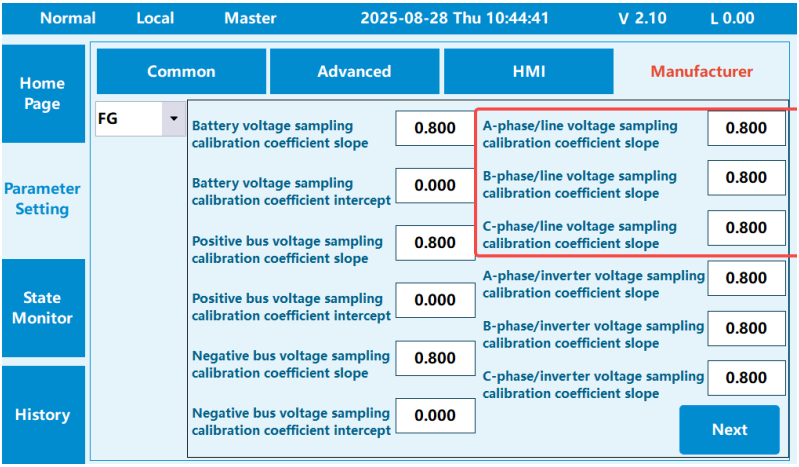
1. Go to "Parameter Setting" > "Manufacturer" > "FG". Set "Upper half bus voltage", "Lower half bus voltage", and "Battery voltage" to 1.
2. Under the first group of battery voltage (such as Udc = 1100 V), click "Standby", measure the upper half-bus voltage U_up between pins 1 and 2 of X20, the lower half-bus voltage U_down between pins 2 and 3 of X20, and the battery voltage U_bat between pins 1 and 3 of X20, and enter the values in the first group of "Measured Value (y)" in the calibration calculator.
3. Record the upper half-bus voltage U_up_dis, lower half-bus voltage U_down_dis, and battery voltage U_bat_dis displayed on the HMI in the first group of "Background Value (x)" in the calibration calculator.
4. Modify the battery voltage. Under the second group of battery voltage (such as Udc = 1400 V), click "Standby", measure the upper half-bus voltage U_up between pins 1 and 2 of X20, the lower half-bus voltage U_down between pins 2 and 3 of X20, and the battery voltage U_bat between pins 1 and 3 of X20, and enter the values in the second group of "Measured Value (y)" in the calibration calculator.
5. Record the upper half-bus voltage U_up_dis, lower half-bus voltage U_down_dis, and battery voltage U_bat_dis displayed on the HMI in the second group of "Background Value (x)" in the calibration calculator.
6. After the preceding operations are complete, the calibration calculator calculates the calibration coefficients k and B. Enter the calculated values as the calibration coefficient slope (k) and calibration coefficient intercept (B) of the calibration parameters (battery voltage, upper half-bus voltage, and lower half-bus voltage) in the FG group.



Calibrating the grid voltage

Check whether the displayed grid voltage deviates from the actual voltage. (Measure the voltage between pins 1 and 2 of X21 as the AB line voltage, between pins 2 and 3 of X21 as the BC line voltage, and between pins 1 and 3 of X21 as the AC line voltage.) If deviations are found, calibrate the PCS as follows:

1. Go to "Parameter Setting" > "Manufacturer" > "FG". Set "A-phase/line voltage sampling calibration coefficient slope", "B-phase/line voltage sampling calibration coefficient slope", and "C-phase/line voltage sampling calibration coefficient slope" to 1.
2. In constant AC power light-load operating mode (such as 10 kW), measure the voltage U_ab between pins 1 and 2 of X21, the voltage U_bc between pins 2 and 3 of X21, and the voltage U_ca between pins 1 and 3 of X21, and enter the measured values in the "Background Value (x)" of the calibration calculator.
3. After the preceding operations are complete, the calibration calculator calculates the calibration coefficient. Enter the calculated values in the following parameters in the FG group: "A-phase/line voltage sampling calibration coefficient slope", "B-phase/line voltage sampling calibration coefficient slope", and "C-phase/line voltage sampling calibration coefficient slope".
   For example, set the calibration coefficient of phase AB voltage sampling to 1. The voltage measured between pins 1 and 2 of X21 is 695 V. Enter the measured value in the "Measured Value (y)" of the calibration calculator. The voltage of phase AB displayed on the HMI is 688 V. Enter this value in the "Background Value (x)" of the calibration calculator. Then, the voltage sampling calibration coefficient is 1.010 (695/688). Enter 1.010 as the value of "A-phase/line voltage sampling calibration coefficient slope" in the FG group.



Calibrating the inverter voltage

CAUTION

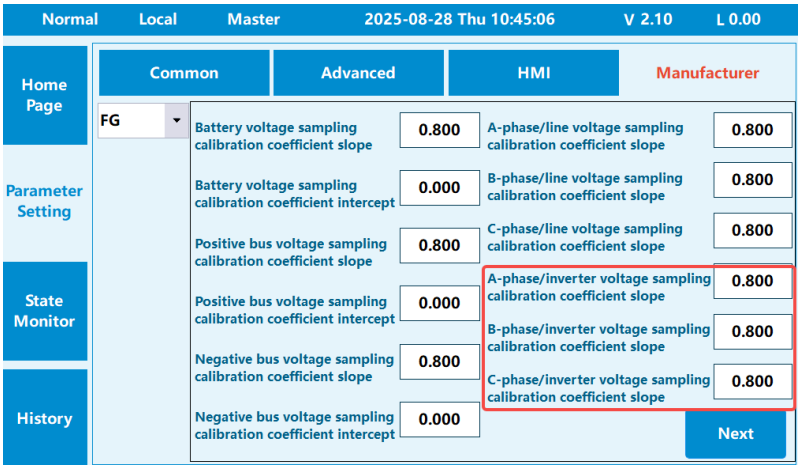
Inverter voltage calibration must be performed after grid voltage calibration.

Check whether the displayed inverter voltage deviates from the actual voltage. (Measure the voltage between pins 1 and 2 of X21 as the AB line voltage, between pins 2 and 3 of X21 as the BC line voltage, and between pins 1 and 3 of X21 as the AC line voltage.)

If deviations are found, calibrate the PCS as follows:

1. Go to "Parameter Setting" > "Manufacturer" > "FG". Set "A-phase/inverter voltage sampling calibration coefficient slope", "B-phase/inverter voltage sampling calibration coefficient slope", and "C-phase/inverter voltage sampling calibration coefficient slope" to 1.
2. In constant AC power light-load operating mode (such as 10 kW), measure the voltage U_a between pins 1 and 2 of X21, the voltage U_b between pins 2 and 3 of X21, and the voltage U_c between pins 1 and 3 of X21, and enter the measured values in the "Background Value (x)" of the calibration calculator.
3. After the preceding operations are complete, the calibration calculator calculates the calibration coefficient. Enter the calculated values in the following parameters in the FG group: "A-phase/inverter voltage sampling calibration coefficient slope", "B-phase/inverter voltage sampling calibration coefficient slope", and "C-phase/inverter voltage sampling calibration coefficient slope".
   For example, set the calibration coefficient of phase AB voltage sampling to 1. The voltage measured between pins 1 and 2 of X21 is 695 V. Enter the measured value in

the "Measured Value (y)" of the calibration calculator. The voltage of phase AB displayed on the HMI is 688 V. Enter this value in the "Background Value (x)" of the calibration calculator. Then, the voltage sampling calibration coefficient is 1.010 (695/688). Enter 1.010 as the value of "A-phase/inverter voltage sampling calibration coefficient slope" in the FG group.



5.7.3 Current Calibration

It is recommended that the current/power calibration be performed after the AC/DC voltage calibration is completed.

If none of the FG parameters is 1, this means that the machine has been calibrated. The calibration range is generally:

Parameter	Calibration Range
DC current (Rectifier)	0.85 to 1.15
DC current (Inverter)	0.85 to 1.15
Rectifier current Iu1, Iv1, Iw1	0.8 to 1.2
Rectifier current Iu2, Iv2, Iw2	0.8 to 1.2
Rectifier current Iu3, Iv3, Iw3, Iu4, Iv4, Iw4	0.95 to 1.05
Rectifier current Iu5, Iv5, Iw5	0.98 to 1.02
Inverter current Iu1, Iv1, Iw1	0.8 to 1.2
Inverter current Iu2, Iv2, Iw2	0.8 to 1.2
Inverter current Iu3, Iv3, Iw3, Iu4, Iv4, Iw4	0.95 to 1.05
Inverter current Iu5, Iv5, Iw5	0.98 to 1.02

CAUTION

If there is still a difference after the modification, you can modify the DC current (inverter) and the rectifier/inverter current coefficient to fine tune, and increase the sampling current calibration coefficient to reduce the actual power (the power on the battery side is reduced). Reducing the sampling current calibration coefficient increases the actual power until the accuracy requirement is met.

You can also contact Wattox service engineers to obtain the Calibration calculator.

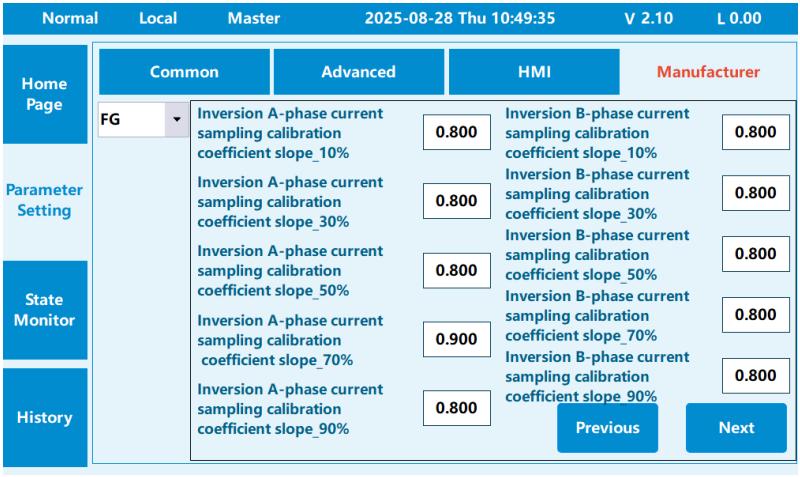
5.7.3.1 Current Calibration for a Single PCS

Preparation before calibration

- Oscilloscope (or power analyzer)
- Three current probes. Considering that the cables on the site are thick, current clamps may not be proper. You'd better use flexible current probes.
- Contact the Wattox engineer to obtain the Calibration calculator.

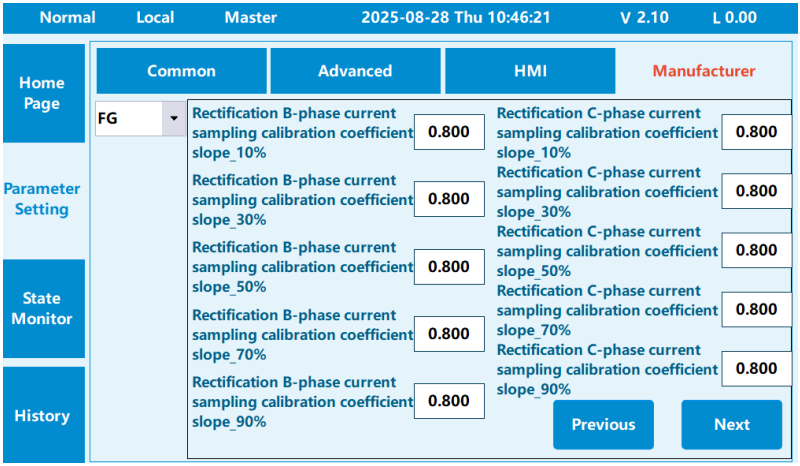
Discharge calibration

1. Connect three current probes to the three-phase power cable.
2. Set the active power to full-load discharge "100% P_n ". Enter the RMS values I_{ab} , I_{bc} , and I_{ca} of the three-phase grid current measured by the site tester in the "Measured Power-Dividing Current" of the calibration calculator.
3. Enter the three-phase inverter inductance current Isa, Isb, and Isc of the master and slave that are displayed on the HMI in the "Inverter Inductance Current of Slave on Host Controller" and "Inverter Inductance Current of Master on Host Controller" of the calibration calculator. The calibration calculator calculates the master calibration coefficient and slave calibration coefficient of phases A, B, and C.
4. Enter the calculated calibration coefficients as values of calibration coefficient slope of inverter phase-A/B/C current sampling_90%, calibration coefficient slope of inverter phase-A/B/C current sampling_70%, calibration coefficient slope of inverter phase-A/B/C current sampling_50%, calibration coefficient slope of inverter phase-A/B/C current sampling_30%, and calibration coefficient slope of inverter phase-A/B/C current sampling_10%.
5. You do not need to perform measurement and calibration for the preceding power segments. The coefficients are the same as those measured under full load. Directly enter the values measured under full load.



Charge calibration

1. Connect three current probes to the three-phase power cable.
2. Set the active power to full-load charge "-100% P_n". Enter the RMS values I_{ab}, I_{bc}, and I_{ca} of the three-phase grid current measured by the site tester in the "Measured Power-Dividing Current" of the calibration calculator.
3. Enter the three-phase inverter inductance current Isa, Isb, and Isc of the master and slave that are displayed on the HMI in the "Inverter Inductance Current of Slave on Host Controller" and "Inverter Inductance Current of Master on Host Controller" of the calibration calculator. The calibration calculator calculates the master calibration coefficient and slave calibration coefficient of phases A, B, and C.
4. Enter the calculated calibration coefficients as values of calibration coefficient slope of rectifier phase-A/B/C current sampling_90%, calibration coefficient slope of rectifier phase-A/B/C current sampling_70%, calibration coefficient slope of rectifier phase-A/B/C current sampling_50%, calibration coefficient slope of rectifier phase-A/B/C current sampling_30%, and calibration coefficient slope of rectifier phase-A/B/C current sampling_10%.
5. You do not need to perform measurement and calibration for the preceding power segments. The coefficients are the same as those measured under full load. Directly enter the values measured under full load.



The oscilloscope is used for the preceding calibration. If another current measuring instrument is used, such as the power analyzer, tester from a third-party test agency, or BMS, replace the current measured by the oscilloscope with the current measured by the third-party tester after the oscilloscope is replaced by the power analyzer.

5.7.3.2 Current Calibration for All PCSs of a Site

The method for calibrating all PCSs of a site is almost the same as that for calibrating a single PCS. All the PCSs are calibrated in a unified manner based on the power or current difference, regardless of the device difference. All the PCSs use the same calibration coefficient.

If a gap still exists after calibration, you can modify the DC (inverter) for fine-tuning. You can use a larger sampling calibration coefficient to reduce the actual power or use a smaller sampling calibration coefficient to increase the actual power, until the gap is close to 0.

1. Set the active power to discharge 10% P_n. Modify the DC (inverter) in group FG (manufacturer parameters) as follows: DC (inverter) = P_real/P_EMS. In this formula, P_real indicates the power measured by Electric Power Research Institute or other high-accuracy test instruments, and P_EMS indicates the power displayed on the EMS.
2. Set the active power to discharge 30% P_n. Modify the inverter current Iu2, Iv2, and Iw2 in group FG (manufacturer parameters) as follows: Iu2 = Iv2 = Iw2 = P_real/P_EMS. In this formula, P_real indicates the power measured by Electric Power Research Institute or other high-accuracy test instruments, and P_EMS indicates the power displayed on the EMS.
3. Set the active power to discharge 50% P_n. Modify the inverter current Iu3, Iv3, and Iw3 in group FG (manufacturer parameters) as follows: Iu3 = Iv3 = Iw3 = P_real/P_EMS. In this formula, P_real indicates the power measured by Electric Power Research Institute

or other high-accuracy test instruments, and P_EMS indicates the power displayed on the EMS.

4. Set the active power to discharge 70% P_n. Modify the inverter current Iu4, Iv4, and Iw4 in group FG (manufacturer parameters) as follows: Iu4 = Iv4 = Iw4 = P_real/P_EMS. In this formula, P_real indicates the power measured by Electric Power Research Institute or other high-accuracy test instruments, and P_EMS indicates the power displayed on the EMS.
5. Set the active power to discharge 90% P_n. Modify the inverter current Iu5, Iv5, and Iw5 in group FG (manufacturer parameters) as follows: Iu5 = Iv5 = Iw5 = P_real/P_EMS. In this formula, P_real indicates the power measured by Electric Power Research Institute or other high-accuracy test instruments, and P_EMS indicates the power displayed on the EMS.
6. Set the active power to charge 10% P_n. Modify the DC (rectifier) in group FG (manufacturer parameters) as follows: DC (rectifier) = P_real/P_EMS. In this formula, P_real indicates the power measured by Electric Power Research Institute or other high-accuracy test instruments, and P_EMS indicates the power displayed on the EMS. (If a DC probe is available, it can be connected to the oscilloscope for the calibration.)
7. Set the active power to charge 30% P_n. Modify the rectifier current Iu2, Iv2, and Iw2 in group FG (manufacturer parameters) as follows: Iu2 = Iv2 = Iw2 = P_real/P_EMS. In this formula, P_real indicates the power measured by Electric Power Research Institute or other high-accuracy test instruments, and P_EMS indicates the power displayed on the EMS.
8. Set the active power to charge 50% P_n. Modify the rectifier current Iu3, Iv3, and Iw3 in group FG (manufacturer parameters) as follows: Iu3 = Iv3 = Iw3 = P_real/P_EMS. In this formula, P_real indicates the power measured by Electric Power Research Institute or other high-accuracy test instruments, and P_EMS indicates the power displayed on the EMS.
9. Set the active power to charge 70% P_n. Modify the rectifier current Iu4, Iv4, and Iw4 in group FG (manufacturer parameters) as follows: Iu4 = Iv4 = Iw4 = P_real/P_EMS. In this formula, P_real indicates the power measured by Electric Power Research Institute or other high-accuracy test instruments, and P_EMS indicates the power displayed on the EMS.
10. Set the active power to charge 90% P_n. Modify the rectifier current Iu5, Iv5, and Iw5 in group FG (manufacturer parameters) as follows: Iu5 = Iv5 = Iw5 = P_real/P_EMS. In this formula, P_real indicates the power measured by Electric Power Research Institute or other high-accuracy test instruments, and P_EMS indicates the power displayed on the EMS.

CAUTION

After the operations are complete, click "Stop", find the discharge function in the F4 group, and discharge the bus voltage to 20 V in off-grid mode. (Note: The bus voltage can only be discharged to 100 V for the plan A model of Star. In this case, you must disconnect the control power and use the discharge resistor or the resistance gear of the multimeter to continue discharging until the voltage is below 10 V to ensure personal safety.)

5.8 Single Unit Function Test

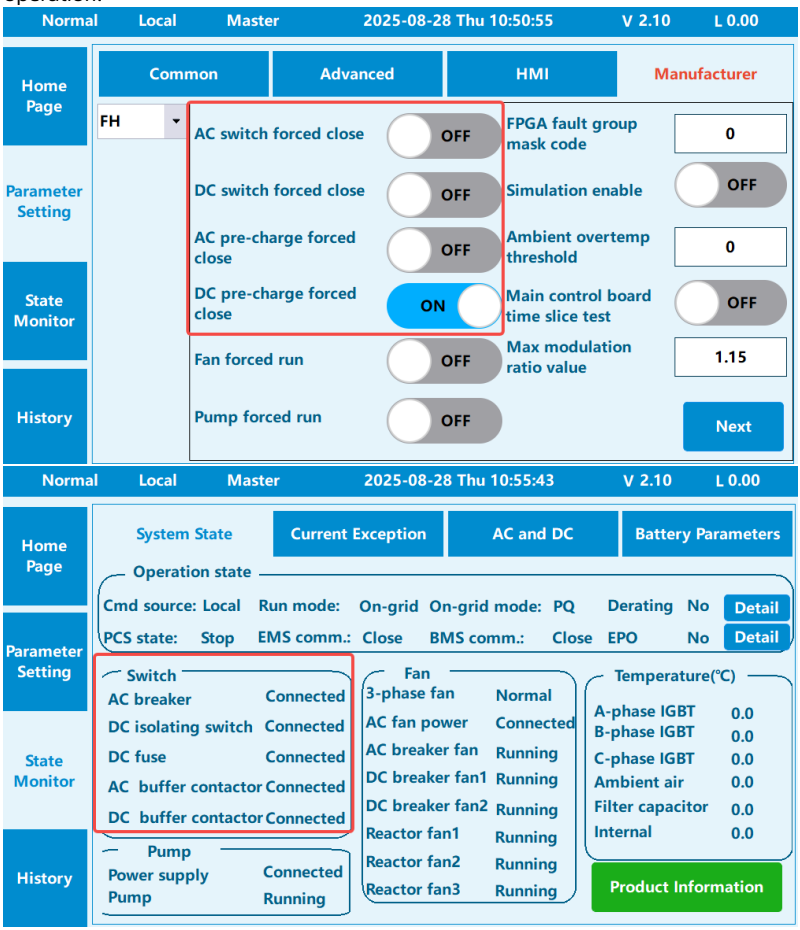
5.8.1 Switch Test

1. Use a multimeter (for the battery side with high voltage, measure against the chassis with the voltage setting) to confirm that no high voltage is present on the DC or AC side.
2. Manually press the open/close button of the AC/DC circuit breaker to release the stored energy in the coil (blue indicates "stored" and white indicates "released"), as shown in the following figure.

Figure 5-10 Switching on/off power storage

3. Close QF4 and QF5 to apply control power, wait approximately 1 minute for control power activation, and check if the AC/DC switches have completed energy storage (accompanied by motor sound, with the storage status changing from white to blue).
4. Turn on the UPS switch in the low-voltage cabinet of the transformer, powering on the HMI screen.
5. On the HMI, go to the "Manufacturer" (FH group) page, and sequentially force-close and open the DC pre-charge contactor, AC pre-charge contactor, DC disconnector, and AC circuit breaker. Verify that the switch feedback status on the state monitor page

matches the actual operation and that the physical device status aligns with the operation.



6. After confirmation, set all switches to the open state.

5.8.2 Fan and Water Pump Operation Test

1. On the HMI, go to the "Manufacturer" (FH group) page, force the water pump to run, and confirm normal operation (listen for pump sound against the chassis). Let the water pump run for 3 minutes to ensure stable operation without vibration.
2. Check the liquid level and pump status on the "State Monitor" page. Verify that the liquid level is normal and the status is running. Turn off the pump after verification.
3. Go to the "Manufacturer" (FE group) page. Force the fans to run. Verify that the large fan on the cabinet top rotates in the correct direction (air is blown in from the top), the

small fan on the back of the control panel rotates normally, and the reactor cooling fan is operating with airflow from the cooling holes.

4. On the "State Monitor" page, check that the fan power is on and all fan states are running. Turn off the fans after verification.

5.8.3 Integrated Off-Grid Test

1. Close the circuit breaker of the DC side battery compartment of the PCS and open the internal high-voltage load switch and disconnector of the AC side box-type transformer. Note that the high-voltage grounding disconnector must be open.
2. Go to the "Manufacturer" page, change "Run Mode" to "Off-Grid" for the master, and set the off-grid voltage to 690 V, as shown in the following figure.

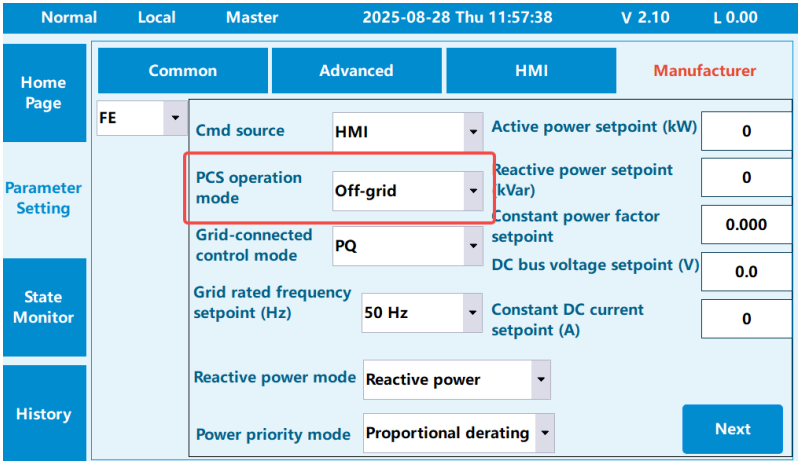


Figure 5-11 Setting the run mode to off-grid

3. On the "Manufacturer" (group F5) page of the HMI, set "Off-grid V/f mode" to "open-loop" for the master and slave, respectively.

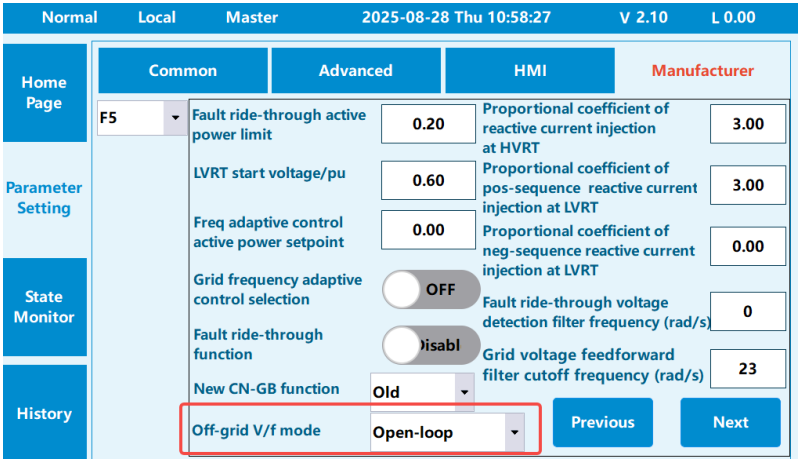


Figure 5-12 Setting the off-grid V/f mode for the master and slave

4. Click "Standby" on the homepage. The upper and lower buses start pre-charging and the voltage increases. Wait until the AC/DC switch is closed.
5. Check whether the upper and lower half-bus voltage is balanced and whether slight deviation exists. If no, click "Start". The device output voltage increases to 690 V. In this case, check whether the upper and lower half-bus voltage gradually deviates, whether the three-phase output voltage is about 690 V, and whether the three-phase current is balanced.
6. After the test is completed, restore the parameters that are modified during the process.

5.9 Charge-Discharge Test

5.9.1 Power Control Test

CAUTION

Test is not allowed when the on-site grid is connected.

1. Set up the test platform (DC side: Battery analog device, AC side: Grid analog device) according to the following figure. Close switches Q1 and Q2.

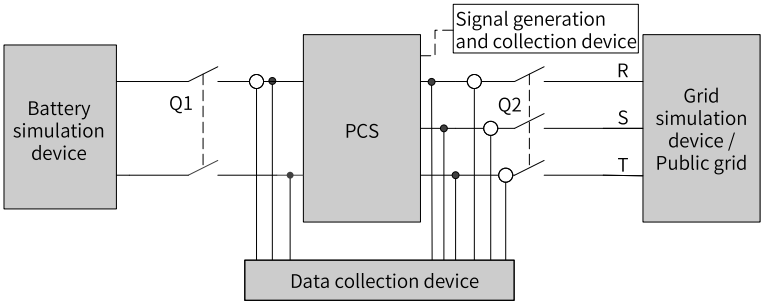
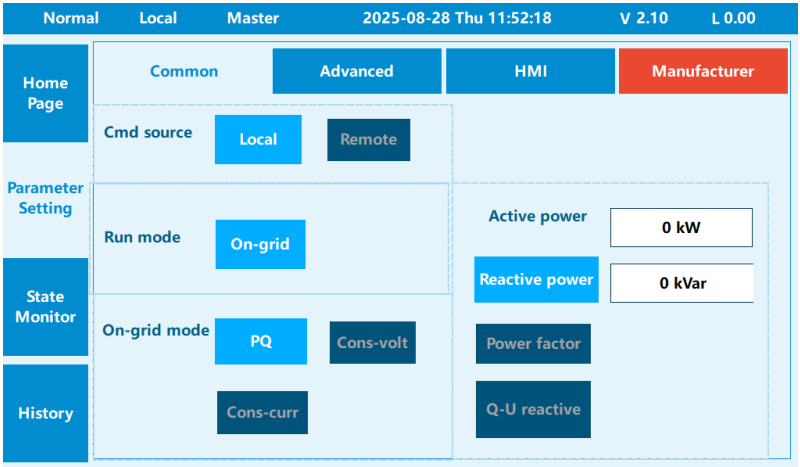
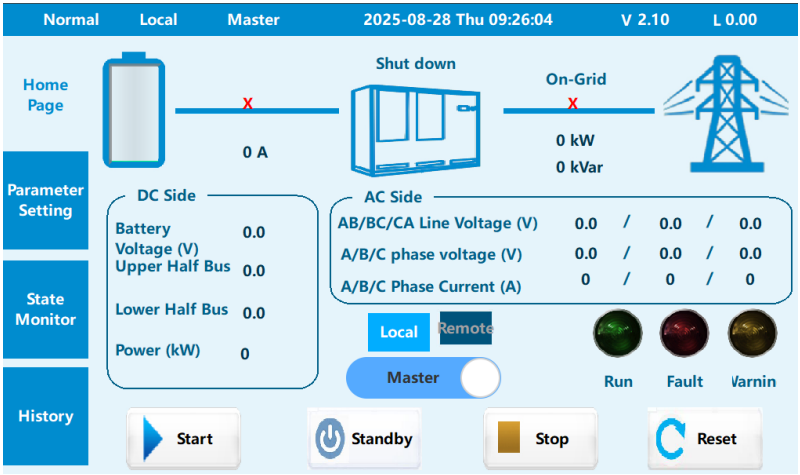


Figure 5-13 Circuit diagram for on-grid detection

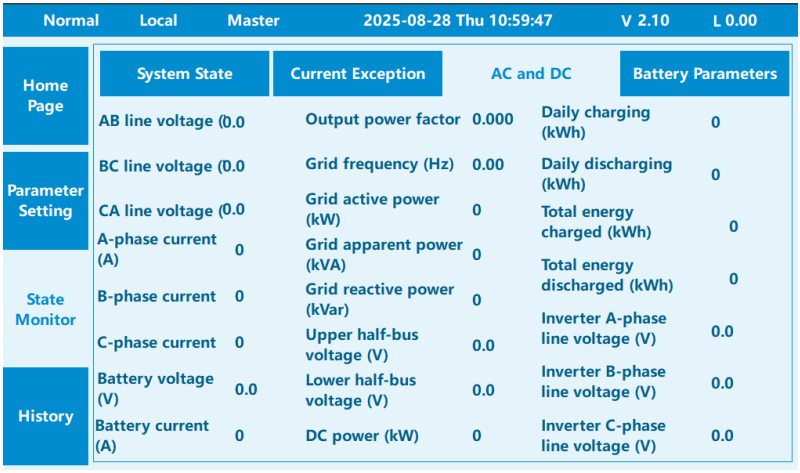
2. Check the communication status in the top-left corner of the HMI homepage to confirm normal communication.
3. Go to "Parameter Setting" > "Common". Set "Run Mode" to "On-Grid" and set "Work Mode" to "Con-curr" (constant current). The active and reactive power defaults are 0.



4. On the homepage, click the "Start" command button and wait for the PCS to start. Confirm the Run indicator is steady green, indicating successful operation.



5. Go to "Parameter Setting" > "Common". Set the active power in 10% increments of rated power, from maximum charging power to maximum discharging power. For each active power range, set the PCS to output maximum inductive and capacitive reactive power, running for 2 minutes at each reactive power point.
6. Go to "State Monitor" > "AC and DC", observe the average grid-side active and reactive power for each active power range. Draw the active power-reactive power characteristic curve.



The characteristic curve shall meet the requirements of the Chinese national standard (GB/T 34120) as shown in the following figure, that is, the active power and reactive power should be dynamically adjustable in the solid and dashed line areas in the figure.

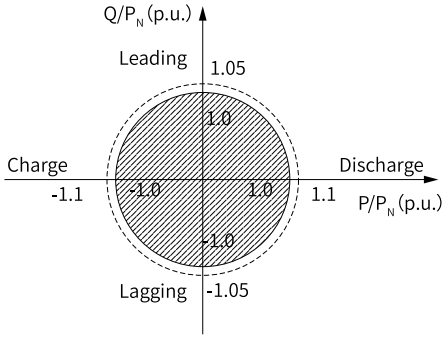


Figure 5-14 Power output range of PCS

Parameter	Name	Value Range	Default Value	Unit	Change Method	Effective Mode
FE.00	Command source	0: HMI 1: EMS 2: Software (Reserved)	1	-	Change at stop	Real time
FE.01	PCS operation mode	0: On-grid mode 1: Off-grid mode	0	-	Change at stop	Real time
FE.02	PCS on-grid control mode	0: PQ mode 1: Constant DC current mode 2: Constant DC voltage mode	0	-	Change at stop	Real time
FE.20	Start command	0: Disabled 1: Send the start command	0	-	Real time	-
FE.25	Active power setpoint	/	0.0	kW	Real time	Real time
FE.26	Reactive power setpoint	/	0.0	kVar	At once	Real time

5.9.2 Active power control

The test standards for active power control are as follows. If all the indicators at the active power setpoint meet the standards, the active power control test passes.

- Response time $t_{p,res} \leq 100$ ms
- Regulation time $t_{p,reg} \leq 300$ms;
- Control deviation $\Delta P\% \leq \pm 1\%$ of the rated power.

1. Set up the test platform as shown in the following figure (DC side: battery simulator, AC side: grid simulator), with a data acquisition device (for example, power analyzer) to measure AC port output power. Close switches Q1 and Q2.

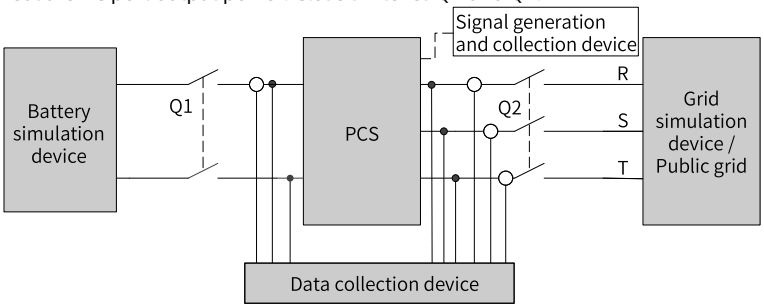
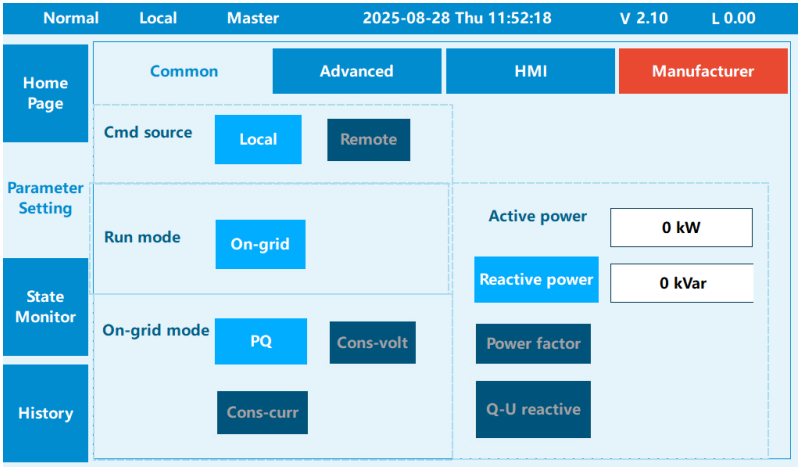
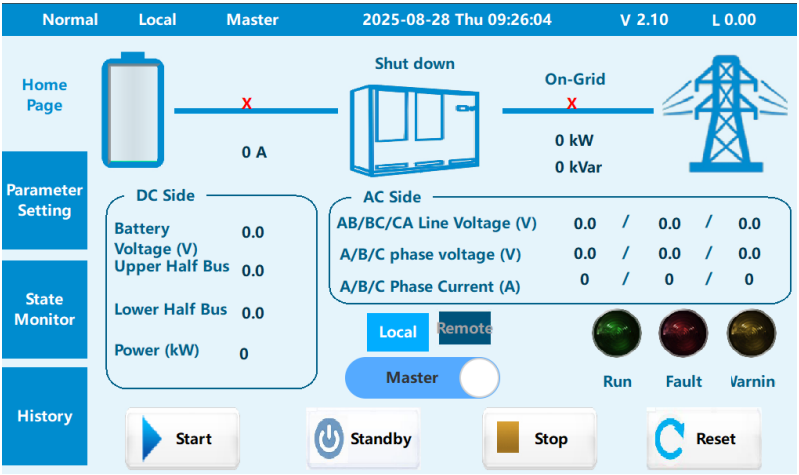


Figure 5-15 Circuit diagram for on-grid detection

2. Check the communication status in the top-left corner of the HMI homepage to confirm normal communication.
3. Go to "Parameter Setting" > "Common". Set "Run Mode" to "On-grid", set "On-grid mode" to "Con-curr" (constant current), and set "Active Power" to 100% Pn.



4. On the homepage, click the "Start" command button and wait for the PCS to start. Confirm the Run indicator is steady green, indicating successful operation.



5. To test the discharge state of the PCS:

a. For discharge condition, set active power as shown below (first switch point from 100% P_n to 80% P_n), running for 2 minutes at each power setpoint.

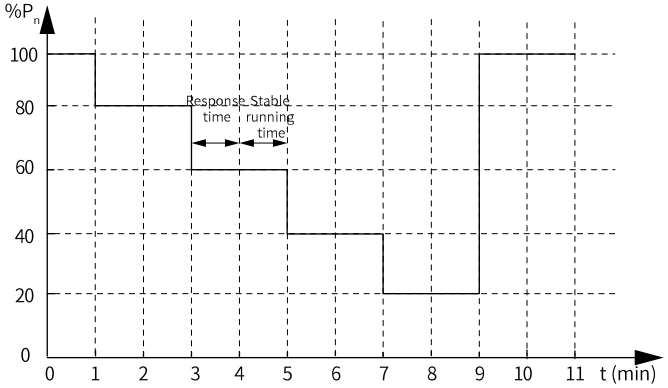


Figure 5-16 Active power control curve in discharge mode

b. Within the running time of 0 to 1 min, use the power analyzer to record the dynamic response waveform during active power switchover. Calculate the response time and regulation time of active power control according to the following figure and formula.

Response time of active power control: $t_{p,res} : t_{p,res} = t_1 - t_0$

Regulation time of active power control: $t_{p,reg} : t_{p,reg} = t_2 - t_0$

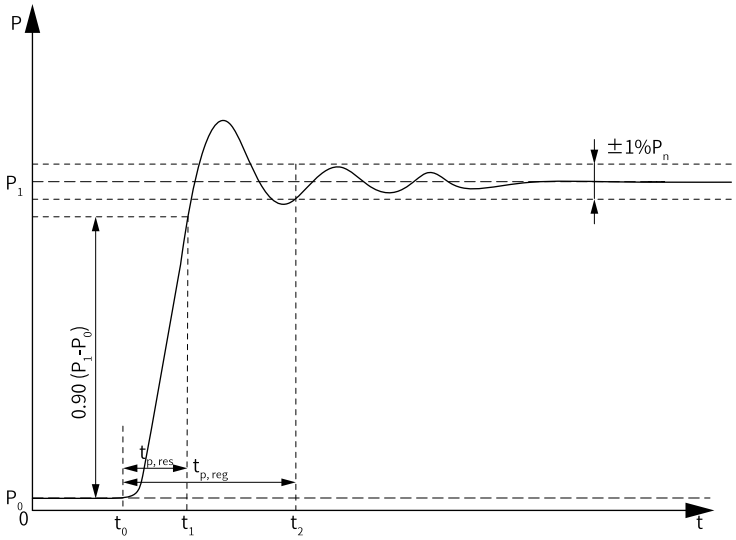


Figure 5-17 Output waveform for active power switching from dynamic state to steady state

c. Within the running time of 1 to 2 min, use the power analyzer to record the average value of the active power of the AC output port. Calculate the deviation of active power control according to the above figure and following formula.
Active power control deviation ΔP%: $\Delta P\% = |P_1 - P_{mes}|/P_n \times 100\%$ (where P_{mes} represents the average measured active power)

6. The charging condition test is consistent with the discharge condition test and set the active power setpoint according to the following figure.

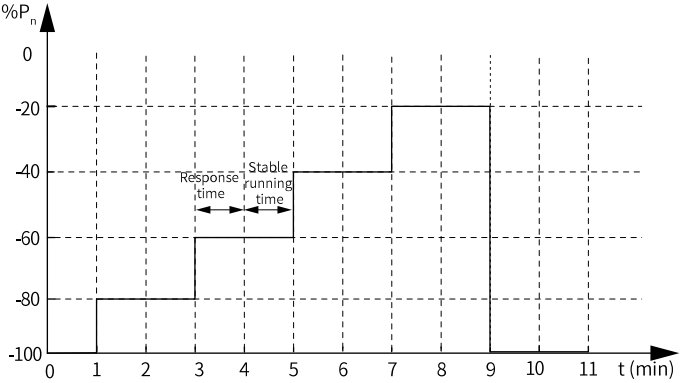


Figure 5-18 Active power control curve in charge mode

7. Charge condition test is the same as the discharge condition test, setting active power to -50% P_n.

5.9.3 Power Factor Control

CAUTION

Test is not allowed when the on-site grid is connected.

The test standards for control deviation are as follows. The power factor control test passes, if the results at all constant power factor setpoint meet the following requirements:

- When the reactive power is greater than or equal to 20% of the rated power, the control deviation is less than ±1% of the rated power.
- When reactive power < 20% of rated power, the control deviation < ±3% of rated power.

1. Set up the test platform (DC side: Battery analog device, AC side: Grid analog device) and data acquisition device (such as power analyzer) correctly according to the following figure to measure the output power of the AC port. Close switches Q1 and Q2.

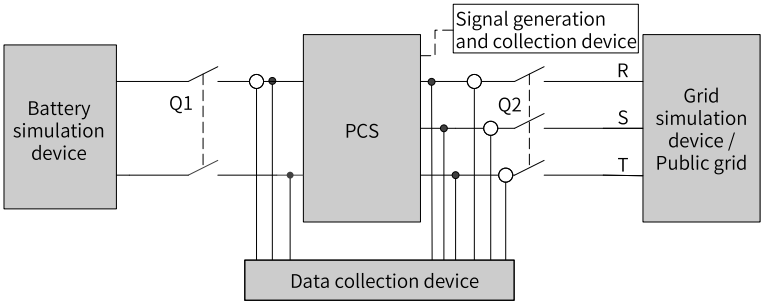


Figure 5-19 Circuit diagram for on-grid detection

2. Check the communication status in the top-left corner of the HMI homepage to confirm normal communication.
3. Go to "Parameter Setting" > "Common". Set "Run Mode" to "On-Grid", set "Work Mode" to "Con-curr" (constant current) and "Power Factor", and set "Active Power" to 50% P_n.

Set the AC port power factor of the PCS as shown below, running for 2 minutes at each power factor setpoint.

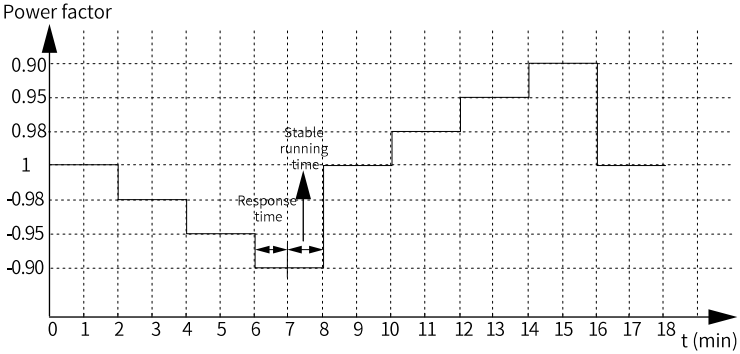
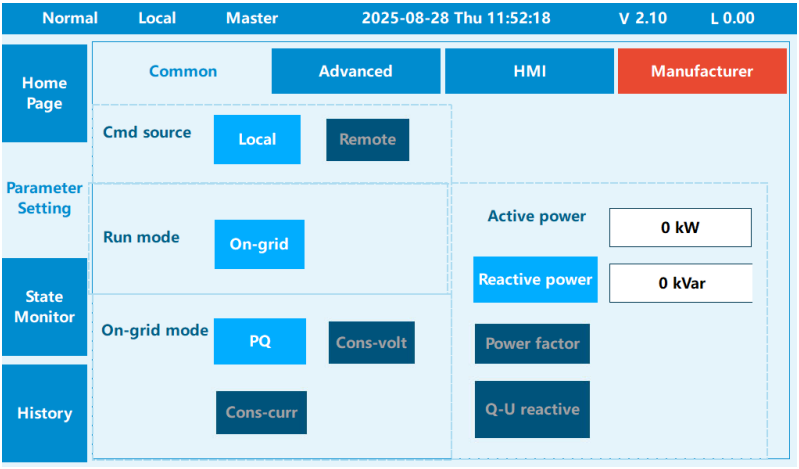


Figure 5-20 Power factor control curve



4. On the homepage, click the "Start" command button and wait for the PCS to start. Confirm the Run indicator is steady green, indicating successful operation.
5. Calculate the target reactive power under the corresponding power factor setpoint when the active power of the PCS is 50% P_n.
6. Within the stable running time of 1 min to 2 min, use the power analyzer to record the average reactive power of the PCS AC port at each power factor setting.
7. Calculate the reactive power control deviation in constant reactive power factor mode and check whether the value meets the requirements.

5.10 Grid Tests

5.10.1Overload capacity

Check whether the PCS overloads, and triggers alarm or protection. If the overload capacity meets the following standards (under rated voltage), the test passes.

- When the AC port current is 110% of the rated current, the PCS runs continuously for more than 10 minutes.
- When the AC port current is 120% of the rated current, the PCS runs continuously for more than 1 minute.

1. Set up the test platform (DC side: Battery analog device, AC side: Grid analog device) and data acquisition device (such as power analyzer) correctly according to the following figure to measure the output power of the AC port. Close switches Q1 and Q2.

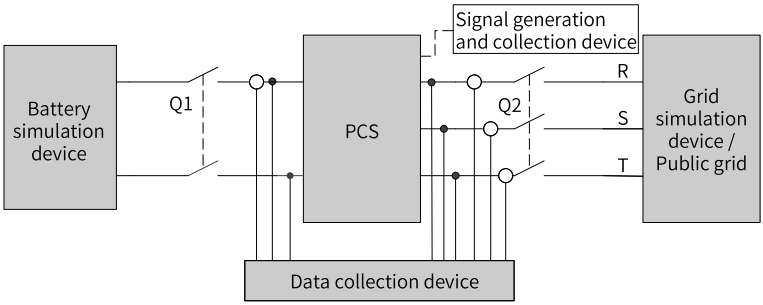
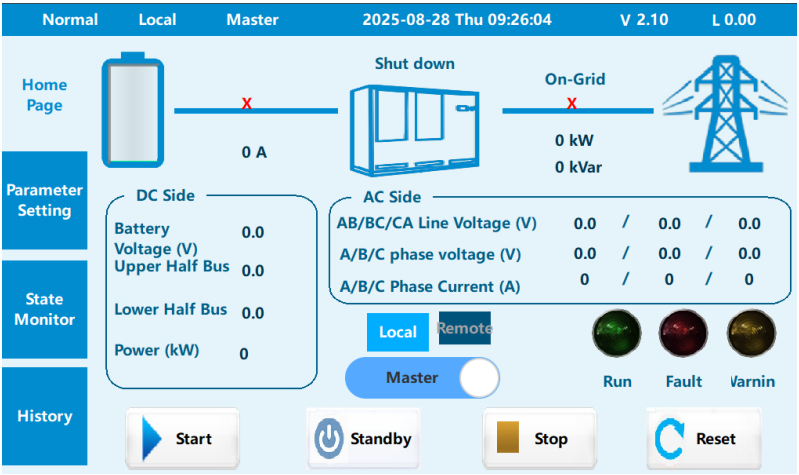
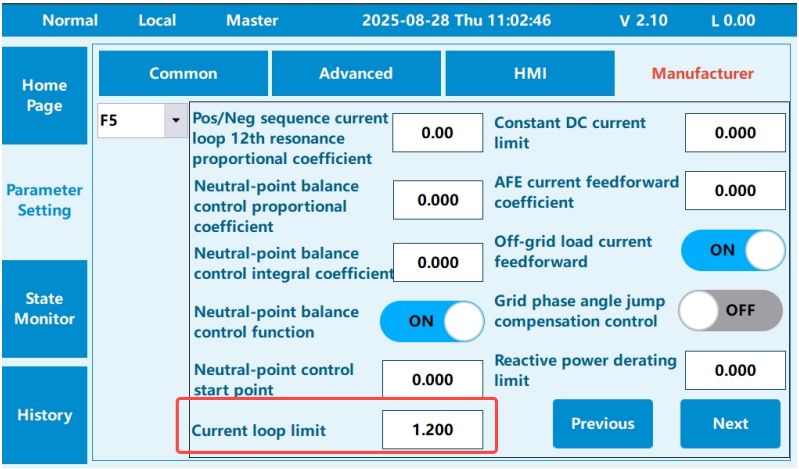


Figure 5-21 Circuit diagram for on-grid detection

2. Check the communication status in the top-left corner of the HMI homepage to confirm normal communication.
3. Go to "Parameter Setting" > "Common". Set "On-grid mode" to "Con-curr" (constant current).
4. On the homepage, click the "Start" command button and wait for the PCS to start. Confirm the Run indicator is steady green, indicating successful operation.



5. Adjust the output voltage of the grid analog device to the rated AC port voltage of the PCS.
6. On the "Common" page, set "Active Power" to 110% of the rated power, running for 10 minutes.
7. Press and hold the "HMI" tab, select "Developer" for "User name", and enter the password to enter the administrator mode. Change "Current loop limit" in the F5 group to 1.2.



8. Set "Active Power" to 120% of the rated power, running for 1 minute.

5.10.2Charge and Discharge Switchover Time

The standard charge/discharge switchover time shall be no greater than 100 ms. The test methods are as follows:

1. Set up the test platform (DC side: Battery analog device, AC side: Grid analog device) and data acquisition device (such as power analyzer) correctly according to the following figure to measure the output power of the AC port. Close switches Q1 and Q2.

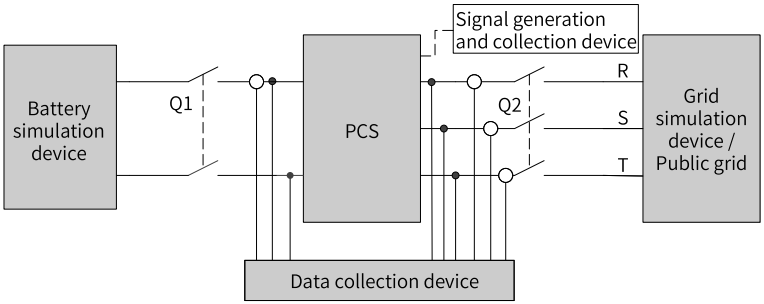
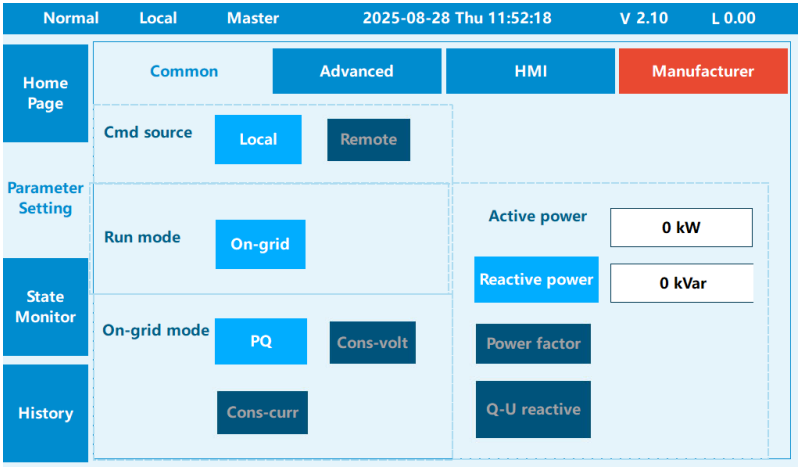
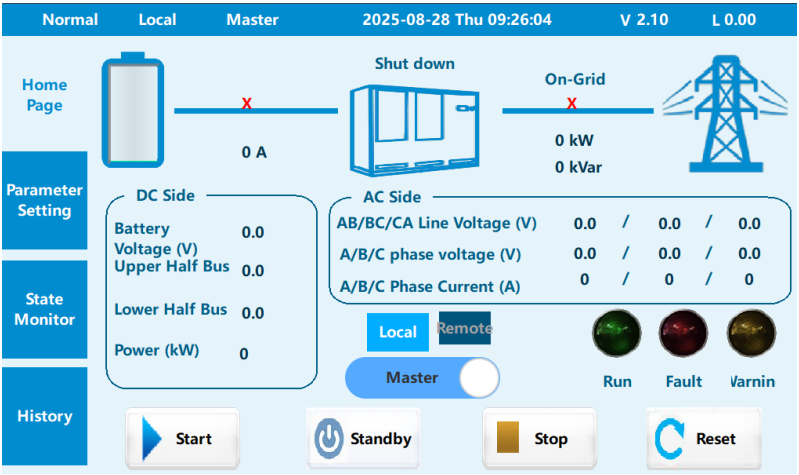


Figure 5-22 Circuit diagram for on-grid detection

2. Check the communication status in the top-left corner of the HMI homepage to confirm normal communication.
3. Go to "Parameter Setting" > "Common". Set “Run mode” to "On-Grid" and “On-grid mode” to "PQ". (The highlighted options are selected.)



4. On the homepage, click the "Start" command button and wait for the PCS to start. Confirm the Run indicator is steady green, indicating successful operation.



5. The following figure is used for the test. Calculate the switchover time interval between the moment the device first reaches 90% of rated power discharging and the moment it first reaches 90% of rated power charging during the transition from the rated power discharge state to the rated power charge state.

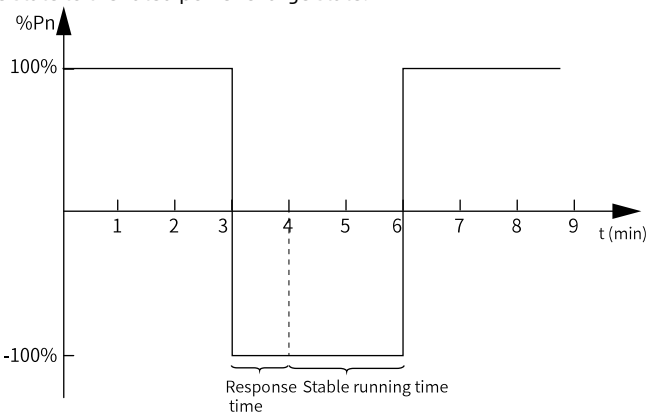


Figure 5-23 Active power setting curve

a. Go to "Parameter Setting" > "Common". Set "Active Power" to 100% P_n, running for at least 3 minutes.

b. Set "Active Power" to -100% P_n, running for at least 3 minutes.

c. Use a power analyzer to record dynamic response waveforms during active power switchover.

5.10.3Harmonic Current

1. Set up the test platform (DC side: Battery analog device, AC side: Grid analog device) and data acquisition device (such as power analyzer) correctly according to the following figure to measure the output power of the AC port. Close switches Q1 and Q2.

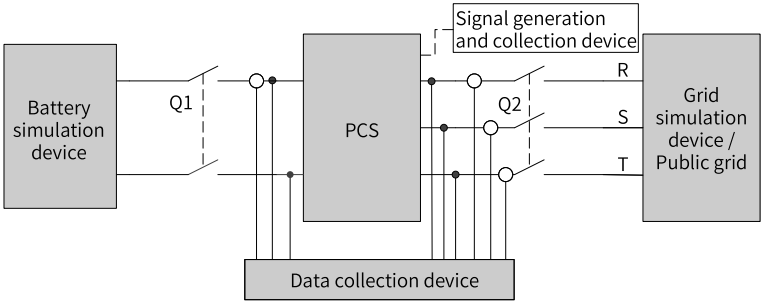
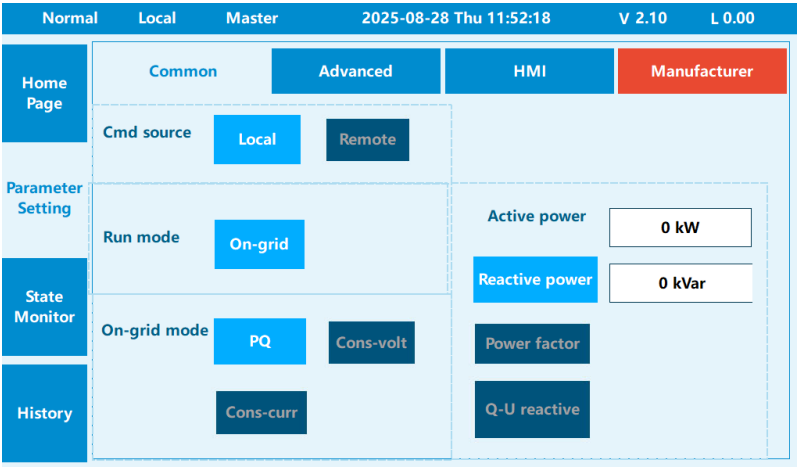
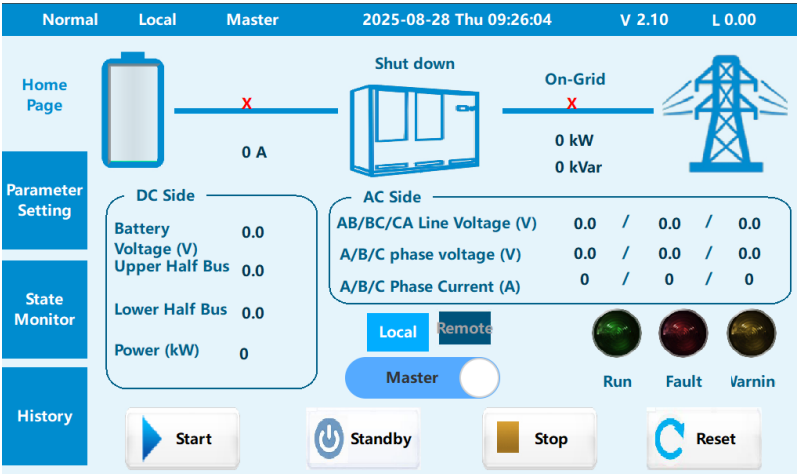


Figure 5-24 Circuit diagram for on-grid detection

2. Check the communication status in the top-left corner of the HMI homepage to confirm normal communication.
3. Go to "Parameter Setting" > "Common". Set "Run Mode" to "On-Grid" and set "Work Mode" to "PQ". (The highlighted options are selected.)



4. On the homepage, click the "Start" command button and wait for the PCS to start. Confirm the Run indicator is steady green, indicating successful operation.



5. Go to "Parameter Setting" > "Common". Set "Active Power" to 33% P_n, 66% P_n, and 100% P_n respectively, running for 10 minutes at each power setpoint.
   Use a power analyzer to detect and record the harmonic current. If the harmonic current limit meet the requirements in the following table, the harmonic current test passes.

Table 5–2 Harmonic current limits

Odd Harmonics	Harmonic Limit	Even Harmonics	Harmonic Limit
3^{rd}~9^{th}	4%I_N	2^{nd}~10^{th}	1%I_N
11^{th}~15^{th}	2%I_N	12^{th}~16^{th}	0.5%I_N
17^{th}~21^{st}	1.5%I_N	18^{th}~22^{th}	0.375%I_N
23^{th}~33^{th}	0.6%I_N	24^{th}~34^{th}	0.15%I_N
35^{th}~49^{th}	0.3%I_N	36^{th}以上	0.075%I_N

6. Test method in charge mode: Set the power to -33% P_n, -66% P_n, and -100% P_n respectively and repeat the preceding steps.

5.10.4HVRT and LVRT

1. Go to the "Parameter Setting" page and click the "Advanced" tab.
2. Configure the parameters as follows:
   - Set "Power change rate setting (%Pn/s)" in the FE group to "10000".
   - Set "Active island detection enable" in the F5 group to "Disable".
   - Set "Grid frequency adaption control" in the F5 group to "Disable".
   - Set "Fault ride-through" in the F5 group to "Enable".
   - Set "Grid voltage feedforward cutt-off frequency" in the F5 group to 1570. (If the value is lower than 1570, contact the R&D engineer. If the value is greater than 1570, change the value to 1570.)

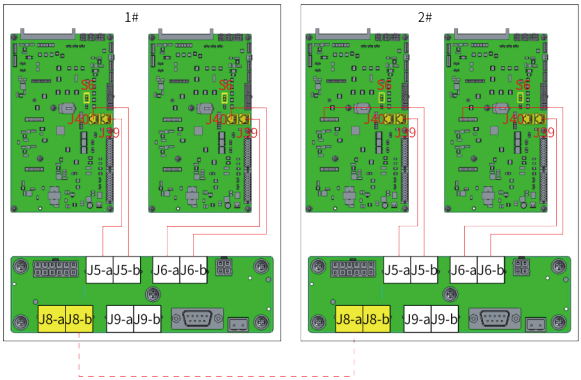
- Set "Mid-point balancing ctrl" in the F5 group to "Enable".
- Set "Wave-generation self adaptation" in the F5 group to "Disable".

5.11 Grid-Connected AC/DC Parallel

This section describes grid-connected AC/DC hardware parallel operations in the DC and AC parallel scenarios.

5.11.1 Hardware Wiring and Dial Commissioning

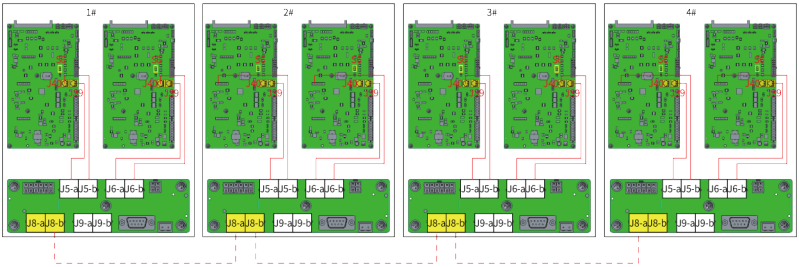
The following figure shows the wiring and dial commissioning of two parallel cabinets.



Commissioning instruction:

- **Solid line in red** indicates electrical wiring inside a cabinet. If a parallel communication error occurs, measure and check whether these cables are correctly connected.
- **Dashed line in red** indicates the LAN cable used to connect two PCS cabinets. Only one LAN cable is required between every two cabinets for the PCSs.
- You must turn the DIP switch S6 of the first cabinet (1#) and the last cabinet (2#) to ON and turn S6 of other communication boards to OFF.
- Pin specifications of the network port for the parallel system are defined by Wattox, and the network port is not a standard port. Do not connect the network port to a computer or switch through a LAN cable.
- If the communication interference issue is hard to resolve, remove the LAN cable from the J39 port on the communication board of the first cabinet (1#) and from the J40 port on the communication board of the last cabinet (2#).

The following figure shows the wiring and dial commissioning of multiple parallel cabinets.



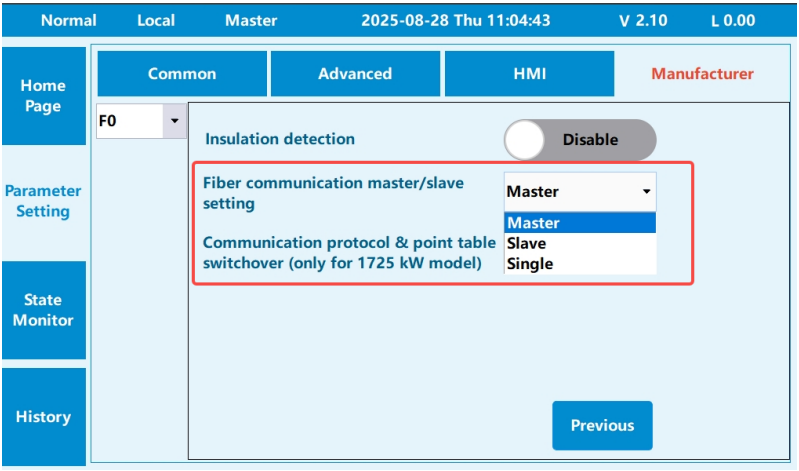
Commissioning instruction:

- **Solid line in red** indicates electrical wiring inside a cabinet. If a parallel communication error occurs, measure and check whether these cables are correctly connected.
- **Dashed line in red** indicates the LAN cable used to connect two PCS cabinets. Only one LAN cable is required between every two cabinets for the PCSs.
- You must turn the DIP switch S6 of the first cabinet (1#) and the last cabinet (4#) to ON and turn S6 of other communication boards to OFF.
- Pin specifications of the network port for the parallel system are defined by Wattox, and the network port is not a standard port. Do not connect the network port to a computer or switch through a LAN cable.
- If the communication interference issue is hard to resolve, remove the LAN cable from the J39 port on the communication board of the first cabinet (1#) and from the J40 port on the communication board of the last cabinet (4#).

5.11.2 Master/slave setting of optical fiber communication in the cabinet

Choose "Parameter Setting" > "Manufacturer" > "F0" and set "Fiber communication master/slave setting" to "Master".

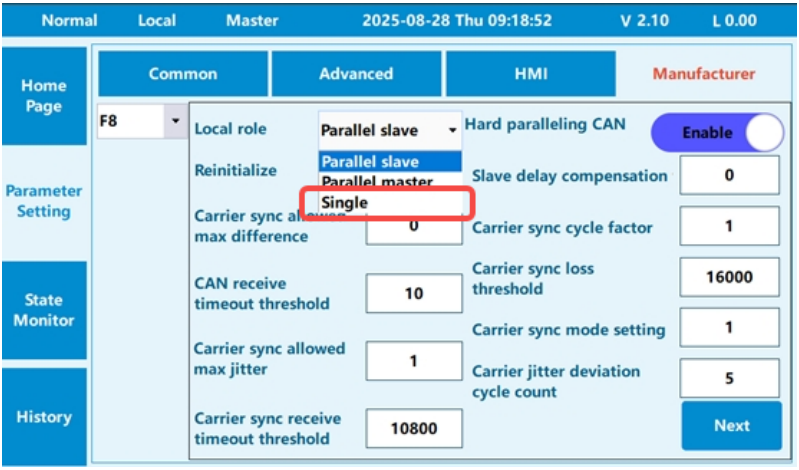
Set the other PCS in the parallel system to "Slave".



5.11.3Setting the Local Role

Choose "Parameter Setting" > "Manufacturer" > "F8" > "F8_00", and set "Local role" to "Parallel master".

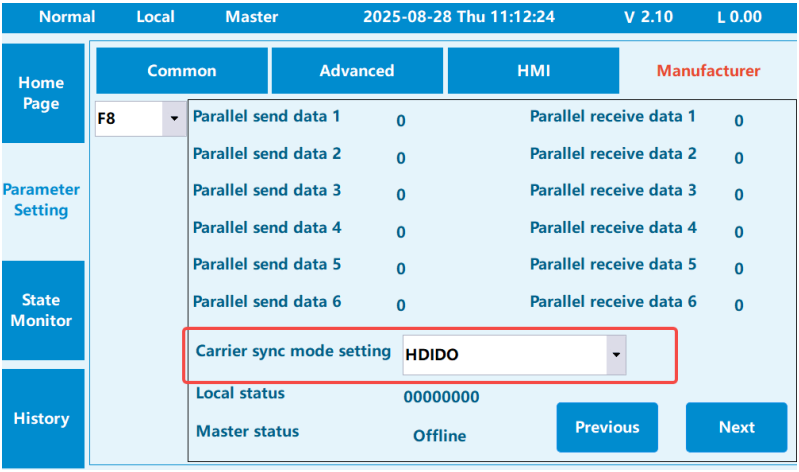
Set the other PCS in the parallel system to "Parallel slave".



5.11.4Setting the Carrier Synchronization Mode

Choose "Parameter Setting" > "Manufacturer" > "F8" > "F8_24", and set "Carrier sync mode" to "HDIDO".

Set the other slave device in the parallel system to "HDIDO" too.

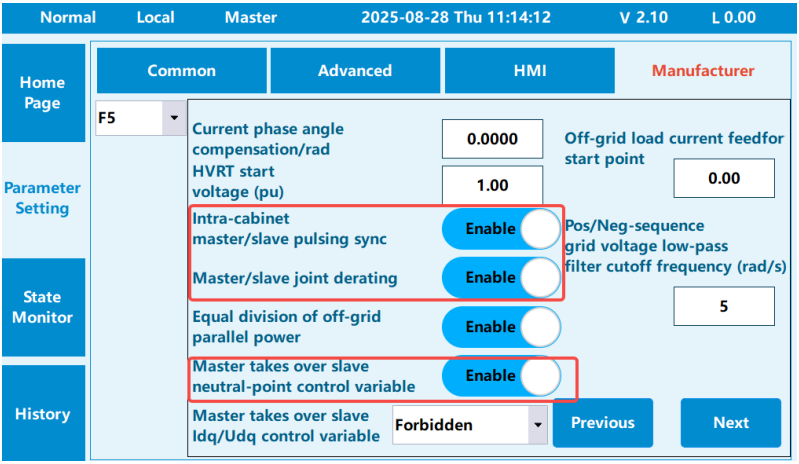


5.11.5Configuring Control Parameters

Choose "Parameter Setting" > "Manufacturer" > "F5" and set "Intra-cabinet master/salve pulsing sync" to "Enable".

Set the other slave device in the parallel system to "Enable" too.

- Choose "Parameter Setting" > "Manufacturer" > "F5", and set "Intra-cabinet master/ salve pulsing sync" to "Enable".
- Choose "Parameter Setting" > "Manufacturer" > "F5", and set "Master/slave joint derating" to "Enable".
- Choose "Parameter Setting" > "Manufacturer" > "F5", and set "Master takes over slave ldq/Udq control variable" to "Enable".

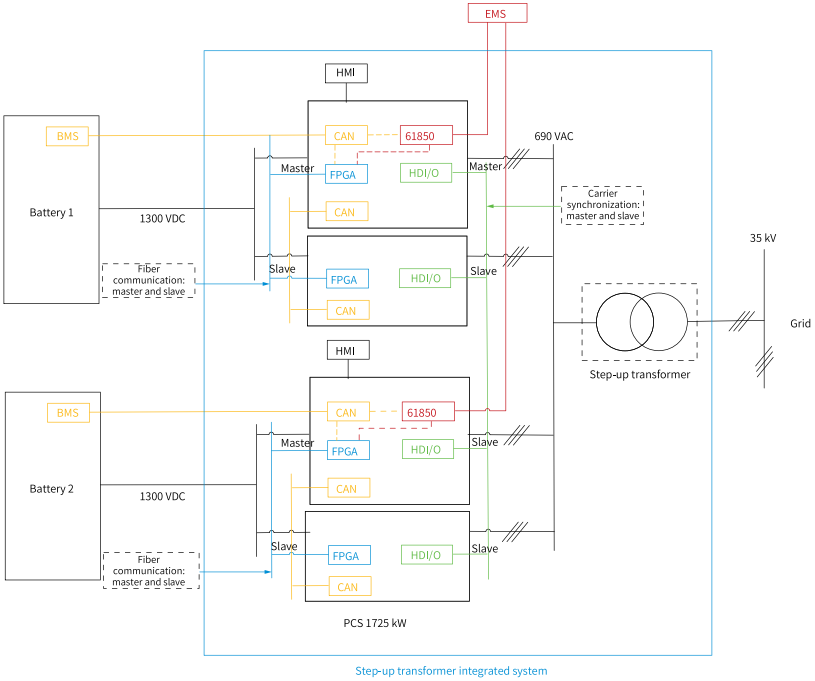


5.11.6Example of Hardware Parallel Settings

On the AC side, four PCSs work in parallel mode. On the DC side, each two PCSs work in parallel mode. #1_PCS and #2_PCS are connected in parallel mode on the DC side and are connected to the same battery. #3_PCS and #4_PCS are connected in parallel mode on the DC side and are connected to the same battery.

Specifically:

- For master/slave role of in-cabinet fiber communication (F0-08): #1_PCS and #3_PCS are set as the master and connected to the EMS or BMS. #2_PCS and #4_PCS are set as the slave.
- For local role in carrier synchronization (F8-00): #1_PCS is set as the master, and #2_PCS, #3_PCS, and #4_PCS are set as the slave.

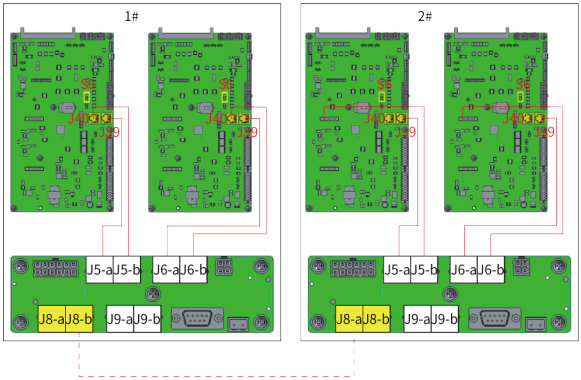


5.12 Grid-Connected AC Parallel

This section describes grid-connection AC parallel operations in the DC separated and AC parallel scenario.

5.12.1Hardware Wiring and Dial Commissioning

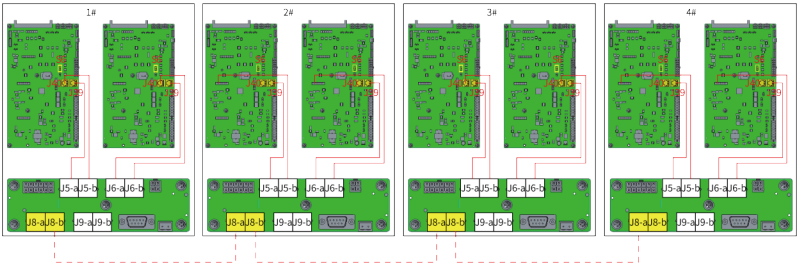
The following figure shows the wiring and dial commissioning of two parallel cabinets.



Commissioning instruction:

- **Solid line in red** indicates electrical wiring inside a cabinet. If a parallel communication error occurs, measure and check whether these cables are correctly connected.
- **Dashed line in red** indicates the LAN cable used to connect two PCS cabinets. Only one LAN cable is required between every two cabinets for the PCSs.
- You must turn the DIP switch S6 of the first cabinet (1#) and the last cabinet (2#) to ON and turn S6 of other communication boards to OFF.
- Pin specifications of the network port for the parallel system are defined by Wattox, and the network port is not a standard port. Do not connect the network port to a computer or switch through a LAN cable.
- If the communication interference issue is hard to resolve, remove the LAN cable from the J39 port on the communication board of the first cabinet (1#) and from the J40 port on the communication board of the last cabinet (2#).

The following figure shows the wiring and dial commissioning of multiple parallel cabinets.



Commissioning instruction:

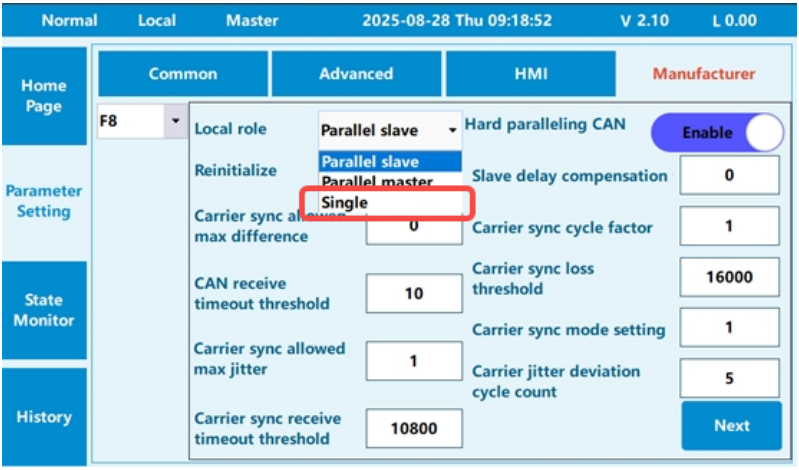
- **Solid line in red** indicates electrical wiring inside a cabinet. If a parallel communication error occurs, measure and check whether these cables are correctly connected.

- **Dashed line in red** indicates the LAN cable used to connect two PCS cabinets. Only one LAN cable is required between every two cabinets for the PCSs.
- You must turn the DIP switch S6 of the first cabinet (1#) and the last cabinet (4#) to ON and turn S6 of other communication boards to OFF.
- Pin specifications of the network port for the parallel system are defined by Wattox, and the network port is not a standard port. Do not connect the network port to a computer or switch through a LAN cable.
- If the communication interference issue is hard to resolve, remove the LAN cable from the J39 port on the communication board of the first cabinet (1#) and from the J40 port on the communication board of the last cabinet (4#).

5.12.2 Setting the Local Role

Choose "Parameter Setting" > "Manufacturer" > "F8" > "F8_00" and set "Local role" to "Parallel master" for one controller of any device.

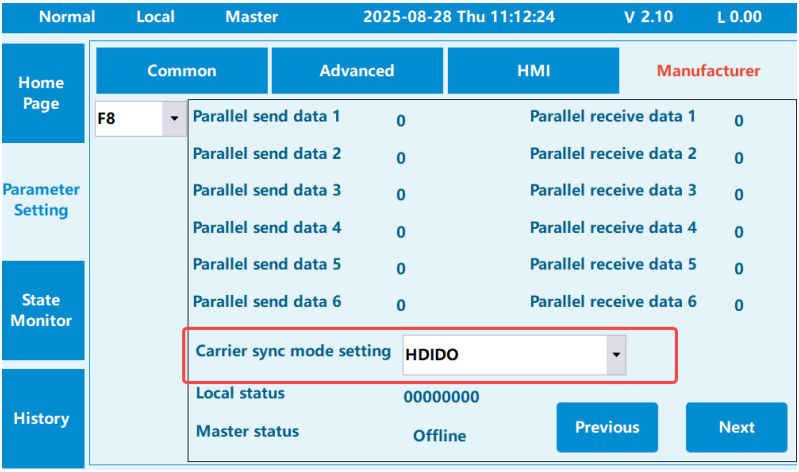
Set the other controller and the controllers of other devices to "Parallel slave".



5.12.3 Setting the Carrier Synchronization Mode

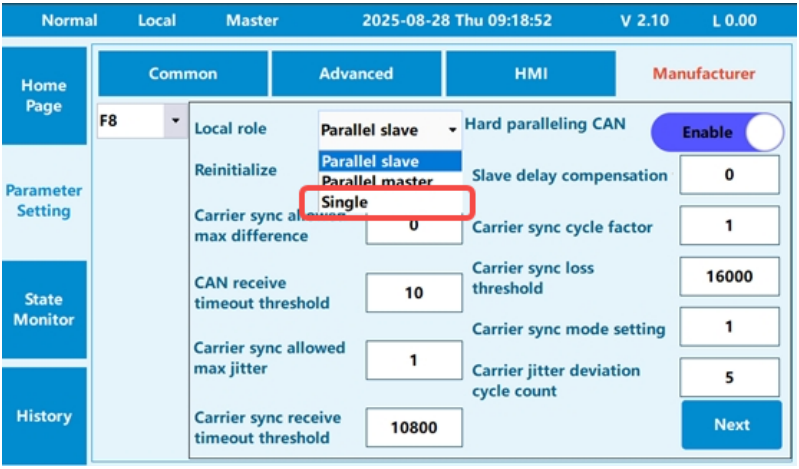
Choose "Parameter Setting" > "Manufacturer" > "F8" and set "Carrier sync mode setting" to "HDIDO".

Set the other slave device in the parallel system to "HDIDO" too.



When you set "Local role" in the F8 group on the "Manufacturer" page to "Parallel master", "Slave compensation delay" must be set to "0".

When you set "Local role" in the F8 group on the "Manufacturer" page to "Parallel slave", "Slave compensation delay" must be set to "-800".



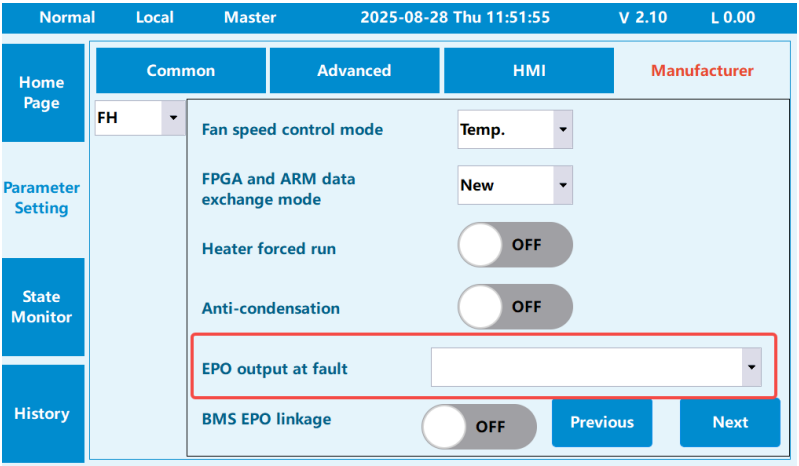
5.13 Configuring Other Parameters

5.13.1EPO Output upon Fault

When a fault occurs on the PCS, the emergency power off (EPO) output function resets the emergency stop DO dry contact signal. This function is disabled upon factory delivery and can be specified based on site requirements.

To configure this function, go to "Parameter Setting" > "Manufacturer" (FH group).

- If you enable this function, the PCS outputs the emergency stop DO in case of emergency stop.
- If you disable this function, the PCS outputs the emergency stop DO in case of any fault.



5.14 Stop

On the left side of the "HMI" page, click "Home Page" and then click the "Stop" command.

If the PCS is powered by an external power supply, disconnect the external power supply.

6 Programming

6.1 Preparations Before Programming

1. Before programming, ensure that the PCS is in the stop state.
2. Remove all cables from network ports of the PCS.
3. Prepare the following tools: SOP-20, one network cable, minUSB-to-USB converter, and USB flash disk.
4. Prepare the software InoLoader.

6.2 Programming Instructions

The ARM and FPGA of the main control board and the ARM of the communication board must be programmed.

Software programming must be performed on both master and slave devices, and the software versions must be consistent.

6.3 ARM Programming

1. Turn on the control power and connect the network cable to the network port of the user adapter board, as shown in the following figure. The PCS executes the programming task for the master and slave devices through network ports J9a and J9b, respectively.

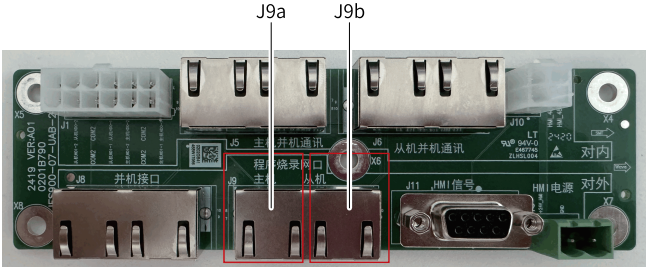
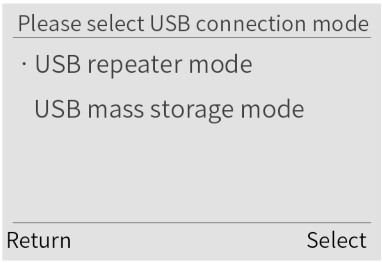


Figure 6-1 Connection port

2. Connect one end of minUSB to the PC and the other end to SOP-20. Select “USB repeater mode" for SOP-20, as shown in the following figure.



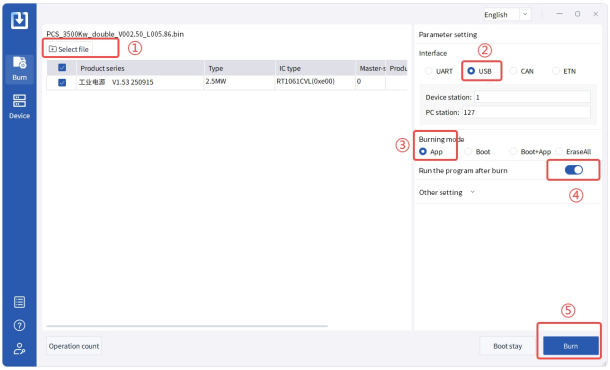
3. Start the programming software InoLoader and perform the following operations:

   a. Select "USB" for "Interface".

   b. Select the FPGA or ARM software to be programmed.

   c. Select "Run the program after burn ".

   d. Click “Burn".



6.4 HMI Programming

6.4.1 Project File Description

The project files are mainly used to upgrade the IT7000 HMI through peripherals such as the USB flash disk.

The upgrade package contains the "firmware.zip" and "project.zip" files.

- firmware.zip: This file is used for HMI firmware upgrade. If the current firmware version of HMI differs from that in the "project.zip" file, the firmware must be upgraded first.
- project.zip: This file is used for HMI project upgrade.

Load the two files to the USB flash disk or SD card. Note that the .zip files do not need to be decompressed.

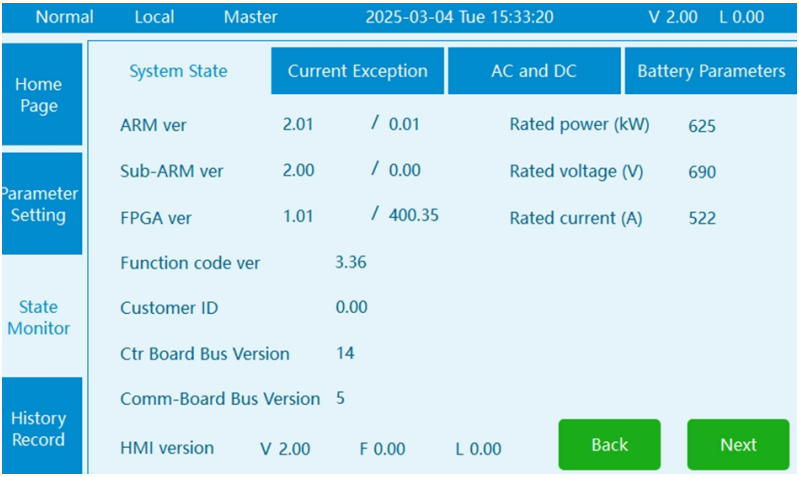
6.4.2 Performing the Upgrade

This section describes how to perform the upgrade by using a USB flash disk.

1. Copy the following files to the USB flash disk.
   a. firmware.zip, used for firmware update
   b. project.zip, used for project update
2. After the HMI is powered on, insert the USB flash disk. The system prompts the control page login page.
3. Check whether the firmware is the latest version 102.V1R2C00SPC33(2023-02-28). If not, update the firmware and project.
   a. Select USB as the mounted device, select the corresponding "firmware.zip", and then click "HMI Project".
   b. Wait for about 2 minutes for the HMI upgrade to complete. After that, the HMI runtime version changes to be the latest.
   c. Reinsert the USB flash disk, select USB as the mounted device, choose the corresponding "project.zip", and click "HMI Project".
   d. The system prompts a message, asking whether you want to update the firmware simultaneously, select "No".
   e. The system prompts a message, indicating that the system is updating the project. After the update, the system automatically restarts and enters the main screen.

6.5 Verification of Update

After the ARM and HMI programs are upgraded, choose "State Monitor" > "System State" to check whether the software version is correct, as shown in the following figure.



7 HMI Operation and Display

7.1 System Menu Architecture

You can view the states of the entire PCS and set some parameters on the HMI. The architecture of the system menu is as follows.

Level I menu	Level II menu	Description
Home page	/	For details, see *"7.2 Home Page" on page 103*.
Parameter Setting	Common	For details, see *"7.3.1 Common" on page 104*.
	Advanced	For details, see *"7.3.2 Advanced" on page 105*.
	HMI	For details, see *"7.3.3 HMI parameter" on page 106*.
	Manufacturer	For details, see *"7.3.4 Factory Parameters" on page 106*.
State Monitor	System State	For details, see *"7.4.1 System state" on page 107*.
	Current Exception	For details, see *"7.4.2 Current Exception" on page 108*.
	AC and DC	For details, see *"7.4.3 AC and DC" on page 108*.
	Battery Parameters	For details, see *"7.4.4 Battery Parameters" on page 109*.
History Record	History	For details, see *"7.5.1 History Exception" on page 110*.
	History Operation	For details, see *"7.5.2 History Operation" on page 110*.
	History State	For details, see *"7.5.3 History State" on page 111*.

7.2 Home Page

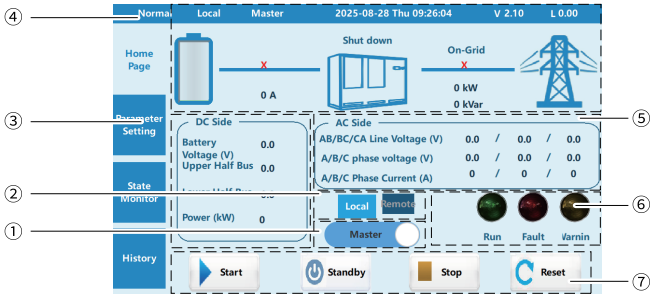


Figure 7-1 HMI page layout

No.	Name	Description
①	Master/Slave switching	Switch between the master and slave PCSs.
③	Command source	Select a command source for the PCS, which can be local or remote.
②	DC side data	Displays the voltage on the DC side of the PCS.
④	Running status	Displays information such as the battery power, charge/discharge mode, grid-connected/off-grid mode, and power flow.
⑤	AC side data	Displays the voltage and current on the three-phase AC side of the PCS.
⑥	Status indicator	Displays the operation, fault, or alarm state of the PCS.
⑦	Control buttons	Control the start, stop and fault reset of the PCS.

- Press "Start". Press "Start" in the acknowledgment box again to start the PCS. If the PCS starts successfully, the RUN indicator lights up.
- The "Stop" operation process is similar.
- When an alarm occurs, the alarm indicator keeps on.
- When a fault occurs, the fault indicator keeps blinking. If the buzzer is turned on, the buzzer beeps upon the fault and the HMI displays a fault message window.
  Press "View" to switch directly to the "Current Exception" page to view the specific fault and silent the buzzer. Press "Close" to silent the buzzer.

7.3 Parameter setting

7.3.1 Common

On the "Common" page, you can configure the command source and operating mode of the PCS.

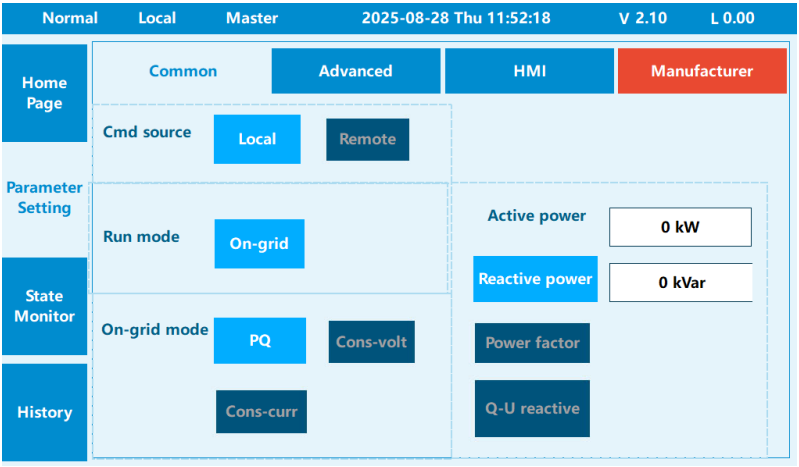
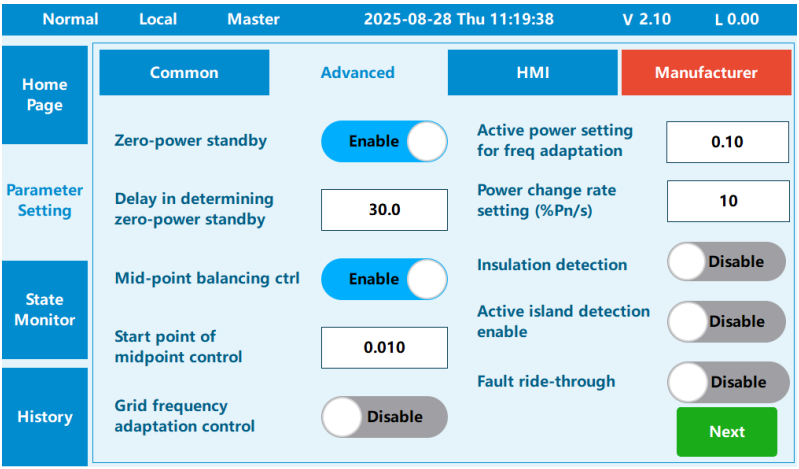


Table 7–1 Configuring common parameters

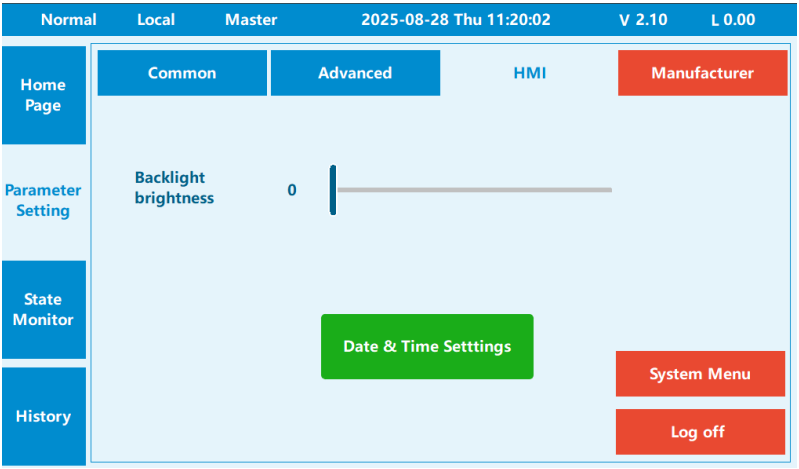
Setting	Item	Description
Command source	Local	PCS parameters are controlled by HMI only.
	Remote	PCS parameters are controlled by EMS only.
Work mode	P/Q	Constant power charge/discharge mode. In this mode, you can set the charging/discharging power.
	Off-grid	Off-grid operation mode. In this mode, you can set the output voltage and frequency.
	Cons-Volt	Constant voltage charge/discharge mode. In this mode, you can set the voltage on the DC side.
	Cons-Curr	Constant current charge/discharge mode. In this mode, you can set the current on the DC side.
	VVSG	The PCS is controlled by the voltage-type virtual synchronous motor.

7.3.2 Advanced



Parameter	Description
Zero-power standby	Specifies whether to enable zero power standby mode.
Delay in determining zero power standby	Specifies the delay in determining zero power standby, which can be set on the host controller.
Midpoint balancing ctrl	Specifies the bus midpoint balance algorithm for adjusting the bus.
Start point of midpoint control	Specifies the home position of the cut-in code of the midpoint balance algorithm, which can be set on the host controller.
Grid frequency adaption control	Specifies whether to enable grid frequency adaptation.
Active power setpoint for frequency adaptation	Specifies the grid frequency adaptation threshold.
Power change rate setting (%Pn/s)	Specifies the power pre-charge rate. For example, if this parameter is set to 100% Pn/s, it takes 1 second for the power to change from 0 to full-load.
Insulation detection	Specifies whether to enable insulation detection.
Active islanding detection enable	Specifies whether to enable active islanding detection.
Fault ride-through	Specifies whether to enable fault ride-through.

7.3.3 HMI parameter



Parameter	Description
Backlight brightness	Adjusts the backlight brightness of the HMI.
Date & Time Settings	Pops up the time settings window, on which you can set the current year, month, day, hour, minute, and second.

7.3.4 Factory Parameters

The "Manufacturer" page is hidden. After you press and hold "HMI" for 3 seconds, the login page is displayed. Enter the password to access the "Manufacturer" page.

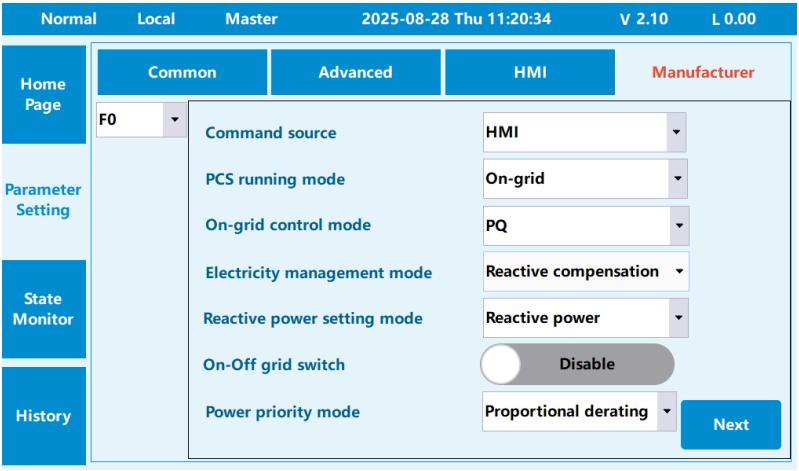


Figure 7-2 Manufacturer parameters of the PCS

7.4 State Monitor

7.4.1 System state

The "System State" page displays the major operation states, devices, and temperature values of the PCS.

You can click "Product Information" to view the current software version, parameter list version, and rated parameters of the PCS.

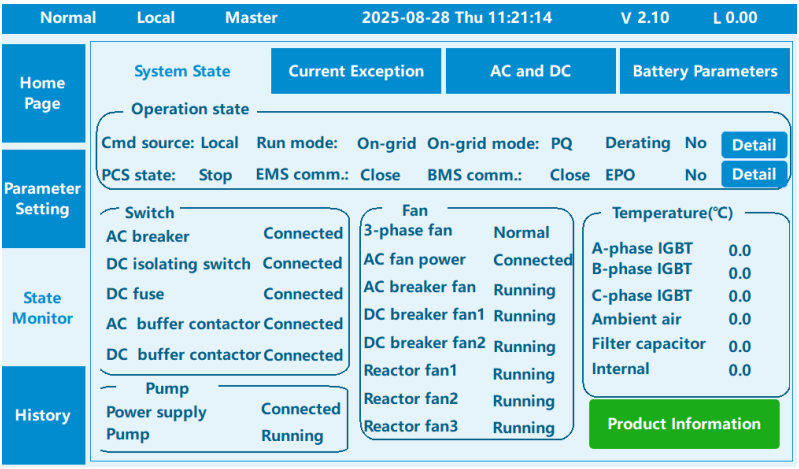


Figure 7-3 System state

7.4.2 Current Exception

The "Current Exception" page displays all exceptions that occur after the PCS is reset last time. The exception information includes the occurrence date/time, code, subcode, and name of the exceptions.

You can view detailed information of the latest 10 exceptions at the bottom of this page, including the possible causes and solutions for the exceptions.

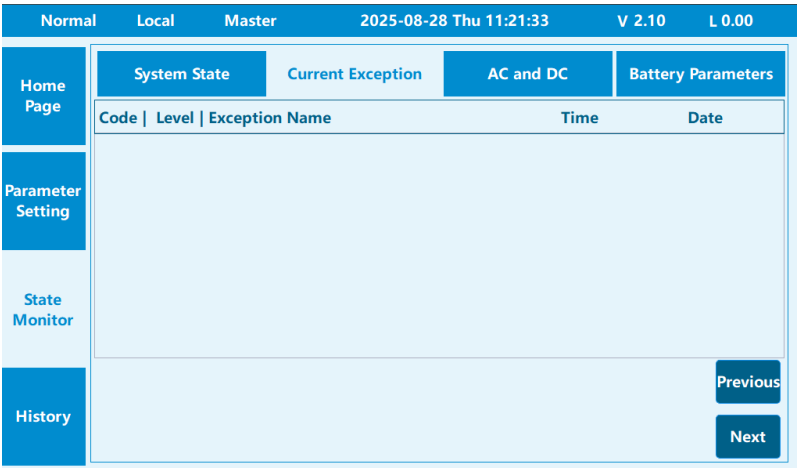


Figure 7-4 Current exception

7.4.3 AC and DC

The "AC and DC" page displays information related to common variables of the PCS on the AC and DC sides.

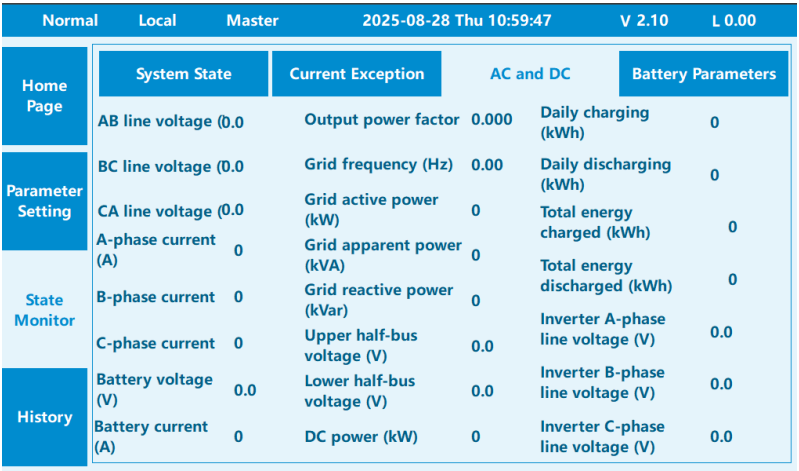


Figure 7-5 AC and DC

7.4.4 Battery Parameters

The "Battery Parameters" page displays battery-related information.

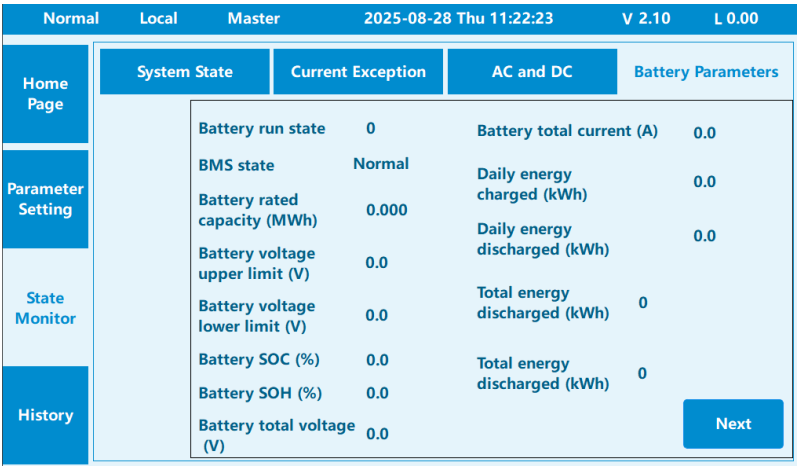


Figure 7-6 Battery parameters

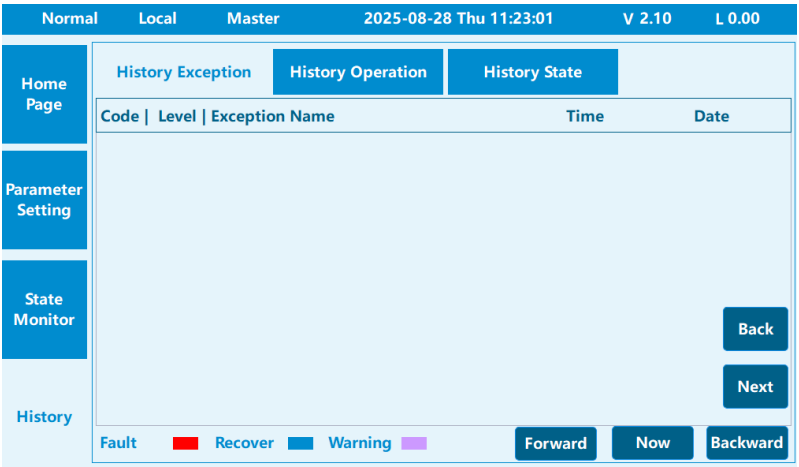
7.5 History

7.5.1 History Exception

The "History Exception" page records the occurrence and recovery time of all the faults and alarms of the PCS.

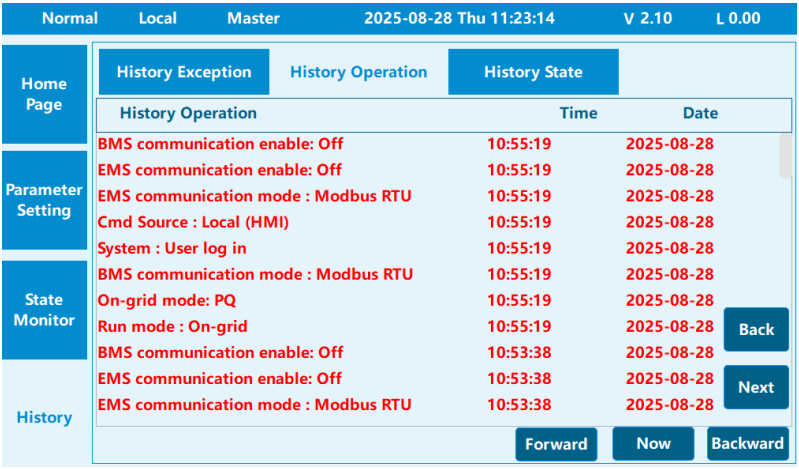
This page can store up to 50,000 records. Excess records will overwrite previous ones.

Faults are displayed by day.



7.5.2 History Operation

The "History Operation" page displays all the operation records of the PCS, including the operation mode, working mode, advanced functions, and restore factory settings.



This page can store up to 50,000 records. Excess records will overwrite previous ones.

7.5.3 History State

The "History State" page displays all the state changes of the PCS, including the operating state, grid state, switch state, and fan state.

This page can store up to 50,000 records. Excess records will overwrite previous ones.

8 Troubleshooting

8.1 Insulation detection function

Overview

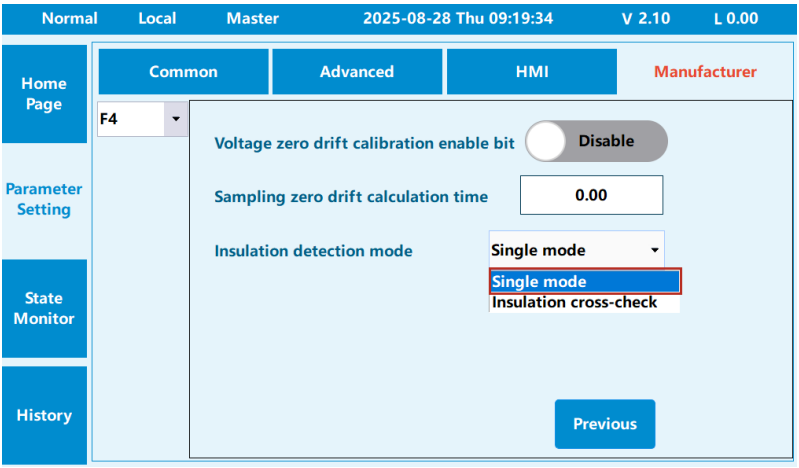
The insulation resistance value is criteria for assessing the insulation performance of electrical systems. It is also critical for the safety of electrical equipment and personnel.

Generally, active or passive insulation detection technology is adopted in the energy field. The unbalanced bridge (passive) method has been widely used because it has high detection accuracy, is easy to operate, and can measure the insulation resistance of the positive bus and negative bus in real time.

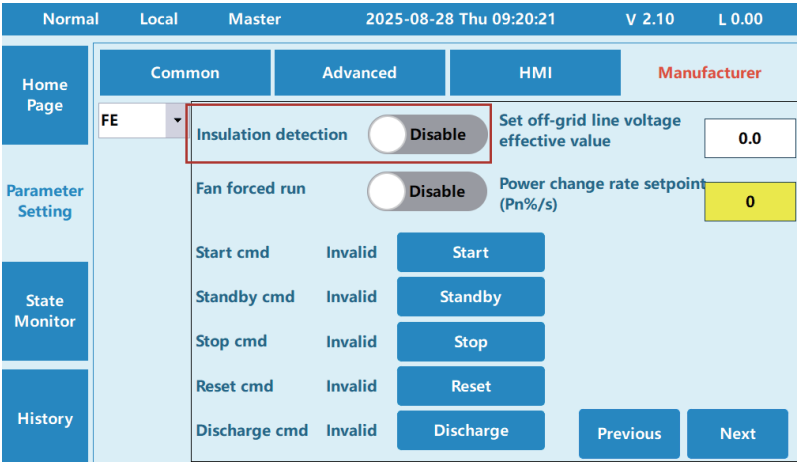
Single and dual DC models adopt the passive insulation detection method. When the battery voltage is greater than 540 V, insulation detection in single mode and mutual insulation detection of master and slave in parallel mode are supported.

Single mode

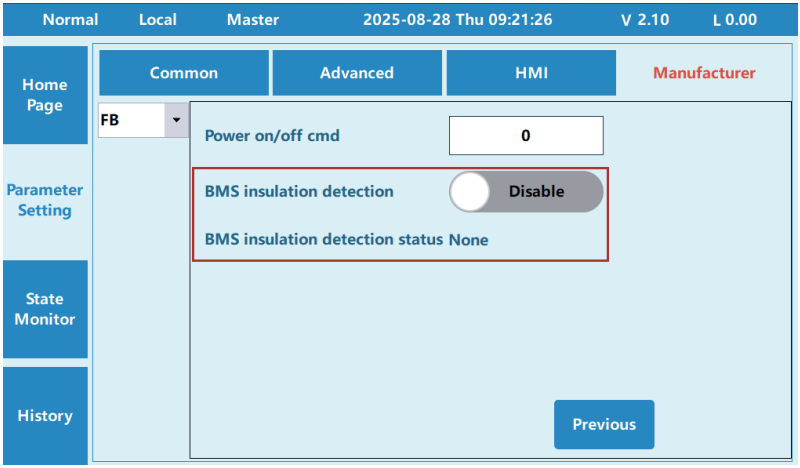
Mode selection: Set the insulation detection mode (F4.19) to 0 (single mode).



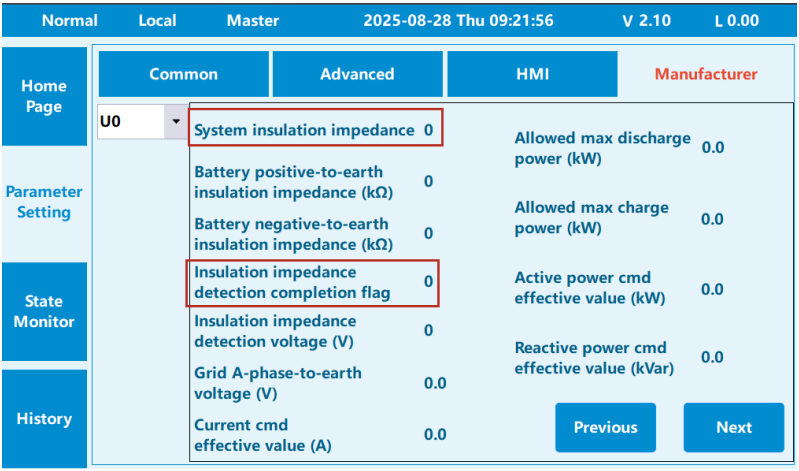
Detection enabling: Set insulation detection (FE.07) to 1 (enable).



BMS detection configuration: If the PCS is connected to the BMS that supports insulation detection, set FB.51 to 1 to enable BMS insulation detection.



Detection result: After insulation detection of the local device is complete, the insulation impedance detection completion flag (U00.82) is set to 1 and the detection result is displayed in U00.73.

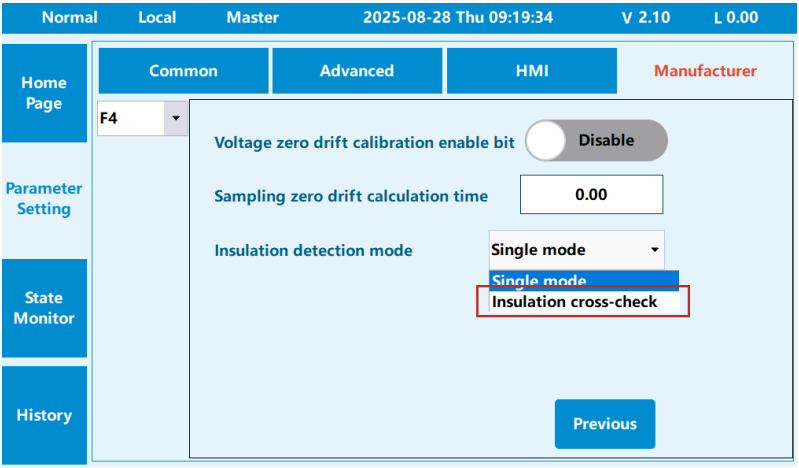


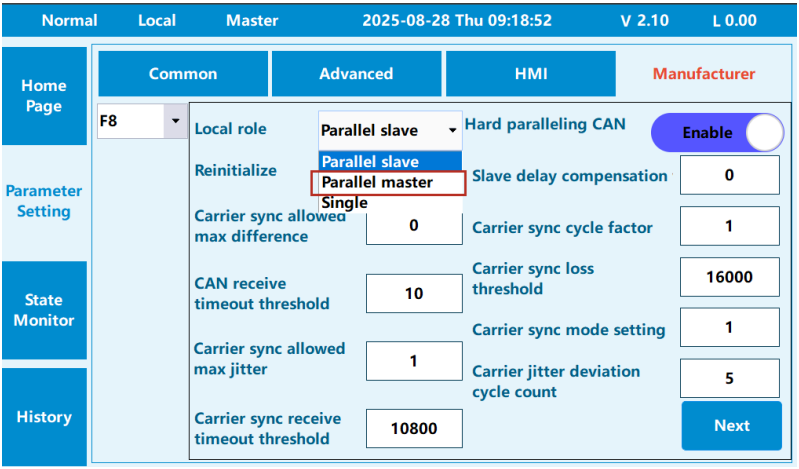
CAUTION

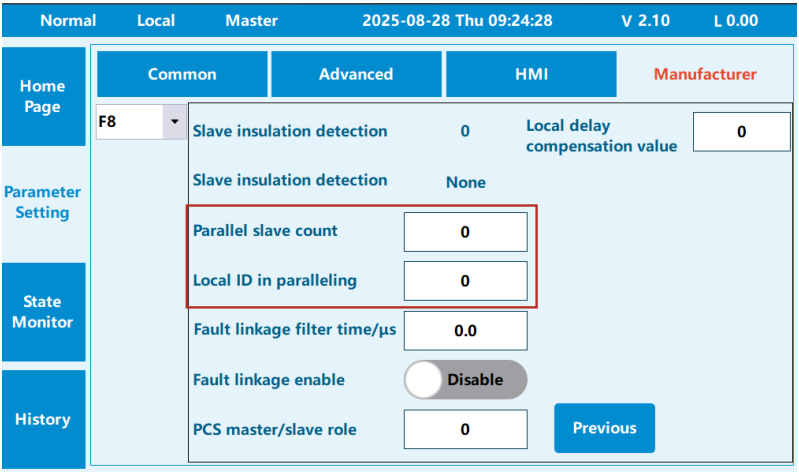
Currently, only the system insulation impedance is displayed. The grounding insulation impedance of the battery positive and negative electrodes is not separately displayed.

Mutual insulation detection

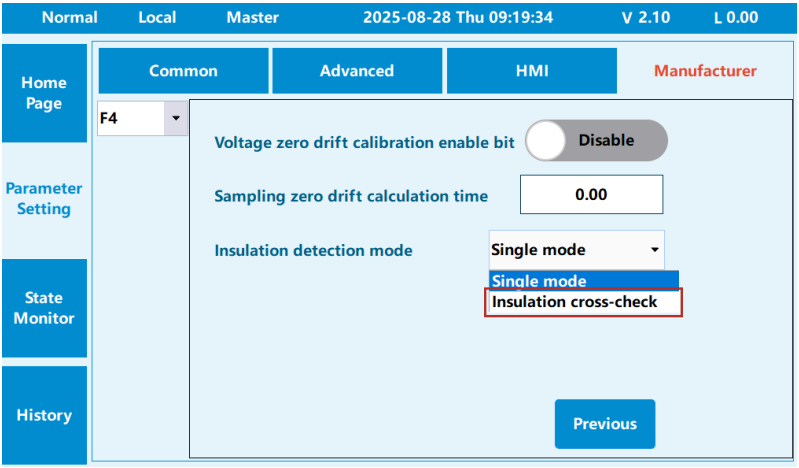
Parallel master configuration: Set the insulation detection mode (F4.19) to 1 (mutual insulation detection), set the Local role (F8.00) to 1 (parallel master), set CAN communication for hardware parallel (F8.26) to 1 (enable), and specify the total number of slaves in parallel mode (F8.41) and the number of the local device in parallel mode (F8.42), which defaults to 0 for the master. If the PCS is connected to the BMS that supports insulation detection, set FB.51 to 1 to enable BMS insulation detection.

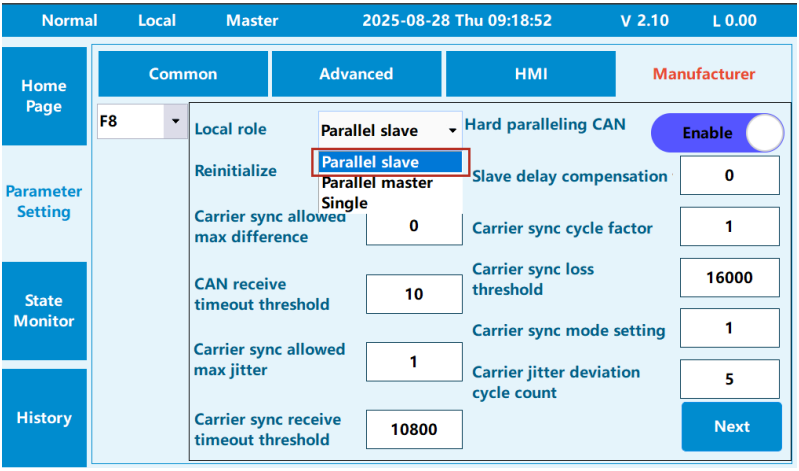

Normal
Local
Master
2025-08-28 Thu 09:19:34
V 2.10
L 0.00
Home Page
Parameter Setting
State Monitor
History
Common
Advanced
HMI
Manufacturer
F4
Voltage zero drift calibration enable bit
Disable
Sampling zero drift calculation time
0.00
Insulation detection mode
Single mode
Single mode
Insulation cross-check
Previous

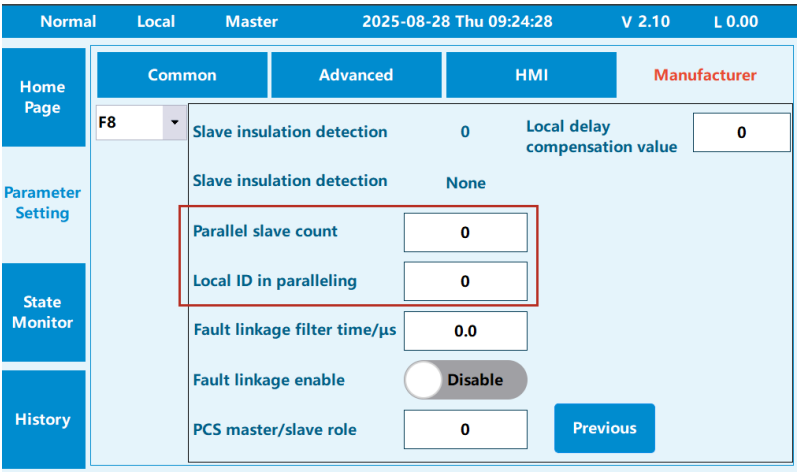

Normal
Local
Master
2025-08-28 Thu 09:18:52
V 2.10
L 0.00
Home Page
Parameter Setting
State Monitor
History
Common
Advanced
HMI
Manufacturer
F8
Local role
Parallel slave
Parallel slave
Parallel master
Single
Reinitialize
Carrier sync allowed max difference
0
CAN receive timeout threshold
10
Carrier sync allowed max jitter
1
Carrier sync receive timeout threshold
10800
Hard paralleling CAN
Enable
Slave delay compensation
0
Carrier sync cycle factor
1
Carrier sync loss threshold
16000
Carrier sync mode setting
1
Carrier jitter deviation cycle count
5
Next



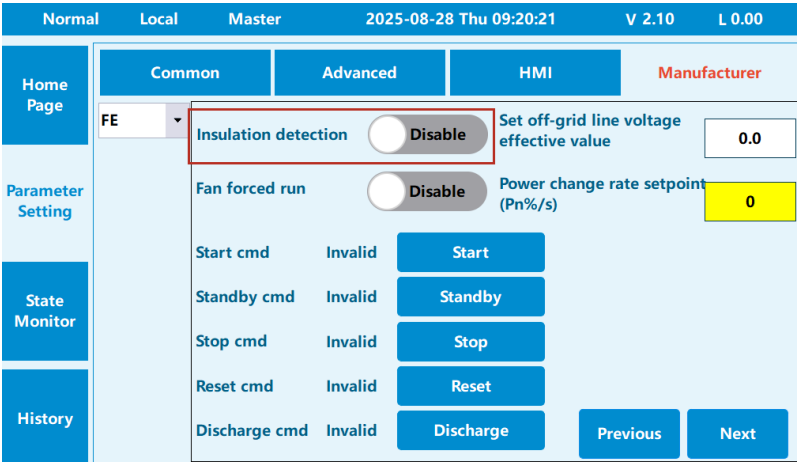
Parallel slave configuration: Set the insulation detection mode (F4.19) to 1 (mutual insulation detection), set the Local role (F8.00) to 0 (parallel slave), set CAN communication for hardware parallel (F8.26) to 1 (enable), and specify the number of the local device in parallel mode (F8.42), which starts from 1 for the slave. If the PCS is connected to the BMS that supports insulation detection, set FB.51 to 1 to enable BMS insulation detection.







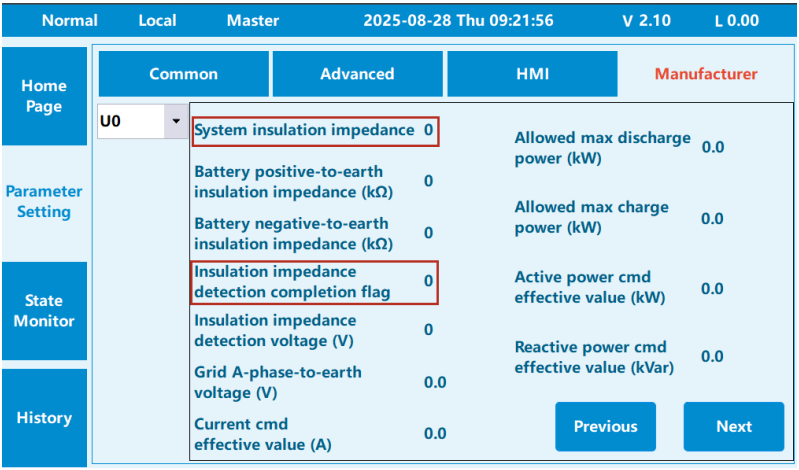
Detection enabling: Set insulation detection (FE.07) to 1 (enable).



CAUTION

If mutual insulation detection is selected for all parallel devices, all the devices start mutual insulation detection only when FE.07 is set to 1 for the parallel master to send a mutual insulation detection command.

Detection result: After insulation detection of the local device is complete, the insulation impedance detection completion flag (U00.82) is set to 1 and the detection result is displayed in U00.73.



CAUTION

Currently, only the system insulation impedance is displayed. The grounding insulation impedance of the battery positive and negative electrodes is not separately displayed.

8.2 Common Troubleshooting

Energy storage battery polarity connected reversely

Possible causes

- The battery is connected to the power conversion system with wrong polarity.
- The battery voltage sampling circuit is abnormal.

Solution

1. Stop and power off all the devices and check whether the battery cables are correctly connected on the DC side.
2. Check whether the detection board battery sampling wires are correctly connected to the J1 (battery positive) and J2 (battery negative) terminals.

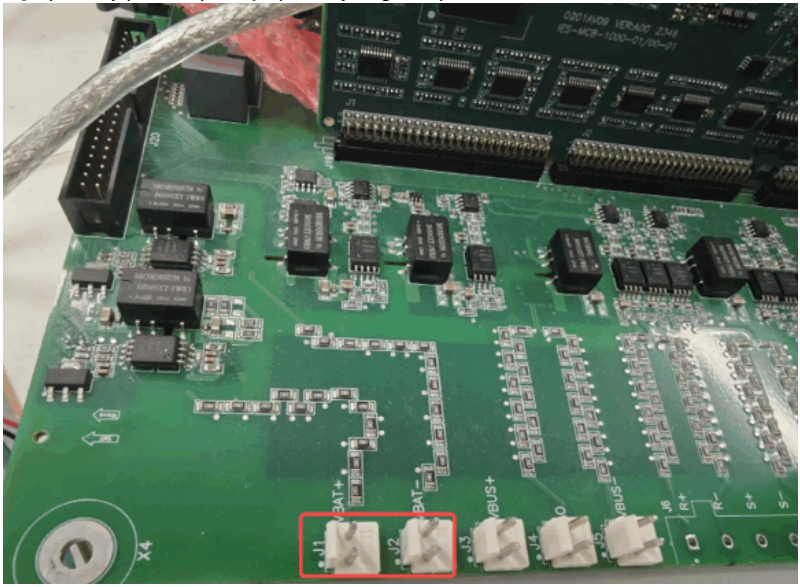


Figure 8-1 Checking the connection status

3. Remove the battery input terminal and measure the voltage of the battery input cable using the DC voltage range of the multimeter.

Use the red lead (positive) of the multimeter to measure the positive input pole (BAT+). Use the black lead (negative) of the multimeter to measure the negative input pole (BAT-).

If negative voltage is measured, the polarity of the connected battery cable is incorrect. In this case, connect the BAT+ cable to the BAT+ terminal and the BAT- cable to the BAT- terminal.

4. Check whether the "Battery voltage polarity reversed protect threshold (V)" parameter in group F6 (Manufacturer) is set to the default value -100.

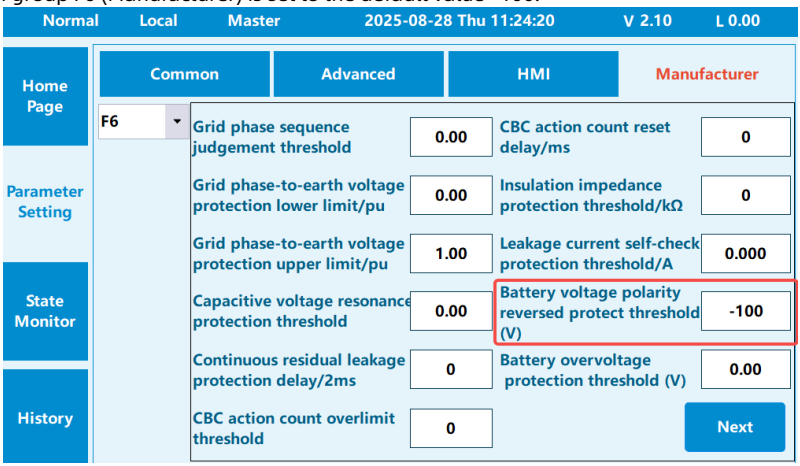


Figure 8-2 Checking manufacturer parameters

5. If the fault persists, collect the fault information (HMI history faults, history operations, and history states), take a photo of the settings of parameters in group E0, and read black box fault data.

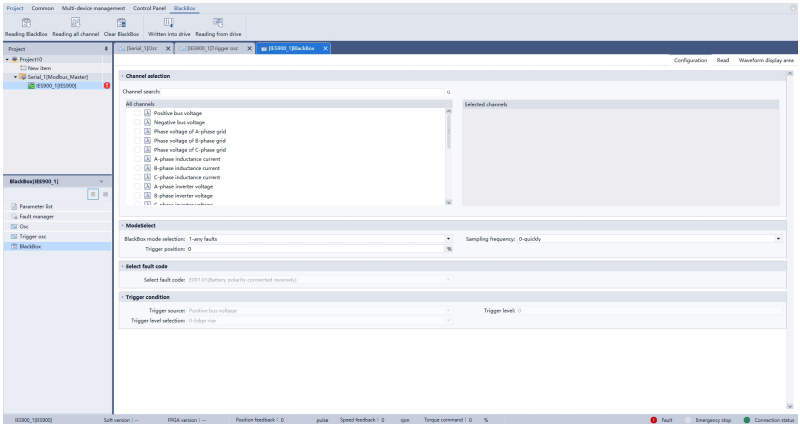


Figure 8-3 Collecting fault information

6. Contact the after-sales service engineer of Wattox to rectify the fault.

Energy storage battery voltage too high

Possible causes

- The battery voltage upon power-on is greater than the allowed value.
- The battery voltage sampling circuit is abnormal.
- The "DC full-bus software overvolt threshold / pu" parameter in group F6 is set to a too small value.

Solution

1. Check the battery cable connections on the DC side (fuse FU5/6 and detection board terminals J1 and J2).
2. Remove the battery input terminal and measure the voltage of the battery input cable using the DC voltage range of the multimeter.
   Use the red lead (positive) of the multimeter to measure the positive input pole (BAT+). Use the black lead (negative) of the multimeter to measure the negative input pole (BAT-). Measure the battery voltage sampling value.
3. Change the value of the "DC full-bus software overvoltage threshold / pu" parameter in group F6 based on the voltage of the externally connected battery.

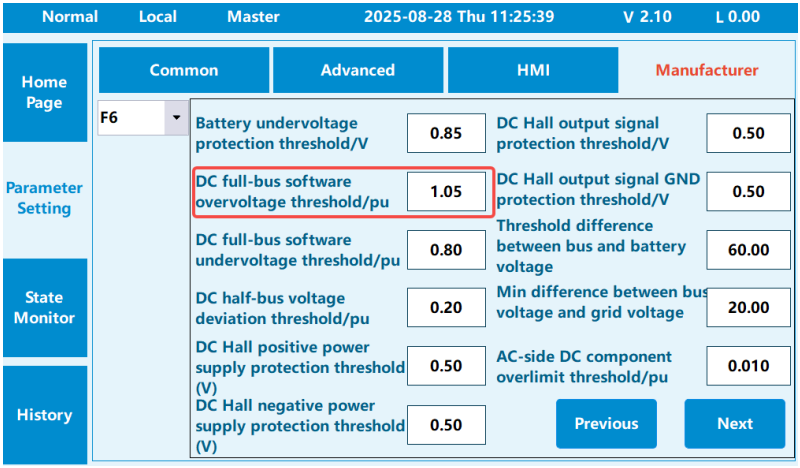


Figure 8-4 Modifying the configuration parameter

4. If the fault persists, collect the fault information (HMI history faults, history operations, and history states), take a photo of the settings of parameters in group E0, and read black box fault data.
5. Contact the after-sales service engineer of Wattox to rectify the fault.

Battery voltage too high

Possible causes

- The battery voltage is greater than the upper limit.
- The battery voltage sampling board is faulty.
- The "Battery overvoltage protection threshold (V)" parameter in group F6 is set to a too small value.

Solution

1. Check the J1 and J2 terminals in the DC voltage sampling circuit.
2. Remove the battery input terminal and measure the voltage of the battery input cable using the DC voltage range of the multimeter.
   Use the red lead (positive) of the multimeter to measure the positive input pole (BAT+). Use the black lead (negative) of the multimeter to measure the negative input pole (BAT-). Measure the battery voltage sampling value.
3. Check whether the "Battery overvoltage protection threshold (V)" parameter (DC voltage upper limit multiplied by a coefficient) in group F6 (Manufacturer parameters) is within the battery voltage range.

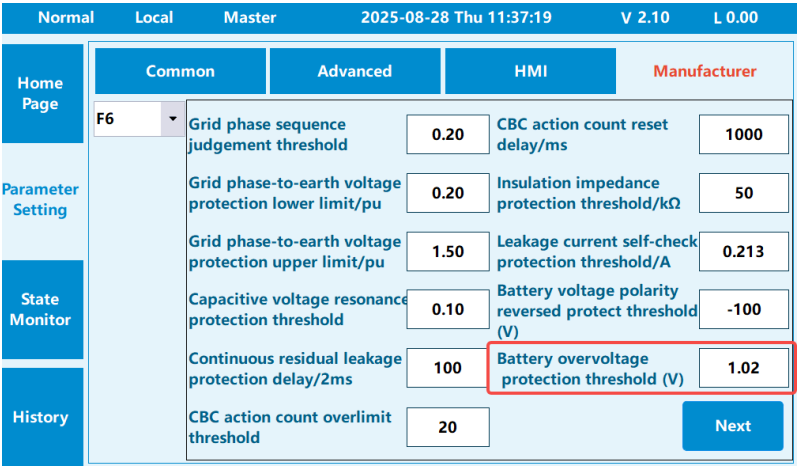


Figure 8-5 Confirming the battery voltage range

4. If the fault persists, collect the fault information (HMI history faults, history operations, and history states), take a photo of the settings of parameters in group E0, and read black box fault data.
5. Contact the after-sales service engineer of Wattox to rectify the fault.

Battery voltage too low

Possible causes

- The battery voltage is less than the battery voltage lower limit.
- The battery voltage sampling board is faulty.
- The "Battery underoltage protection threshold (V)" parameter in group F6 is set to a too large value.

Solution

1. Check the J1 and J2 terminals in the DC voltage sampling circuit.
2. Remove the battery input terminal and measure the voltage of the battery input cable using the DC voltage range of the multimeter.
   Use the red lead (positive) of the multimeter to measure the positive input pole (BAT+). Use the black lead (negative) of the multimeter to measure the negative input pole (BAT-). Measure the battery voltage sampling value.
3. Check whether the "Battery undervoltage protection threshold (V)" parameter (DC voltage lower limit multiplied by a coefficient) in group F6 (Manufacturer parameters) is within the battery voltage range.

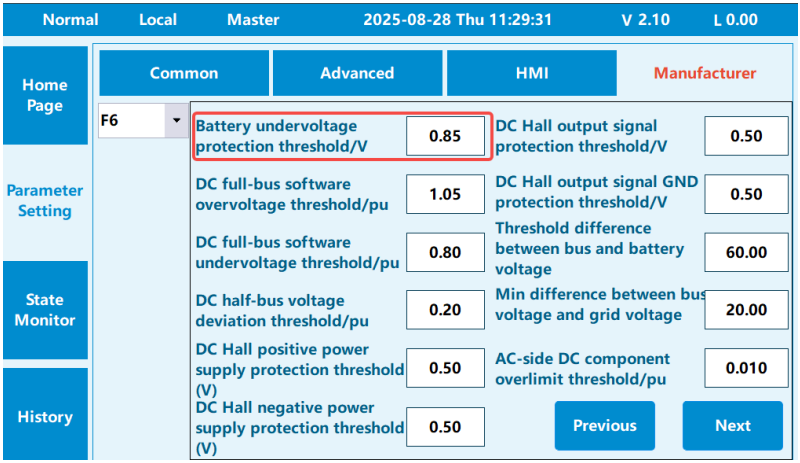


Figure 8-6 Confirming the battery voltage range

4. If the fault persists, collect the fault information (HMI history faults, history operations, and history states), take a photo of the settings of parameters in group E0, and read black box fault data.
5. Contact the after-sales service engineer of Wattox to rectify the fault.

BMS system fault

Possible causes

An exception occurs on the battery. (The fault signal is directly received from the BMS and displayed on the PCS.)

Solution

1. Check whether the "BMS state" parameter on the "Battery Parameters" page is a BMS fault.

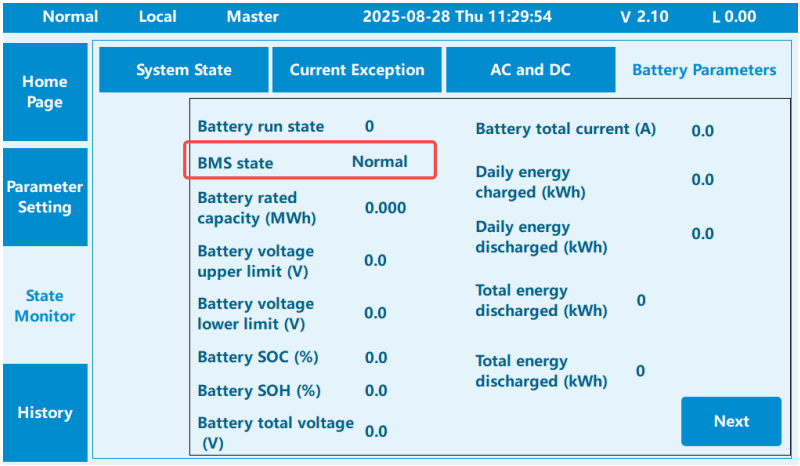


Figure 8-7 Confirming the battery parameter

2. If yes, contact the BMS manufacturer to rectify the fault.

IGBT A-phase fault

Possible causes

- The output circuit of the PCS is grounded or short-circuited.
- The IGBT module is damaged.
- The control hardware is faulty.

Solution

1. Take photos to record the HMI history faults, history operations, history states, and black box faults, and read black box data.

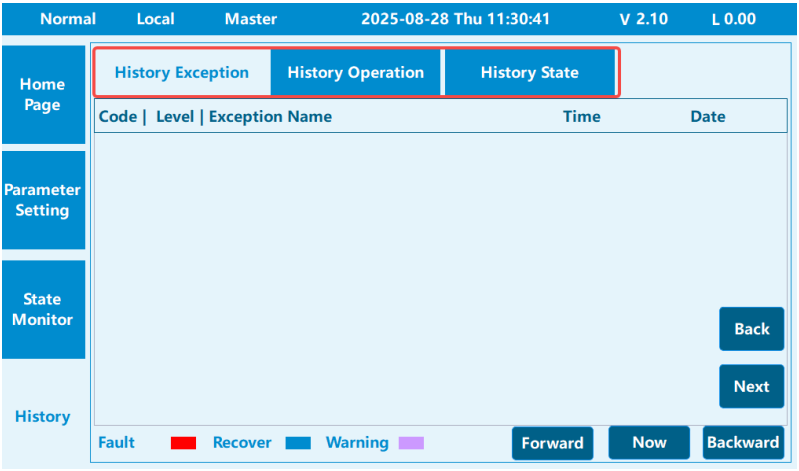


Figure 8-8 Viewing history data

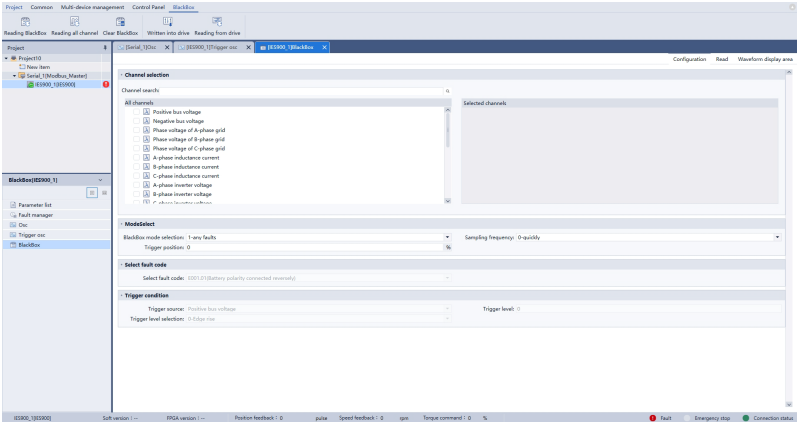


Figure 8-9 Reading black box data

2. Check whether a short circuit occurs on the AC side and whether the drive flat cable is firmly connected.
3. Check whether the three-phase IGBT module is working properly.
4. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

IGBT C-phase fault

Possible causes

IGBT-OT takes effect because the IGBT A-, B-, or C-phase temperature exceeds the temperature protection threshold.

Solution

1. Take photos to record the HMI history faults, history operations, history states, and black box faults, and read black box data.

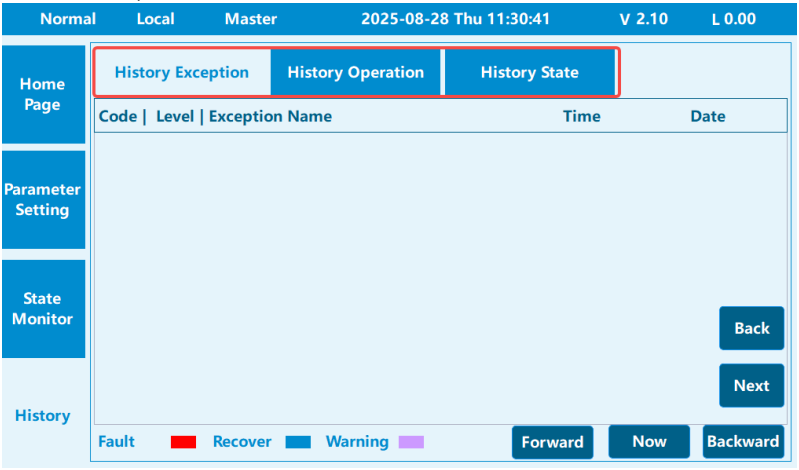


Figure 8-10 Viewing history data

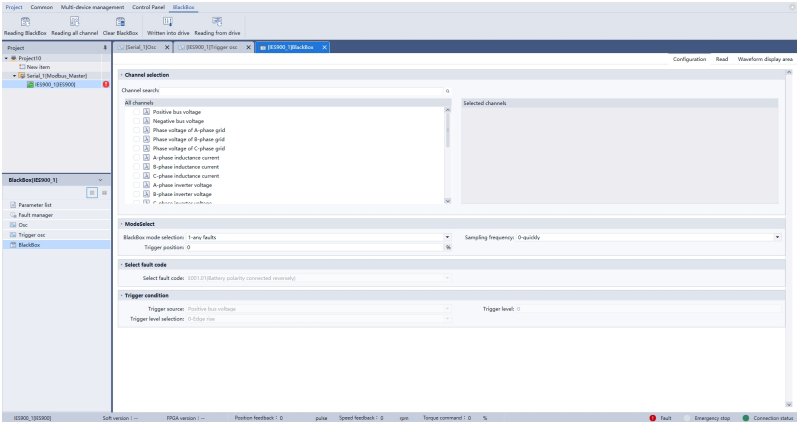


Figure 8-11 Reading black box data

2. Check whether the fan and water pump of the device are working properly.
3. Check whether the three-phase IGBT module is working properly.
4. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

CBC overcurrent

Possible causes

- The load is too large, and too many CBC actions are performed.
- A short circuit occurs on the AC side of the PCS, having an impact on the load.

Solution

1. Take photos to record the HMI history faults, history operations, history states, and black box faults, and read black box data.

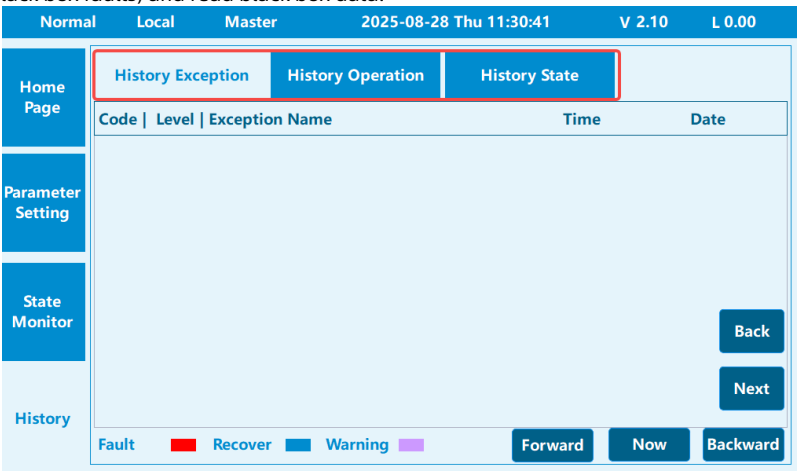


Figure 8-12 Viewing history data

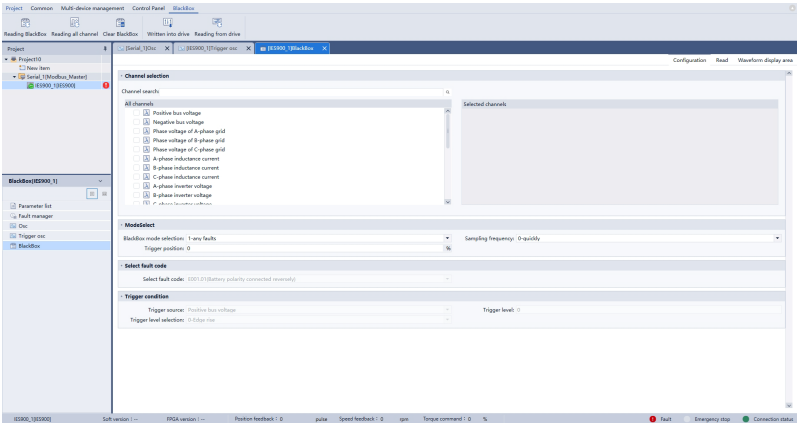


Figure 8-13 Reading black box data

2. Check whether a shirt circuit occurs on the AC side.
3. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Grid AC overvoltage

Possible causes

- The "Level-1 overvoltage coefficient/pu" parameter in group F6 is set to a too small value.
- The actual grid voltage is out of the specified range.
- The three-phase input voltage sampling circuit is abnormal.

Solution

1. Check whether the "Level-1 overvoltage coefficient/pu" parameter in group F6 (Manufacturer parameters) is set to 1.1.

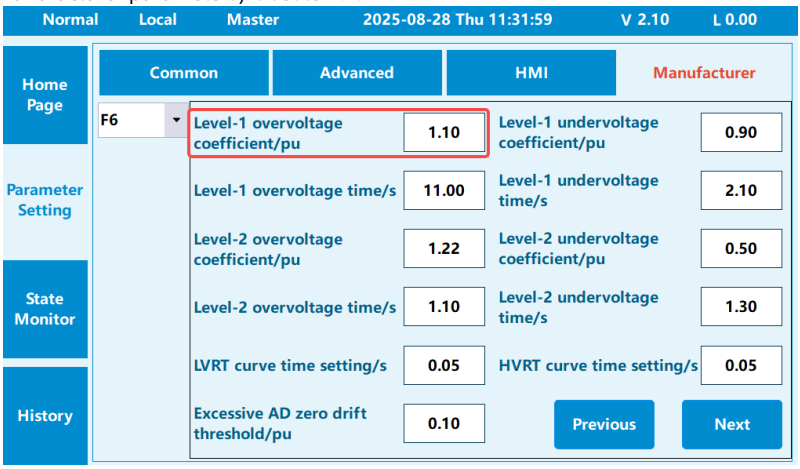


Figure 8-14 Check grid overvoltage (level 1) coefficient

2. Check whether the grid voltage range matches the that on the device nameplate.
3. Check whether the J6 and J35 terminals of the three-phase input voltage sampling circuit are properly connected and whether the three-phase voltage sampling value is correct.
4. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Grid AC undervoltage

Possible causes

- The "1-level undervolt coeff/pu" parameter in group F6 is set to a too large value.
- The actual grid voltage is out of the specified range.
- The three-phase input voltage sampling circuit is abnormal.

Solution

1. Check whether the "1-level undervolt coeff/pu" parameter in group F6 (Manufacturer parameters) is set to 0.9.

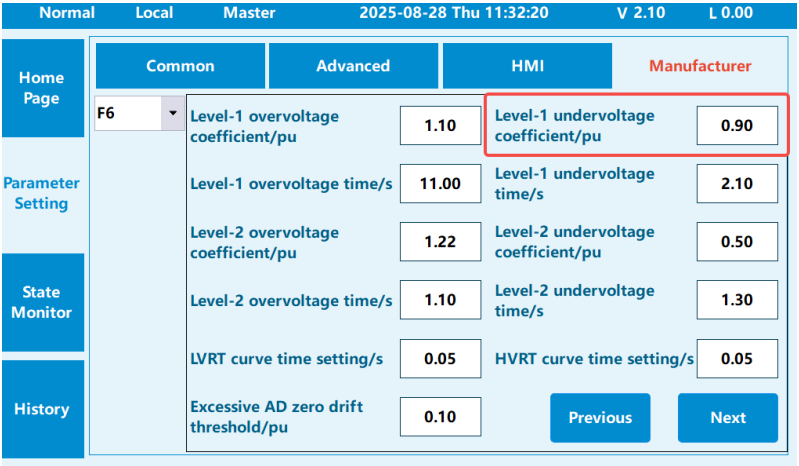


Figure 8-15 Check grid undervoltage (level 1) coefficient

2. Check whether the grid voltage range matches the that on the device nameplate.
3. Check whether the J6 and J35 terminals of the three-phase input voltage sampling circuit are properly connected and whether the three-phase voltage sampling value is correct.
4. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Grid AC overfrequency

Possible causes

- The grid voltage frequency is higher than the value of the "CN-GB level-1 over-freq protection threshold" parameter in group F6.
- The grid is powered off.

Solution

1. Check whether the "CN-GB level-1 over-freq protection threshold" parameter in group F6 (Manufacturer parameters) is set to 50.50.

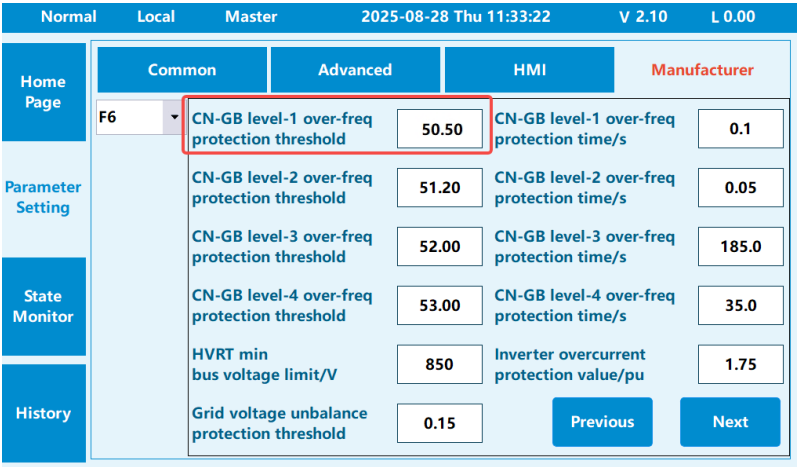


Figure 8-16 Confirming the manufacturer parameter

2. Check whether the power grid voltage is normal.
3. Check the grid record data on the MV side collected in the EMS background.
4. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Grid AC underfrequency

Possible causes

- The grid voltage frequency is lower than the value of the "CN-GB level-1 under-freq protection threshold" parameter in group F6.
- The grid is powered off.

Solution

1. Check whether the "CN-GB level-1 under-freq protection threshold" parameter in group F6 (Manufacturer parameters) is set to 48.5.

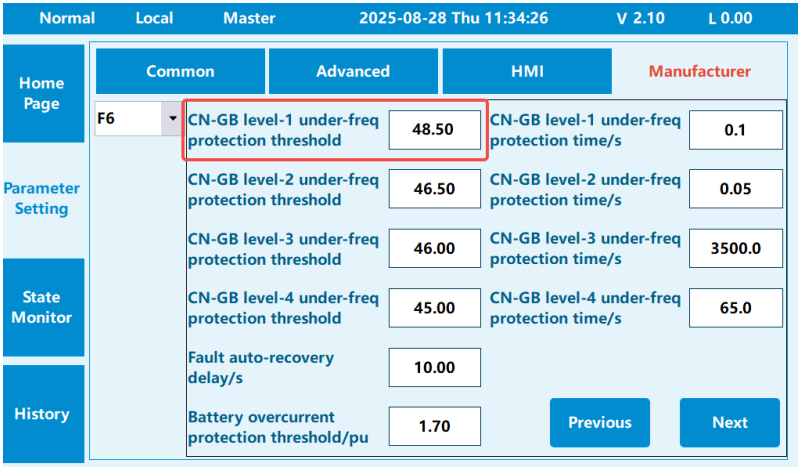


Figure 8-17 Confirming the manufacturer parameter

2. Check whether the power grid voltage is normal.
3. Check the grid record data on the MV side collected in the EMS background.
4. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Grid voltage reversed

Possible causes

The phase sequence in the three-phase grid connection is incorrect.

Solution

1. Check whether the phase sequence in the three-phase grid connection is correct.
2. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Phase lock error

Possible causes

A grid voltage phase lock error occurs.

Solution

1. Take photos to record the HMI history faults, history operations, history states, and black box faults, and read black box data.

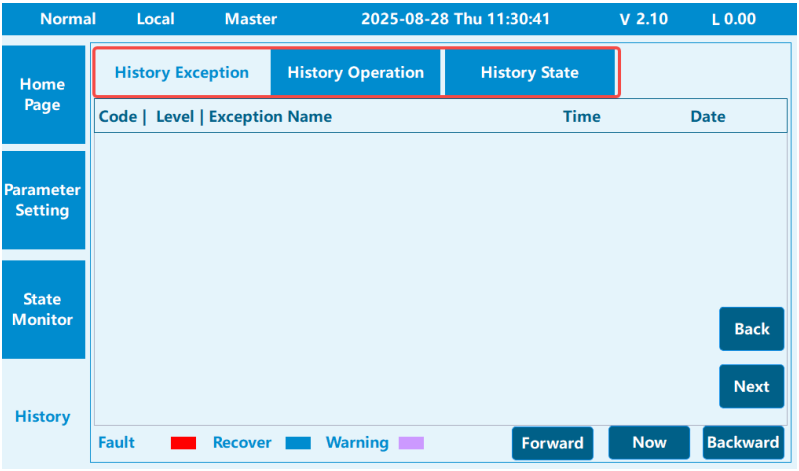


Figure 8-18 Viewing history data

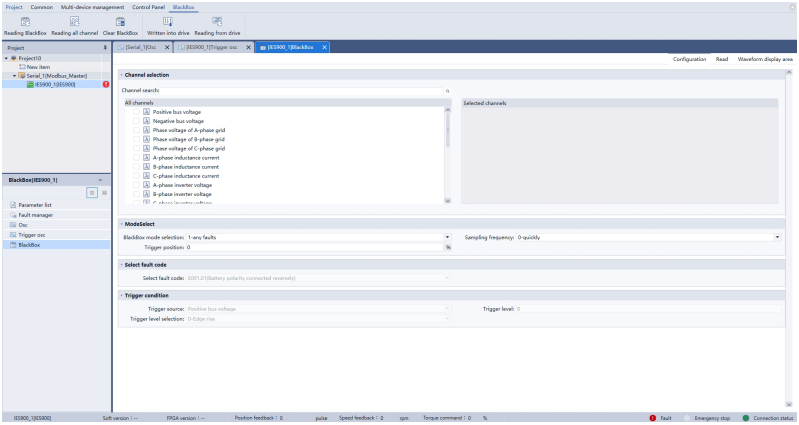


Figure 8-19 Reading black box data

2. Check whether the phase sequence in the three-phase grid connection is correct.
3. Check whether the grid voltage sampling circuit is disconnected or loosely connected.
4. Check whether the A/B/C three-phase output voltage is normal.
5. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Grid voltage imbalance

Possible causes

- The three-phase grid voltage imbalance is higher than the value of the "Grid voltage unbalance protection threshold" parameter in group F6.

- Grid input phase loss occurs.
- Grid voltage resonance occurs.

Solution

1. Take photos to record the HMI history faults, history operations, history states, and black box faults, and read black box data.

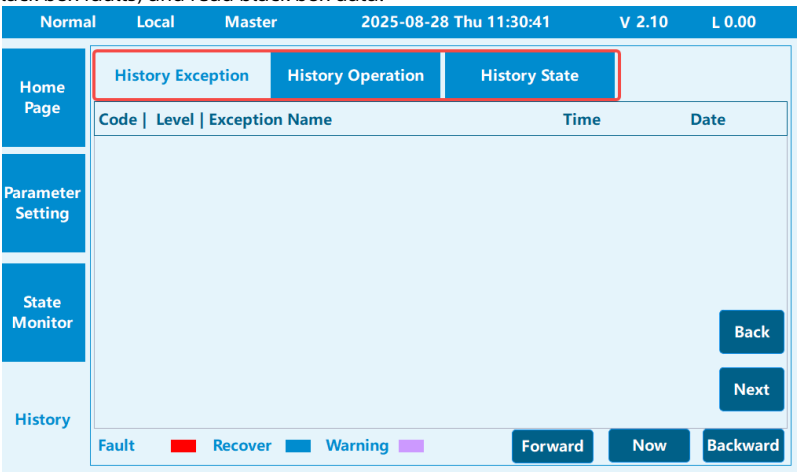


Figure 8-20 Viewing history data

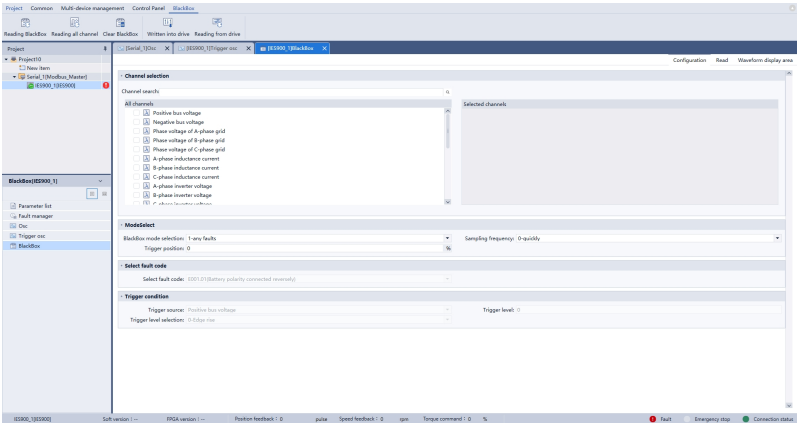


Figure 8-21 Reading black box data

2. Check whether the three-phase voltage is balanced and whether phase loss occurs.
3. Check whether the value of the "Grid voltage unbalance protection threshold" parameter in group F6 is 0.15.

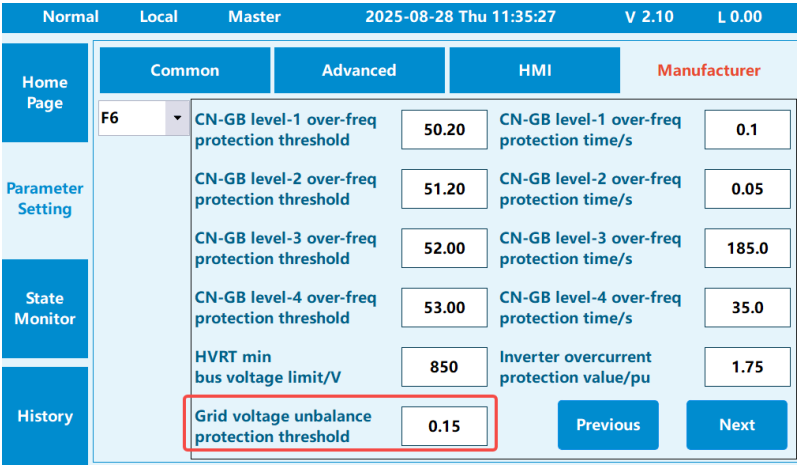


Figure 8-22 Confirming the manufacturer parameter

4. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Grid voltage mismatch

Possible causes

The amplitude or frequency of the three-phase grid voltage exceeds the allowed range before startup.

Solution

1. Check whether the amplitude of the three-phase grid voltage or frequency exceeds the allowed range of the device.
2. Check whether the grid voltage sampling circuit is disconnected or loosely connected.
3. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Islanding fault

Possible causes

- The grid is disconnected, or the device runs with load separately.
- The grid is weak, and factory defaults of the islanding parameters are not proper.

Solution

1. Take photos to record the HMI history faults, history operations, history states, and black box faults, and read black box data.

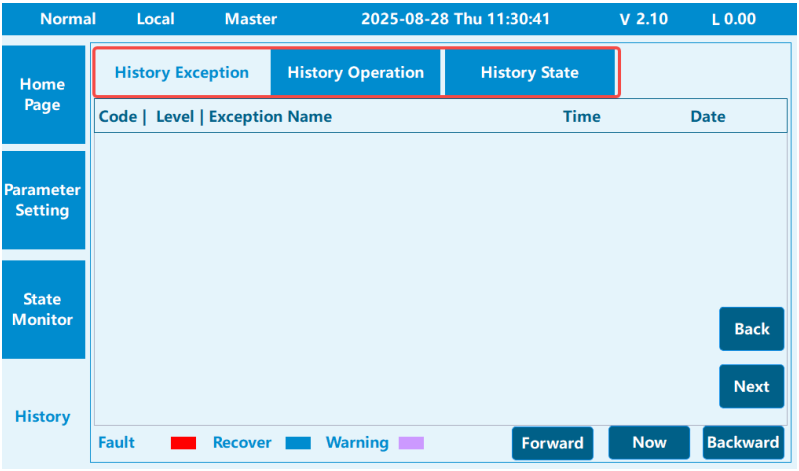


Figure 8-23 Viewing history data

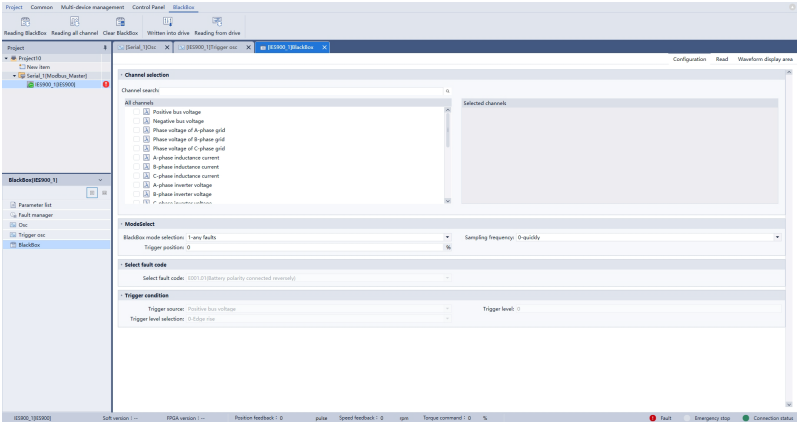


Figure 8-24 Reading black box data

2. Check whether islanding parameters F5.126 to F5.132 use factory defaults.
3. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Bus soft start fault

Possible causes

- The pre-charge contactor fails.
- The air switch of the pre-charge circuit is open.
- The battery voltage is too low.
- The bus voltage sampling cable is disconnected.

- The pre-charge resistor is faulty.

Solution

1. Take photos to record the HMI history faults, history operations, history states, and black box faults, and read black box data.

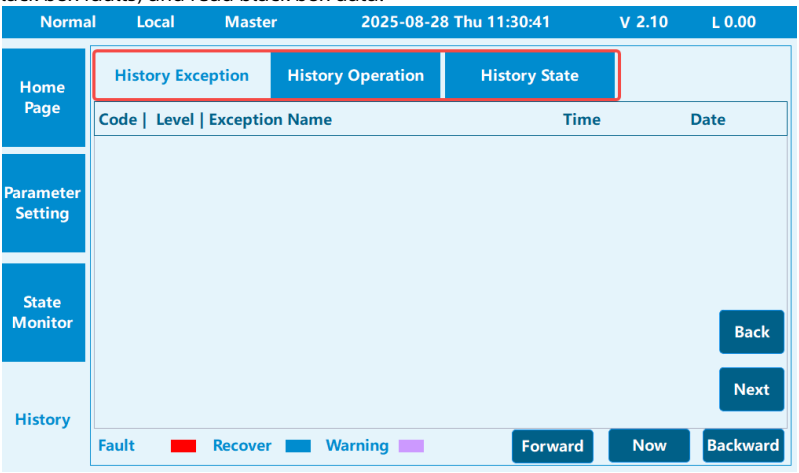


Figure 8-25 Viewing history data

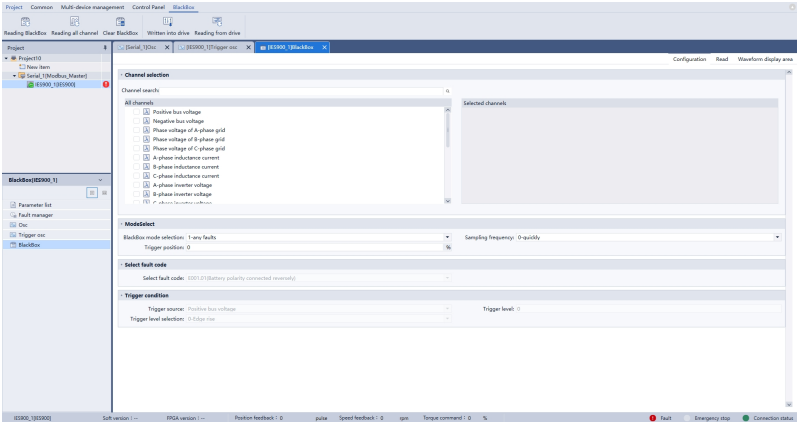


Figure 8-26 Reading black box data

2. Check whether the pre-charge contactor is working properly.
3. Check whether the battery voltage matches the model.
4. Check whether the bus voltage sampling cable is connected correctly and whether disconnection or loose connection occurs.
5. Check whether the pre-charge resistor is working properly.

6. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Low insulation resistance

Possible causes

The insulation resistance is lower than the threshold.

Solution

1. Check whether the value of the "Insulation impedance protection threshold/kΩ" parameter in group F6 (Manufacturer parameters) is lower than 50.

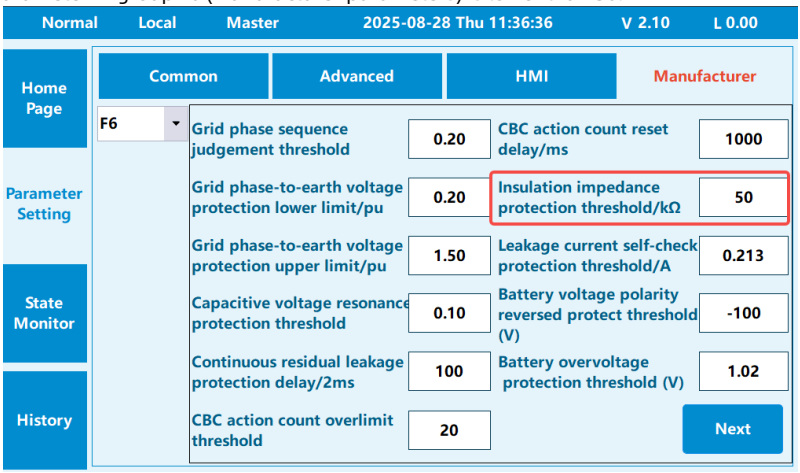


Figure 8-27 Confirming the manufacturer parameter

2. Check the grounding insulation resistance of the PCS main circuit and perform an insulation test to confirm the insulation resistance.
3. Check the master and slave relationship of the PCSs and whether insulation detection is enabled for both PCSs in parallel mode. (Insulation detection cannot be simultaneously enabled for both PCSs in parallel mode.)
4. Replace the insulation detection board.
5. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Insulation detection fault

Possible causes

The insulation detection switch command is different from the feedback.

Solution

1. Check whether the insulation resistance detection contactor can be normally switched on.

2. Check whether the feedback signal of the insulation resistance detection contactor is normal.
3. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

IGBT over-temperature

Possible causes

- The temperature of the IGBT module is too high. The over-temperature threshold varies based on the model.
- The water pump and AC fan work improperly.
- The temperature sensor is damaged.
- The load is too large during off-grid operation.

Solution

1. Check whether the filter screen on the air inlet of the device is blocked and whether there is a heat source around the air inlet.
2. Check whether all the fans and water pumps inside the device work properly.
3. Check whether the IGBT over-temperature threshold matches the device model.
4. Collect the fault information (HMI history exception, history operations, and history states), take a photo of the settings of parameters in group E0, and read black box fault data.

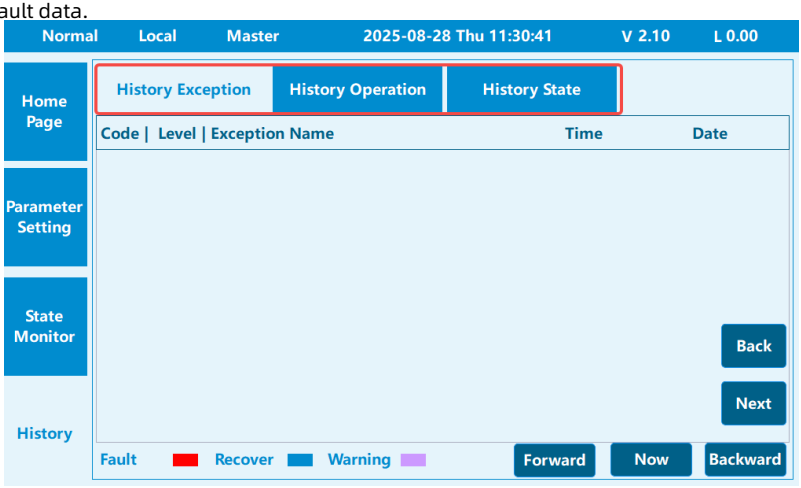


Figure 8-28 Viewing history data

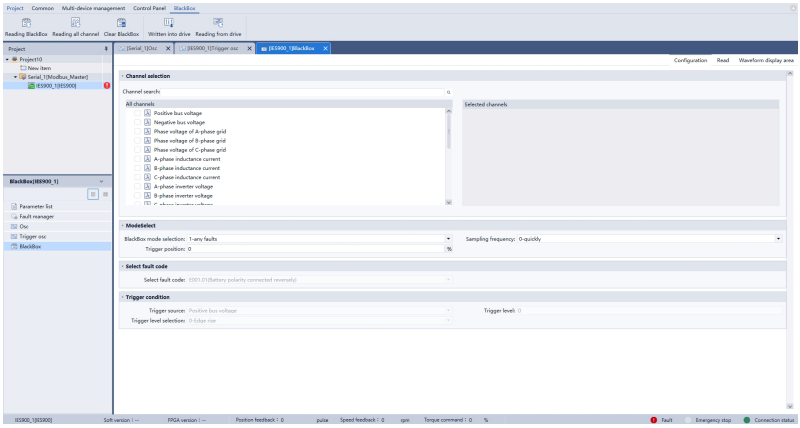


Figure 8-29 Reading black box data

5. Check whether the device load is too large during off-grid operation.
6. Measure and compare the NTC resistance of pins 1 and 10, pins 2 and 11, and pins 3 and 12 on the NTC sampling cable terminal J27 (communication board) of the IGBT module. In normal cases, the resistance of the three phases is similar.
   - Water pump port: M1 connected to pin 13, 14, 15, 16, 17, 18, 19, or 20 of X8
   - AC fan power port: FAN1 or FAN2 connected to pin 1, 2, 3, 4, 5, 6, 7, or 8 of X8
   - AC fan regulating port: FAN1 connected to pin 25 or 26 of X8 and FAN2 connected to pin 23 or 24 of X8
7. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

IGBT low-temperature

Possible causes

- The temperature of the IGBT module is too low.
- The NTC temperature sensor is damaged.
- The flat cable is not connected.

Solution

1. Check whether the ambient temperature is lower than the low temperature protection threshold.
2. Measure and compare the NTC resistance of pins 1 and 10, pins 2 and 11, and pins 3 and 12 on the NTC sampling cable terminal J27 (communication board) of the IGBT module. In normal cases, the resistance of the three phases is similar.
3. Check whether the flat cable is properly connected.
4. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Parallel communication fault

Possible causes

The communication between the control board and other PCSs fails.

Solution

1. Check the value of the "Local role" parameter in group F8 (Manufacturer parameters) on the HMI. Confirm that at least one master and one slave are available.

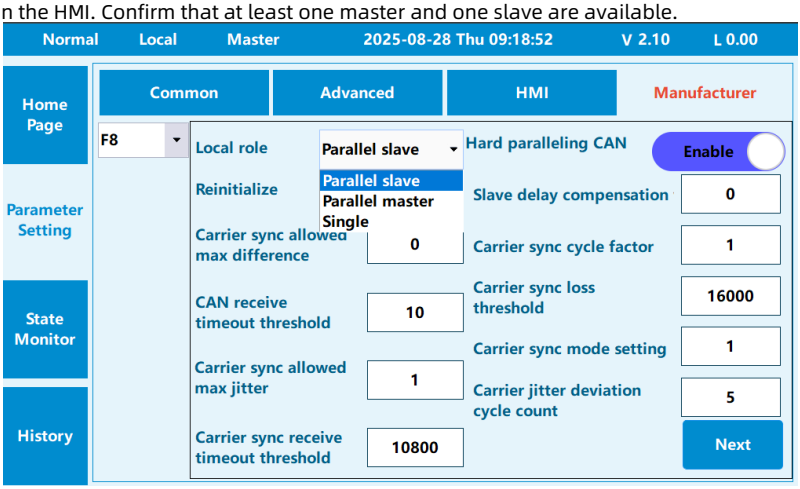


Figure 8-30 Confirming the manufacturer parameter

2. Check whether the DIP switch S6 is ON.
3. Check the CAN communication cable for the parallel system according to the electrical wiring.
4. Replace the flat cable and check whether the fault persists.
5. Replace the communication board and check whether the fault persists.
6. Replace the detection board and check whether the fault persists.
7. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Carrier synchronization fault

Possible causes

The carrier synchronization cable is incorrectly connected.

Solution

1. Check the value of the "Local role" parameter in group F8 (Manufacturer parameters) on the HMI. Confirm that at least one master and one slave are available.

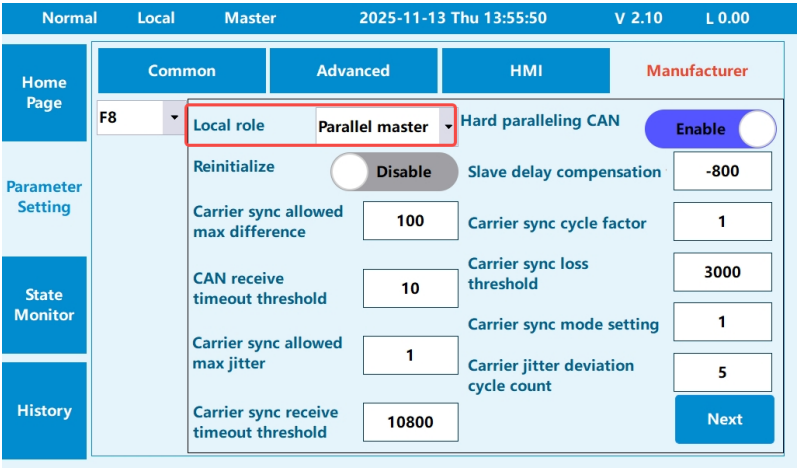


Figure 8-31 Confirming the manufacturer parameter

2. Check the sequence of the carrier synchronization cable according to the electrical wiring.
3. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

PCS association fault

Possible causes

A fault occurs in the HDI/HDO parallel system.

Solution

1. Check whether another fault except the PCS association fault occurs on the parallel device.
2. Rectify the fault and check whether the PCS association fault is automatically reset.
3. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Internal communication fault

Possible causes

The internal CAN communication is abnormal.

Solution

1. On the HMI, choose "State Monitor" > "System State> "Product Information" to check the software version of the primary and secondary ARM. Confirm that the software version of both the primary ARM and secondary ARM is read.

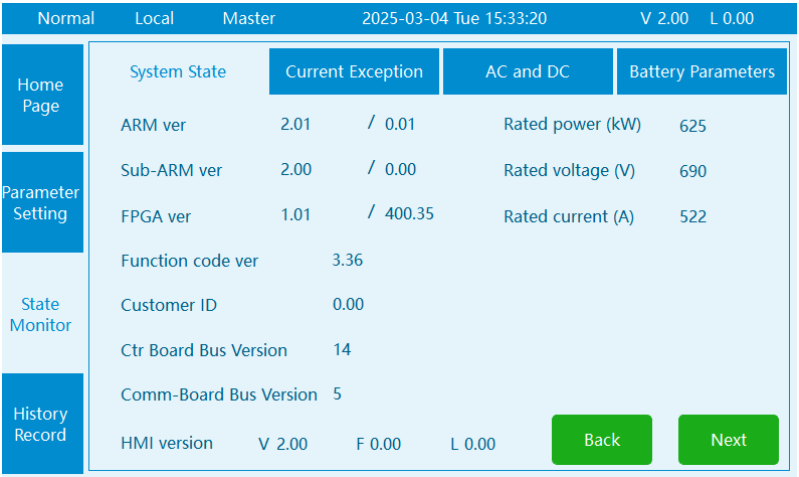


Figure 8-32 Confirming the manufacturer parameter

2. Check whether the flat cable between the detection board and communication board is properly connected.
3. Replace the flat cable and check whether the fault persists.
4. Replace the communication board and check whether the fault persists.
5. Replace the detection board and check whether the fault persists.
6. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

HMI communication fault

Possible causes

The HMI is disconnected from the user adapter board.

Solution

1. Check whether the communication cable between the HMI and user adapter board is connected properly.
2. Power on the device again and remove and insert the communication cable or replace the communication cable.
3. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

BMS communication fault

Possible causes

The BMS communication connection timed out.

Solution

1. Check whether the communication connection circuit is disconnected.

2. Check whether the DIP switches (CAN-S4 and RS485-S1) of the communication board are ON.
3. Check whether BMS communication protocol and Modbus protocol manufacturer are correctly configured.

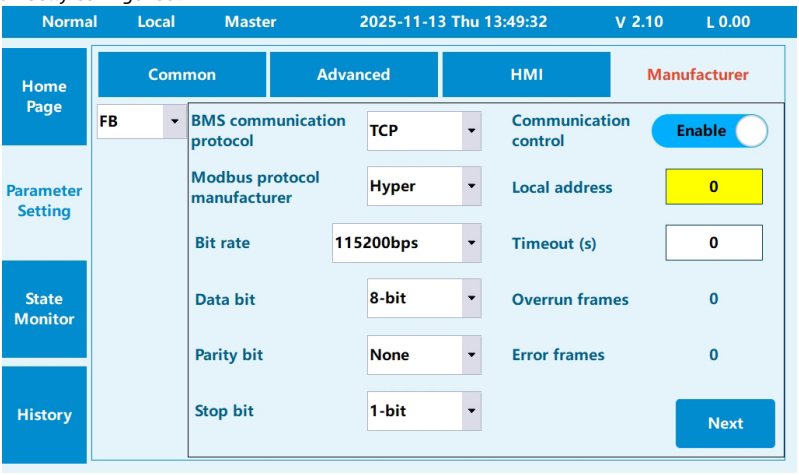


Figure 8-33 Confirming the manufacturer parameter

4. Check whether the local address is correctly configured.
5. Check whether the master and slave use the same data format.
6. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

EMS communication fault

Possible causes

EMS communication is disconnected.

Solution

1. Check whether the "EMS communication protocol" parameter in group FA is correctly configured.

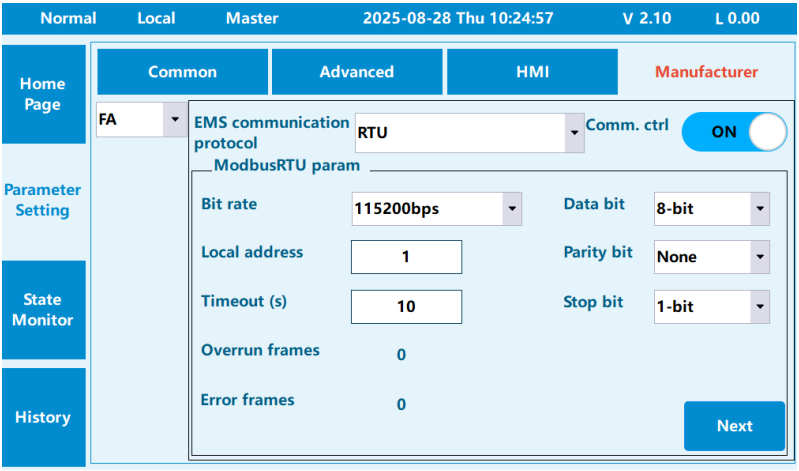


Figure 8-34 Confirming the manufacturer parameter

2. Check whether the local address is correctly configured.
3. Check whether the external EMS communication circuit is normal.
4. Turn the control power off and then on, and remove and insert the LAN cable.
5. Check whether the expansion card and main control board are connected firmly. If not, press the cable tightly.
6. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Three-phase fan fault

Possible causes

- Three-phase fan locked-rotor occurs.
- The power cable of the three-phase fan is not connected.
- A fan power phase sequence error occurs.

Solution

1. Check whether the fan power phase sequence is correct. (Check whether the fan is running in the correct direction in forced mode.)
2. Check whether a power phase loss occurs on the operating fan and whether the voltage of the three phases is 400 VAC.
3. Change the value of the "Active damping proportional coefficient" parameter in group F5 to 0.350 when external power supply is available.

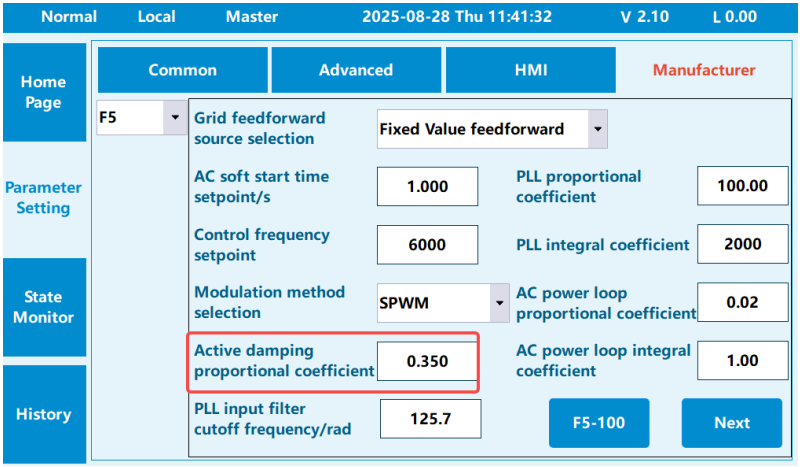


Figure 8-35 Confirming the manufacturer parameter

4. Check whether fluctuations occur in case of a grid voltage fault.
5. Check whether the total harmonics distortion (THD) of the fan's supply voltage is too large.
6. Check whether the fan power contactor KM4 is switched on.
7. Check whether the fan power circuit is disconnected.
8. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

Master-slave fault

Possible causes

- The software versions of the master and slave are inconsistent.
- Failed to identify the master and slave.
- The master and slave roles are conflicting.
- The active or reactive power or DC power of the master and salve is inconsistent.

Solution

1. Choose "State Monitor" > "System State" > "Product Information"and check the software version of the master and slave. Confirm that the software versions of the master and slave are consistent.

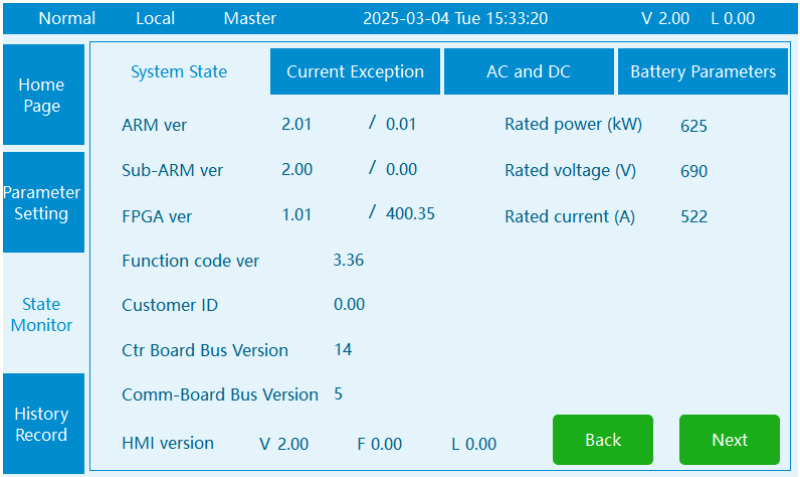


Figure 8-36 Confirming the manufacturer parameter

2. Check the value of the "Local role" parameter in group U0 (Manufacturer parameters) and confirm that the master and slave roles meet the actual situation.
3. Check whether the power delivered for the master and slave is consistent.
4. If the fault persists, contact the after-sales service engineer of Wattox to rectify the fault.

9 Appendix

9.1 Burning the IEC 61850 Communication Program

1. If the SD card is used for the first time, format the SD card.
   a. Prepare the following devices and software: SD card, card reader, PC (Windows 7 operating system), software for kernel upgrade (robotsystool.exe).
   b. Double-click to execute robotsystool.exe.
   c. Select the drive letter of the SD card for "USB" and click "Format".
2. Check whether the software is the latest version.
   If no, use the IT7000ubi folder for last archived software to replace the folder with the same name in the SD card.
3. Copy the eight files in the PCB software folder to the SD card, uninstall the SD card, and then remove the SD card.
4. Short-connect the jumper cap to the start port of the SD card, insert the SD card, connect the board to the power supply, and then wait for 90s for the upgrade to complete. Power off to restart the board.
5. Modify the IP address of the PC.
6. Remotely commission SecureCRT.exe.
7. Choose "File" > "Quick connection".
8. In the "Quick connection" dialog box, select "Telnet" for "Protocol".
9. Set the hostname to 192.168.1.217 and click "Connect".
10. On the interface that is displayed, enter "root" next to "carrier login:" and press "Enter".
11. The "Password" field is displayed. Press "Enter".
12. Enter "cd /users/carrier/cidFiles" and press "Enter". Then, enter "vi IES1000.cid" and press "Enter".
13. Check the version information.

9.2 DAID Use Rules

The following table describes the DAID use rules. Field 1 defines data functions. Field 2 defines communication services used by the data. Field 3 defines data type conversion and accuracy. Fields 4 and 5 define data mapping of the parameter number.

DAID	Value	Description	
Field 1	1	DA indicates telemetering and telesignaling data.	-
	2	DA indicates the time of telemetering and telesignaling data.	-
	3	DA indicates teleadjustment and telecontrol data.	-
	4	Fixed value group	The values cannot be changed.
	5	DA indicates (quick) teleadjustment and telecontrol data.	-
	Others	Reserved	-
Field 2	0	MMS data	-
	1	MMS and GOOSE data	-
	Others	Reserved	-
Field 3	0	DA data type	Timer data type
	10		float→U16, no decimal place
	11		float→U16, one decimal place
	12		float→U16, two decimal places
	13		float→U16, three decimal places
	20		float→S16, no decimal place
	21		float→S16, one decimal place
	22		float→S16, two decimal places
	23		float→S16, three decimal places
	30		BOOL
	40		INT32
	50		INT32U
	60		INT16U
	70		INT16
	80		INT8U
	90		INT8
	Others		Reserved
Field 4	index	High bit in the parameter number	0x20F0 (parameter number: 0x20F0)
Field 5	subindex	Low bit in the parameter number	1 (parameter number: 0)

9.3 Power Unit Measurement Guide

CAUTION

During the check, reduce the bus voltage to 0 V before you measure all the IGBT modules.

Tools

No.	Tools	Quantity
1	Multimeter	1
2	Screwdriver	1

Power unit

For details about the power unit topology, see *"Figure 9-1 " on page 150* . For details about pins on the power unit, see *"Figure 9-2 " on page 151* .

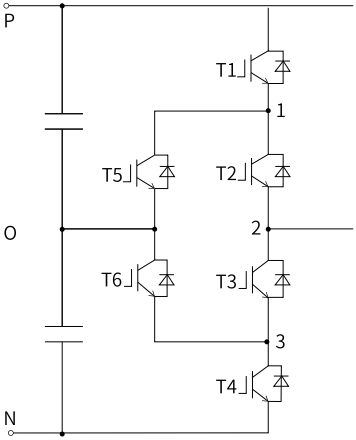


Figure 9-1 Power unit topology

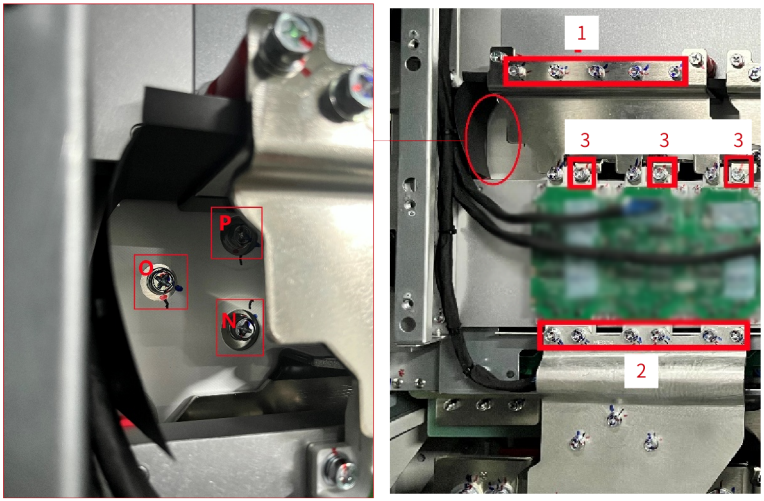


Figure 9-2 Pins on the power unit

Note

Only phase A is marked in the figure, and phase B and phase C are to the right. Three phases form a group. All the three phases must be measured during the power unit measurement.

IGBT test

Use the diode gear of the multimeter to check whether the IGBT module of tubes T1 to T6 of the same phase is damaged. The following uses phase A as an example.

- T1: Connect the red pen to point 1 and the black pen to point P and measure the voltage. A voltage value of about 0.3 V indicates normal.
- T2: Connect the red pen to point 2 and the black pen to point 1 and measure the voltage. A voltage value of about 0.3 V indicates normal.
- T3: Connect the red pen to point 3 and the black pen to point 2 and measure the voltage. A voltage value of about 0.3 V indicates normal.
- T4: Connect the red pen to point N and the black pen to point 3 and measure the voltage. A voltage value of about 0.3 V indicates normal.
- T5: Connect the red pen to point O and the black pen to point 1 and measure the voltage. A voltage value of about 0.3 V indicates normal.
- T6: Connect the red pen to point 3 and the black pen to point O and measure the voltage. A voltage value of about 0.3 V indicates normal.

9.4 Terminology

Term	Description
PCS	Power conversion system
BMS	Battery management system
EMS	Energy management system
61850	IEC 61850 communication protocol
PQ mode	Power control mode
VF mode	Voltage/frequency control mode
Parallel operation	Carrier synchronization connection
MMS network	IEC 61850 information network
GOOSE network	IEC 61850 control network
IEDname	Naming information of the PCS on the IEC 61850 network
CID file	Configuration file of the IED instance for the IEC 61850 configuration process
SCD file	File containing device information of the whole station used in the IEC 61850 configuration process

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